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Padala

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(54) **SYSTEMS, DEVICES, AND METHODS FOR REDUCING HEART VALVE REGURGITATION**

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See application file for complete search history.

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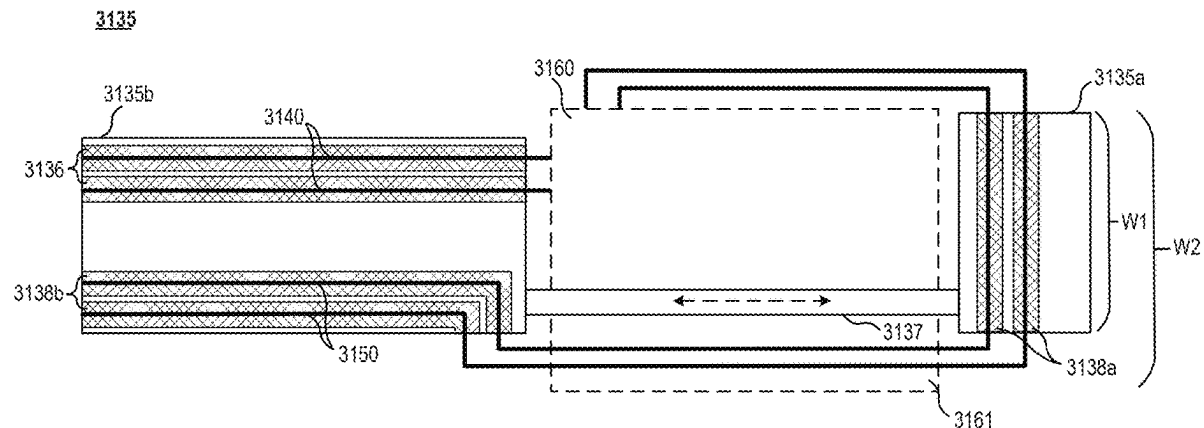
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ABSTRACT

Embodiments described herein relate to an implant delivery system for delivering an implant for reducing heart valve regurgitation. The implant delivery system may include an implant catheter disposed in an inner lumen of a guide catheter. The implant catheter may include one or more hypotubes disposed therein, each hypotube configured to receive an elongate member such as a braided tether. A distal end of the implant catheter may be coupled to an implant holder configured to receive the implant. The implant holder may define one or more channels, each channel configured to receive a portion of a respective elongate member. The elongate members configured to couple the implant to the implant holder and transition the implant between configurations. The implant configured to be disposed around a portion of a leaflet of a heart valve to improve coaptation of the heart valve.

31 Claims, 47 Drawing Sheets



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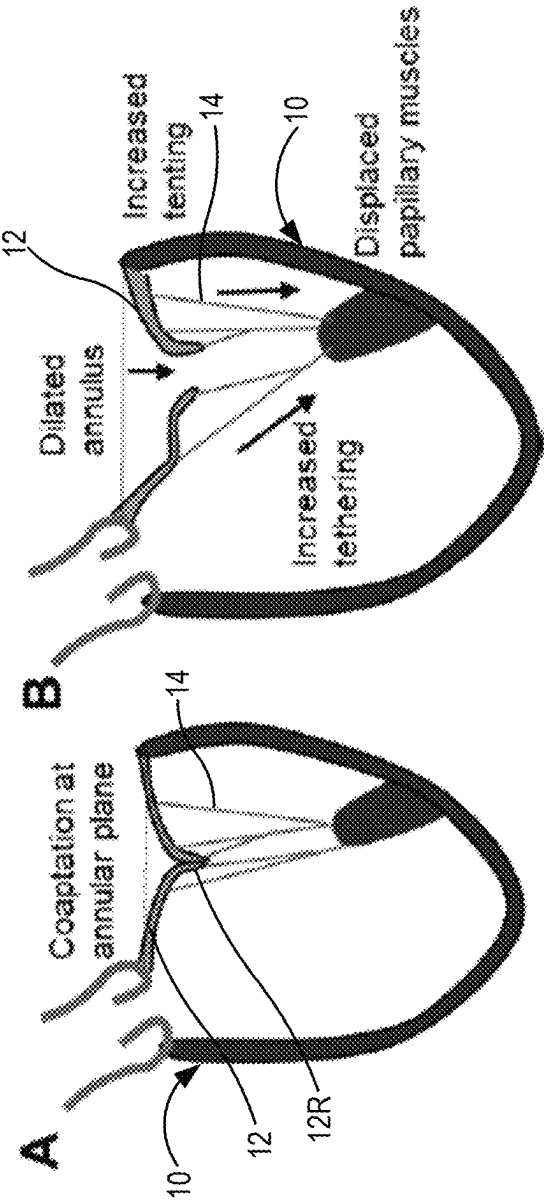


FIG. 1A

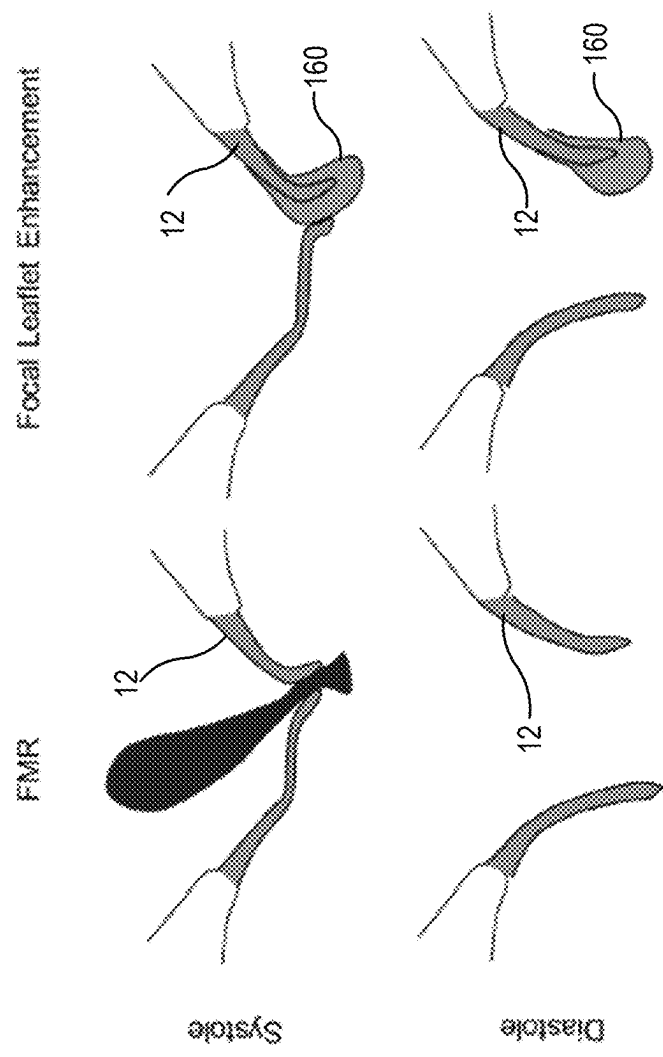


FIG. 1B

100 

GC Actuator 112	Guide Catheter 110
DC Actuator 122	Delivery Catheter 120
IC Actuator 132	Implant Catheter 130
AT Actuator 142	Atrial Tether 140
VT Actuator 152	Ventricular Tether 150
	Implant Holder 135
	Implant 160

FIG. 2A

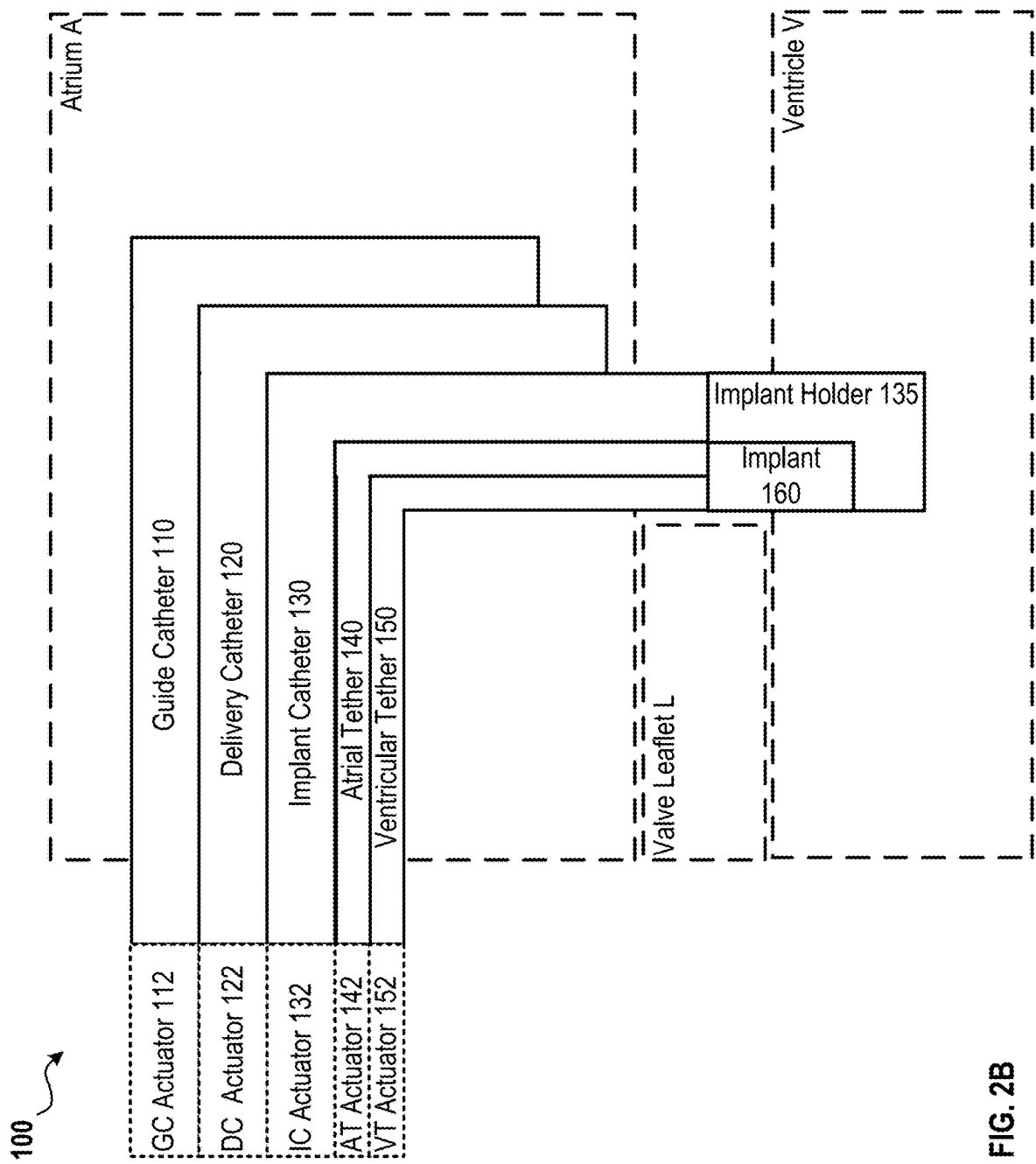
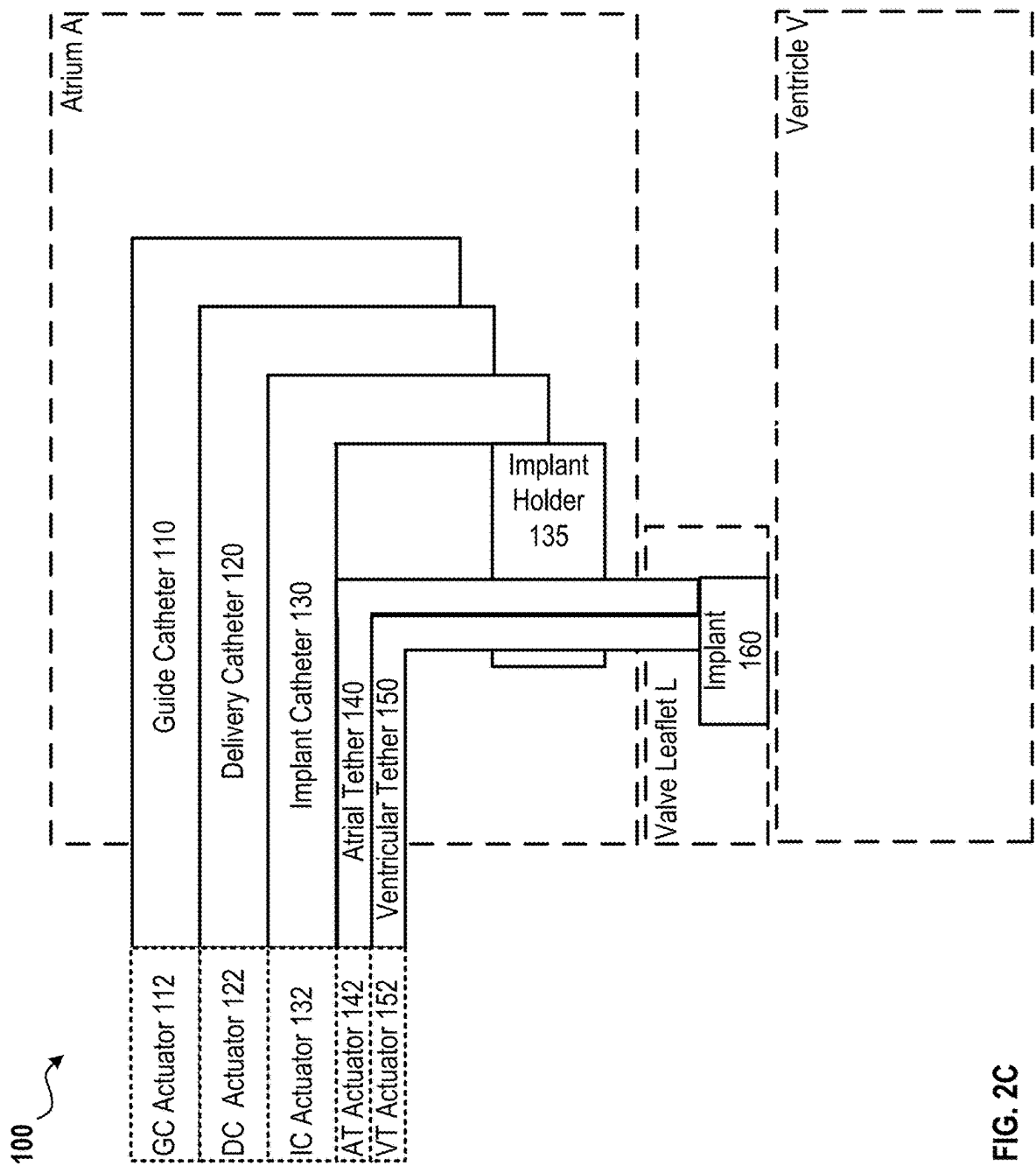


FIG. 2B



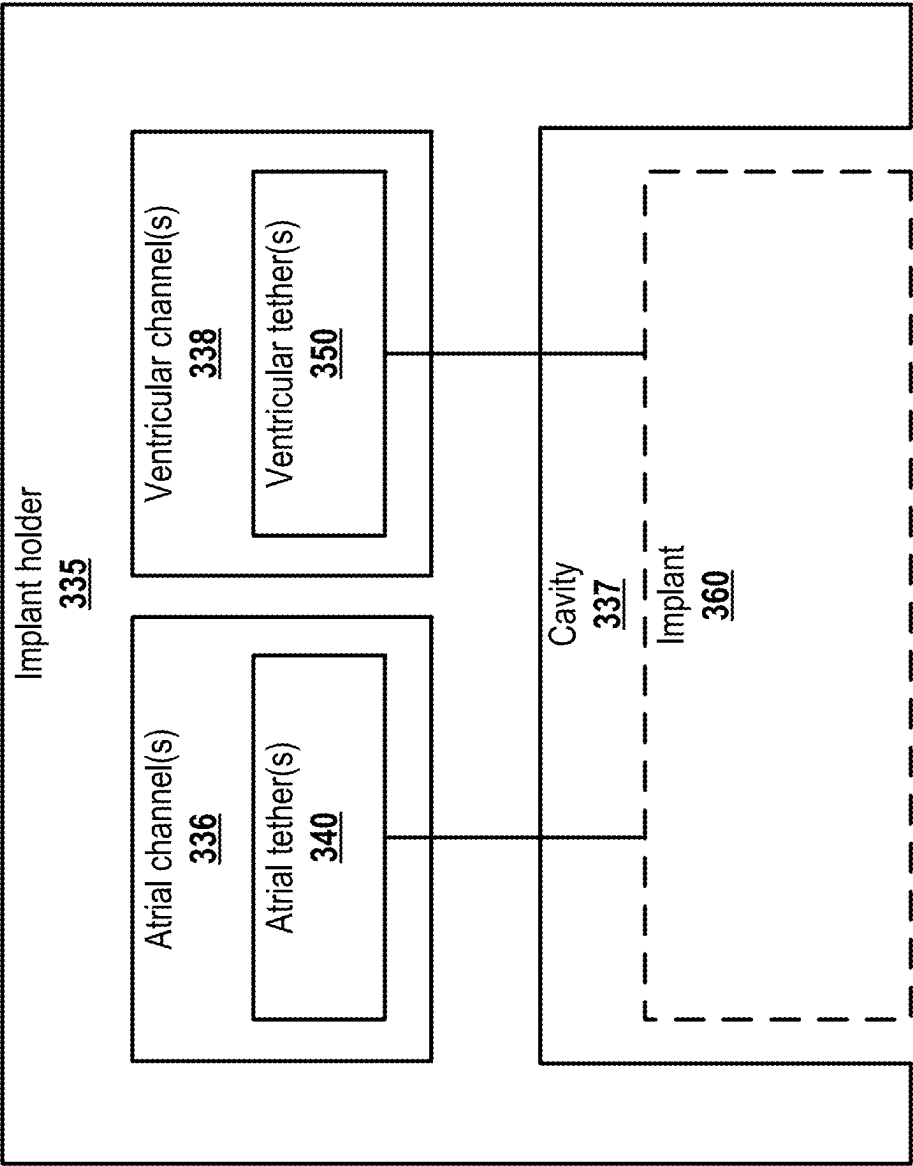


FIG. 3

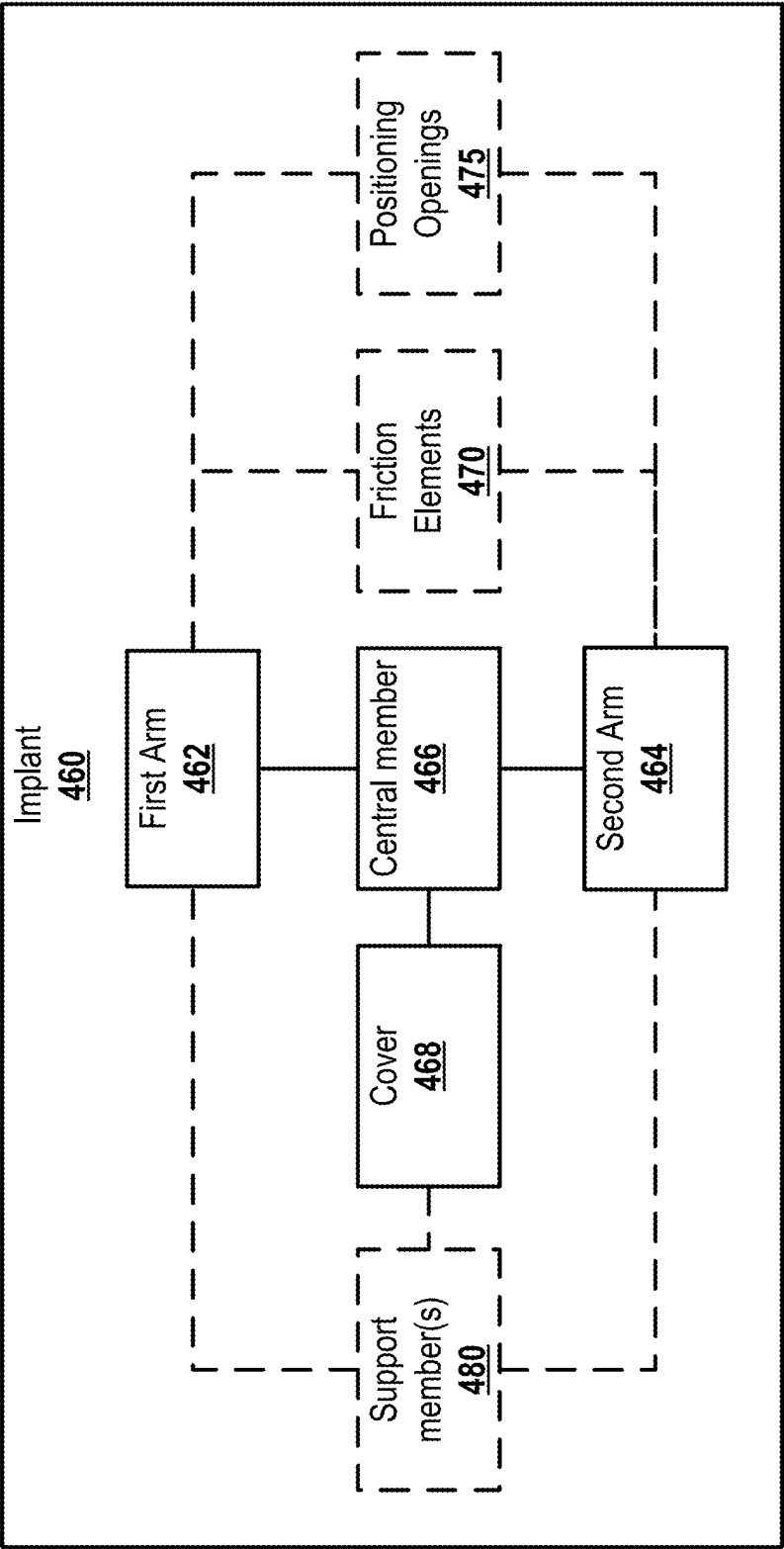


FIG. 4

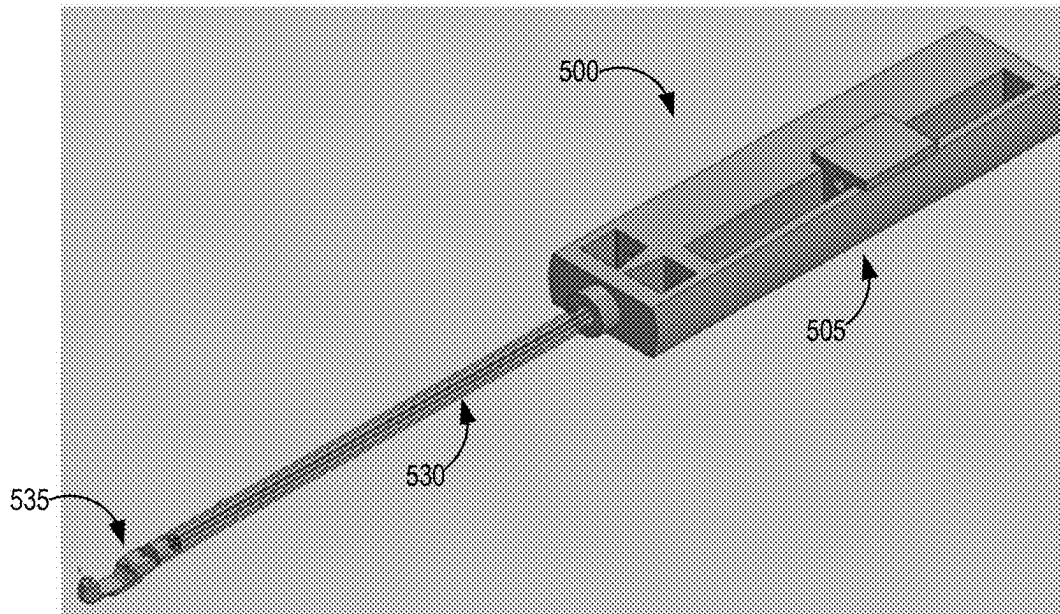


FIG. 5A

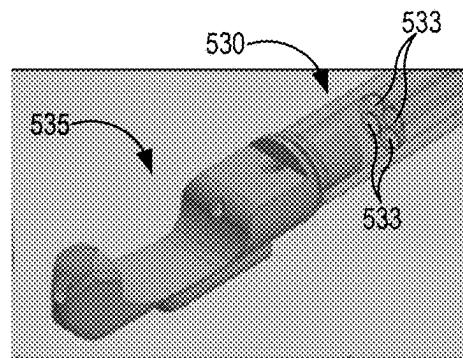


FIG. 5B

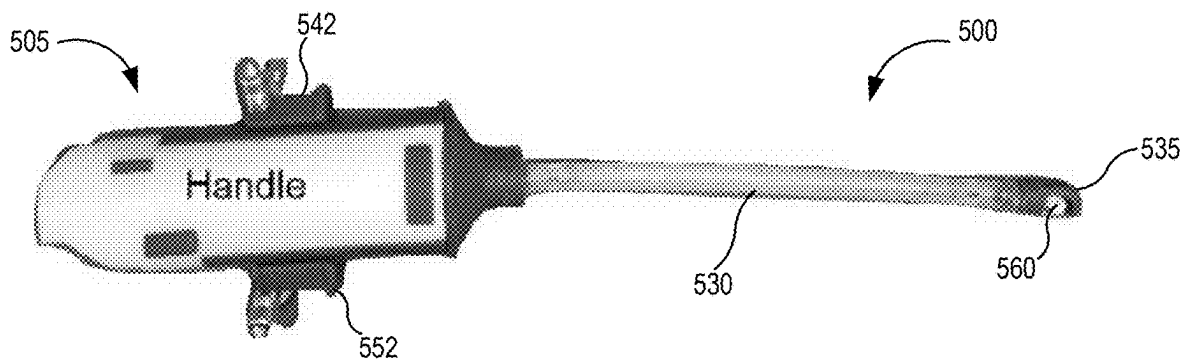
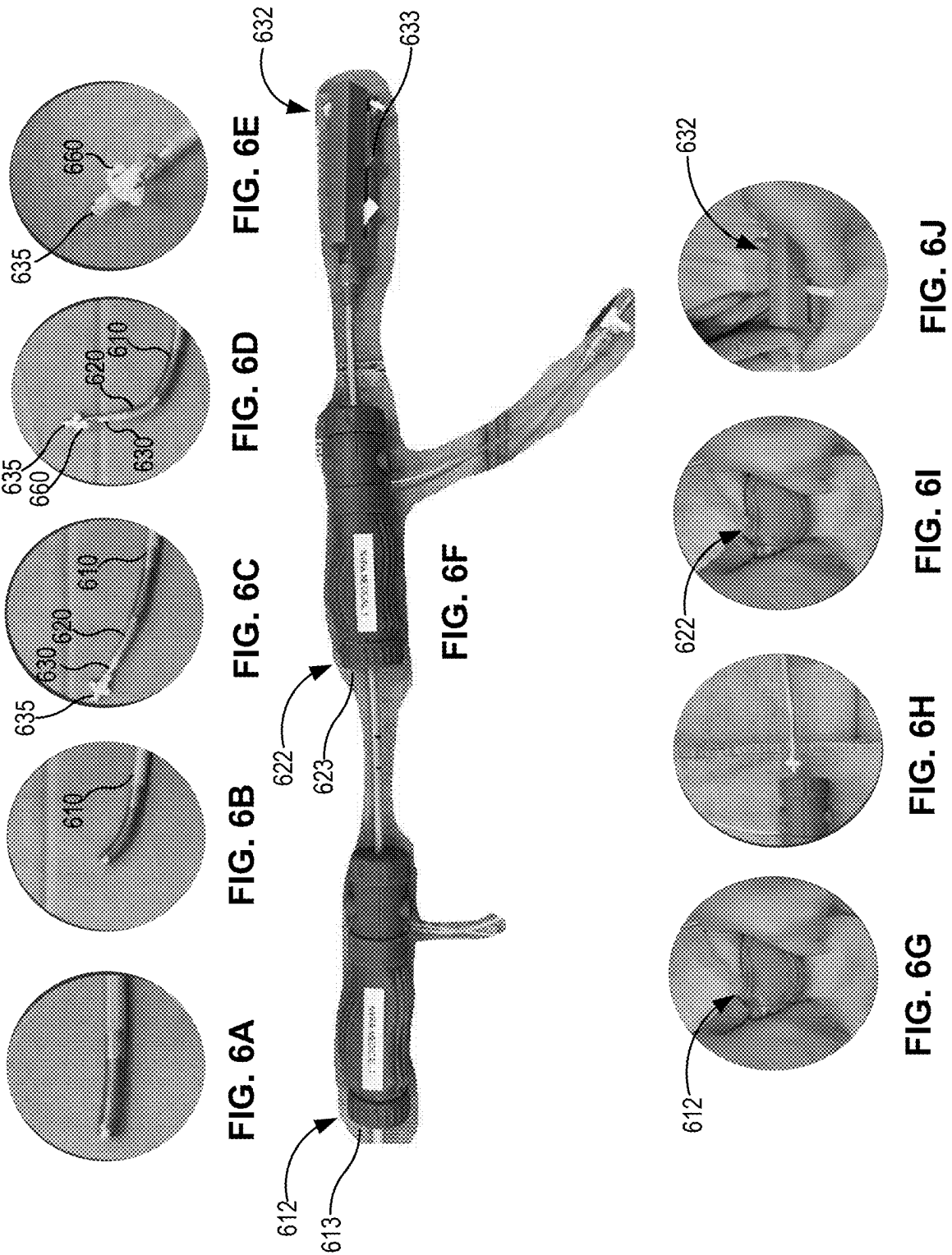


FIG. 5C



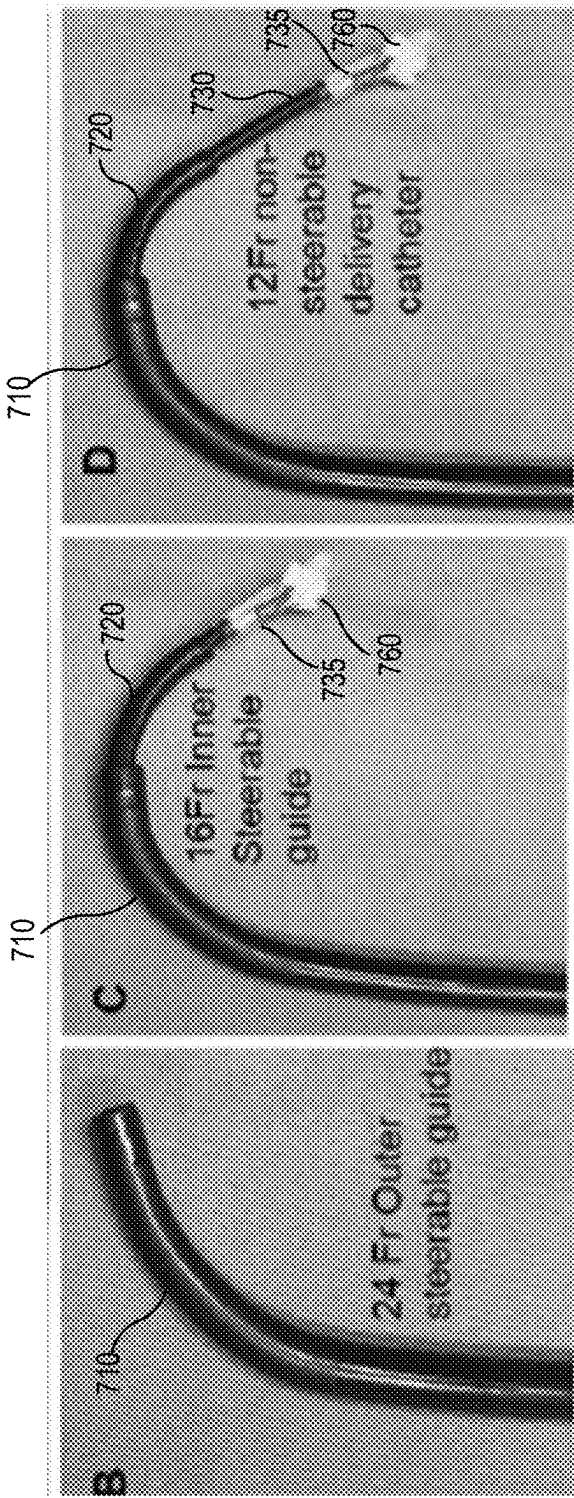
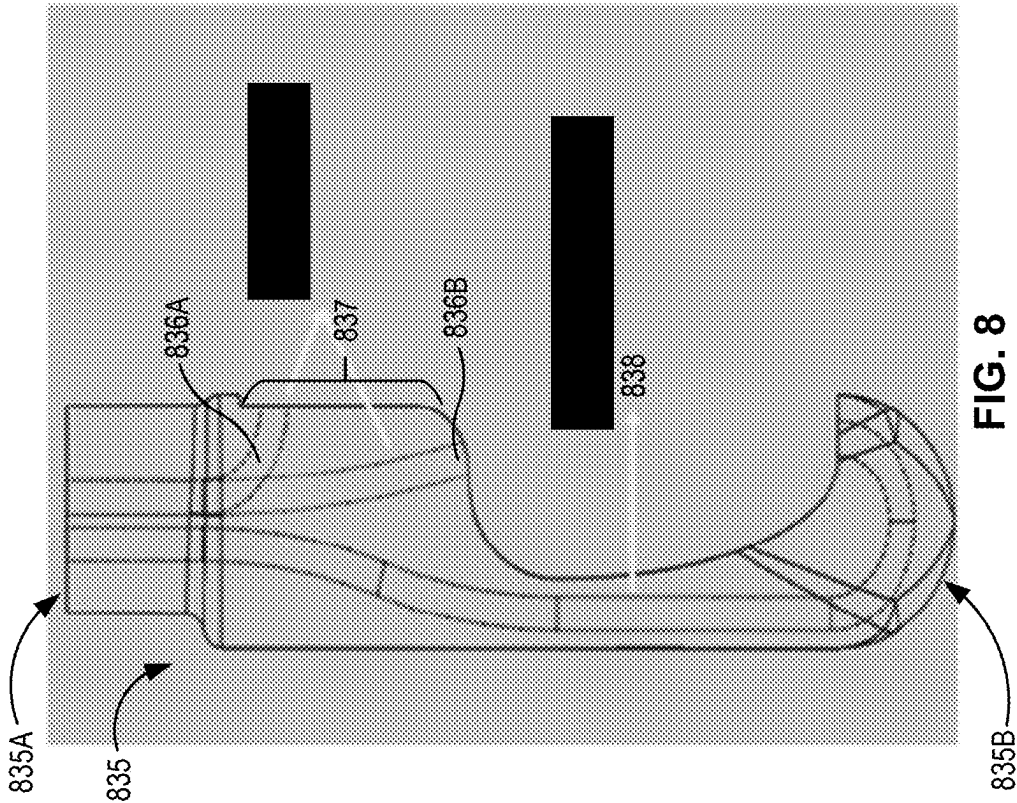


FIG. 7C

FIG. 7B

FIG. 7A



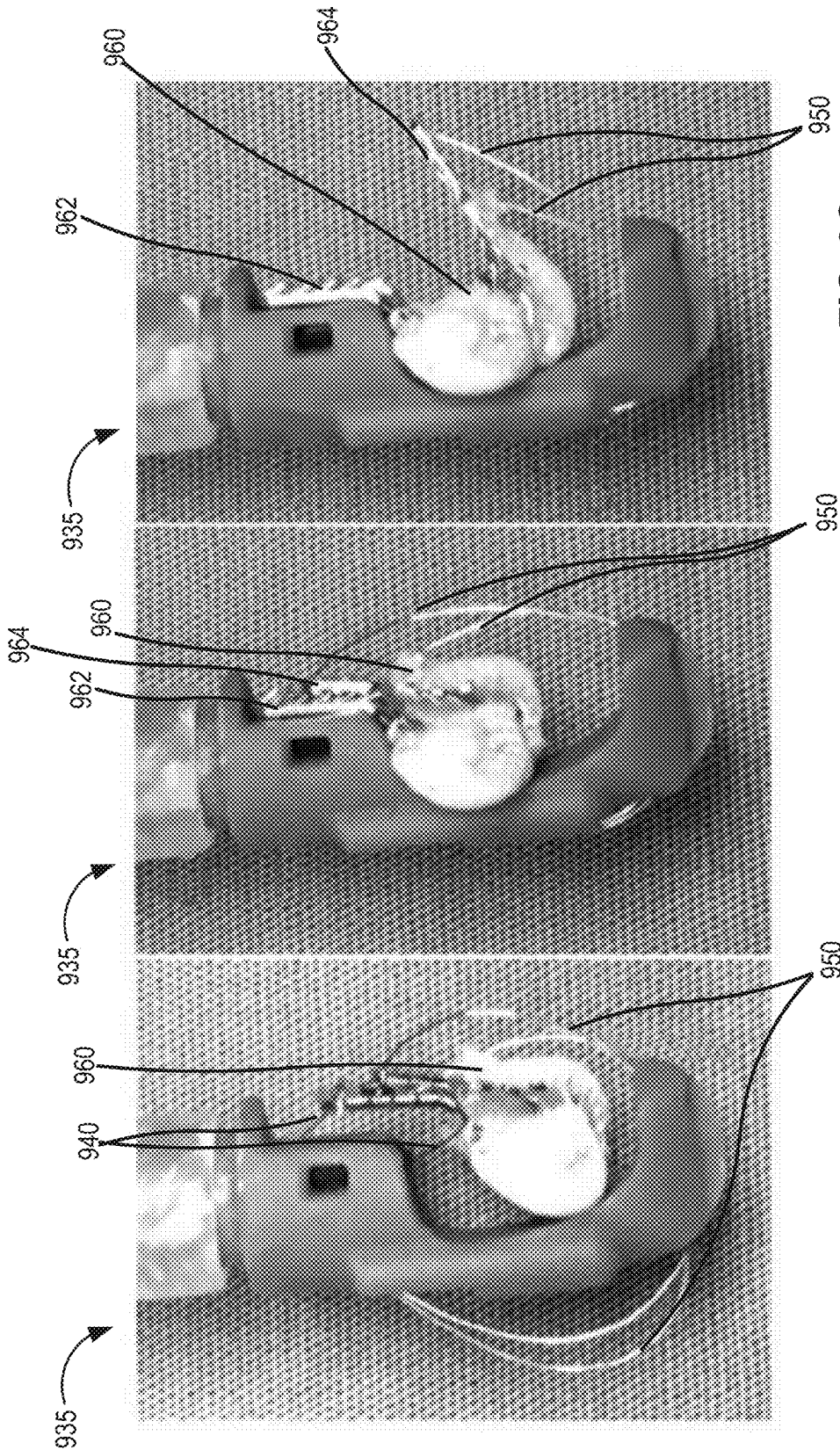


FIG. 9C

FIG. 9B

FIG. 9A

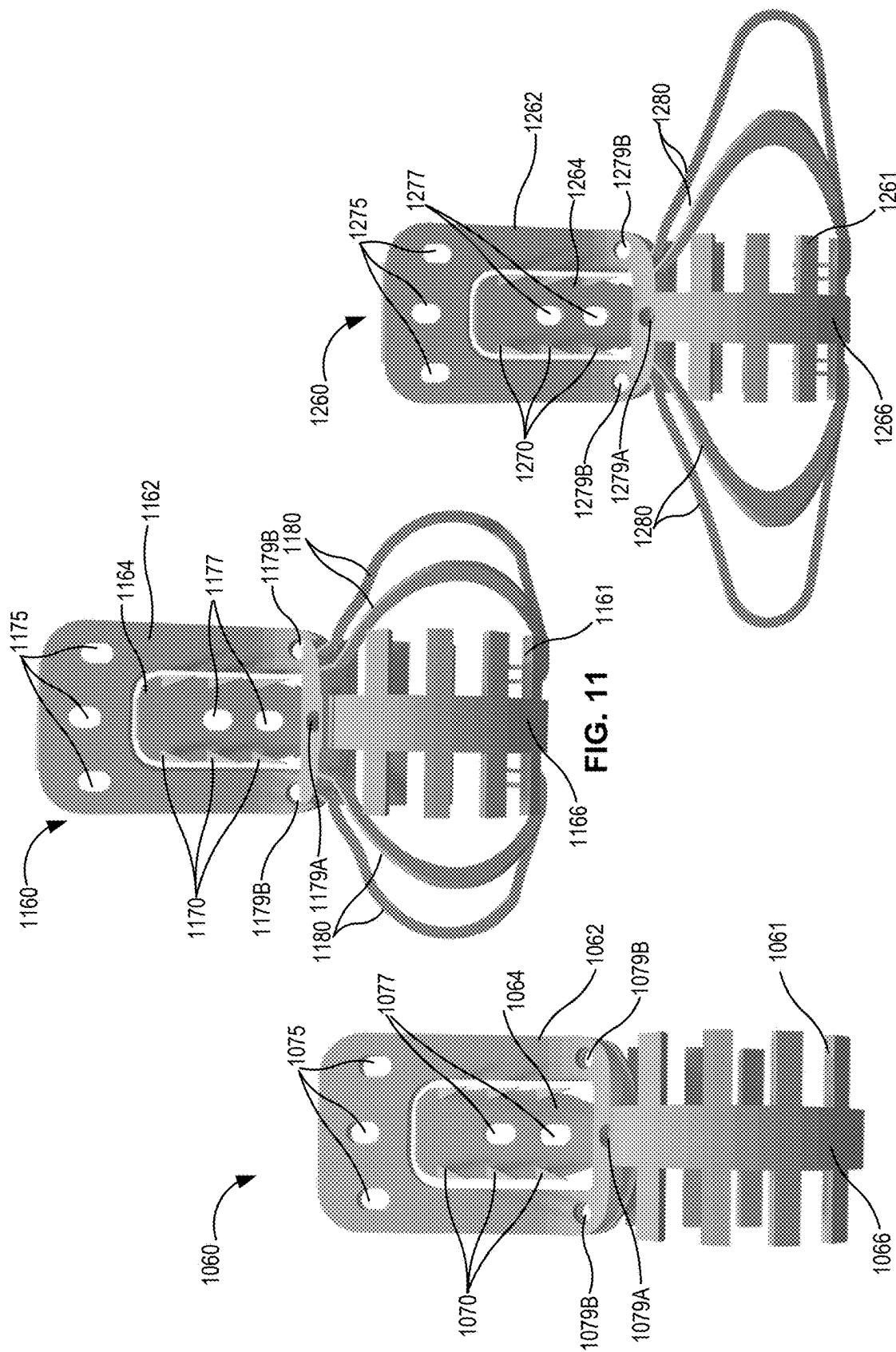


FIG. 10

FIG. 11

FIG. 12

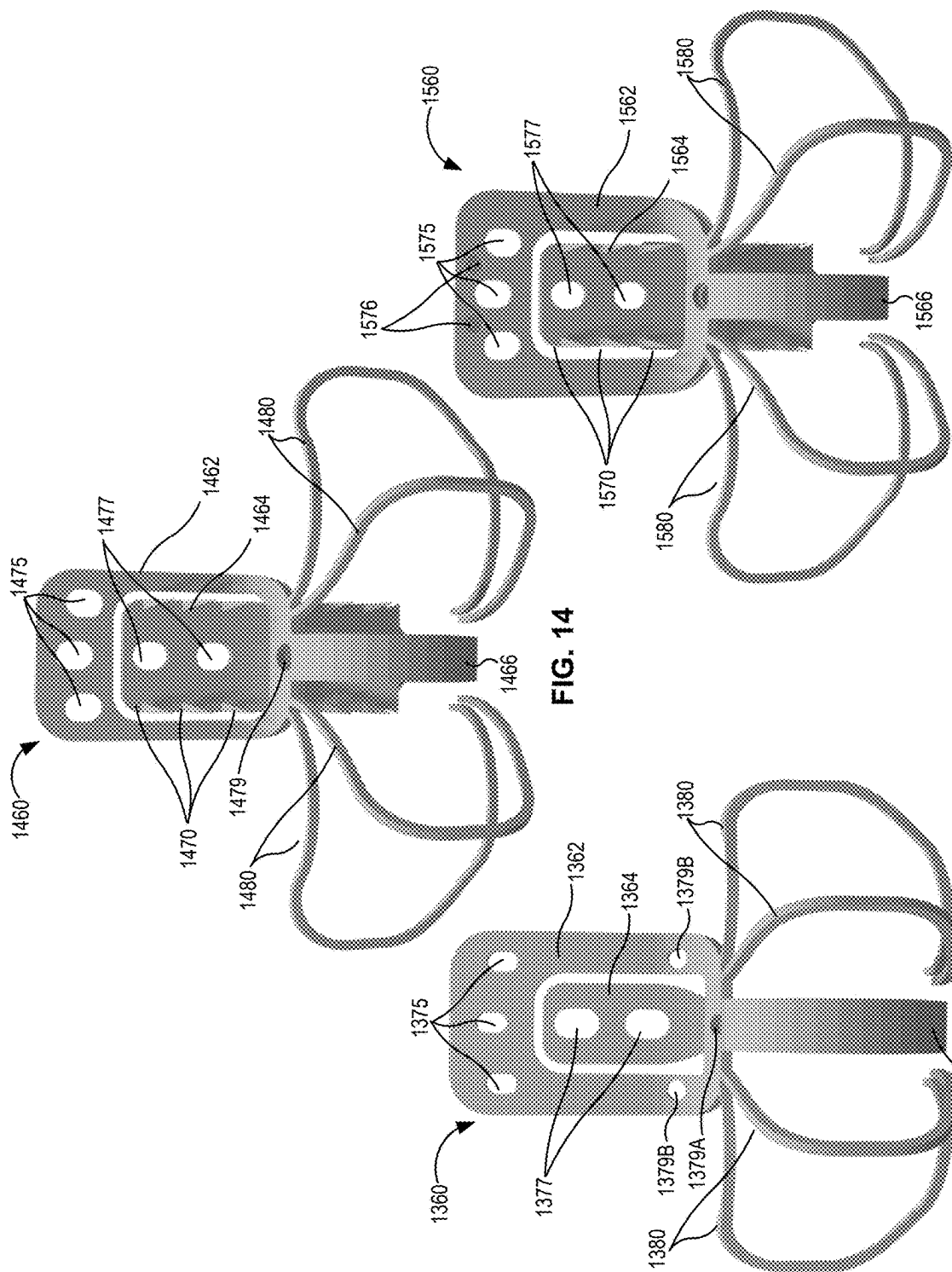


FIG. 13

FIG. 14

FIG. 15

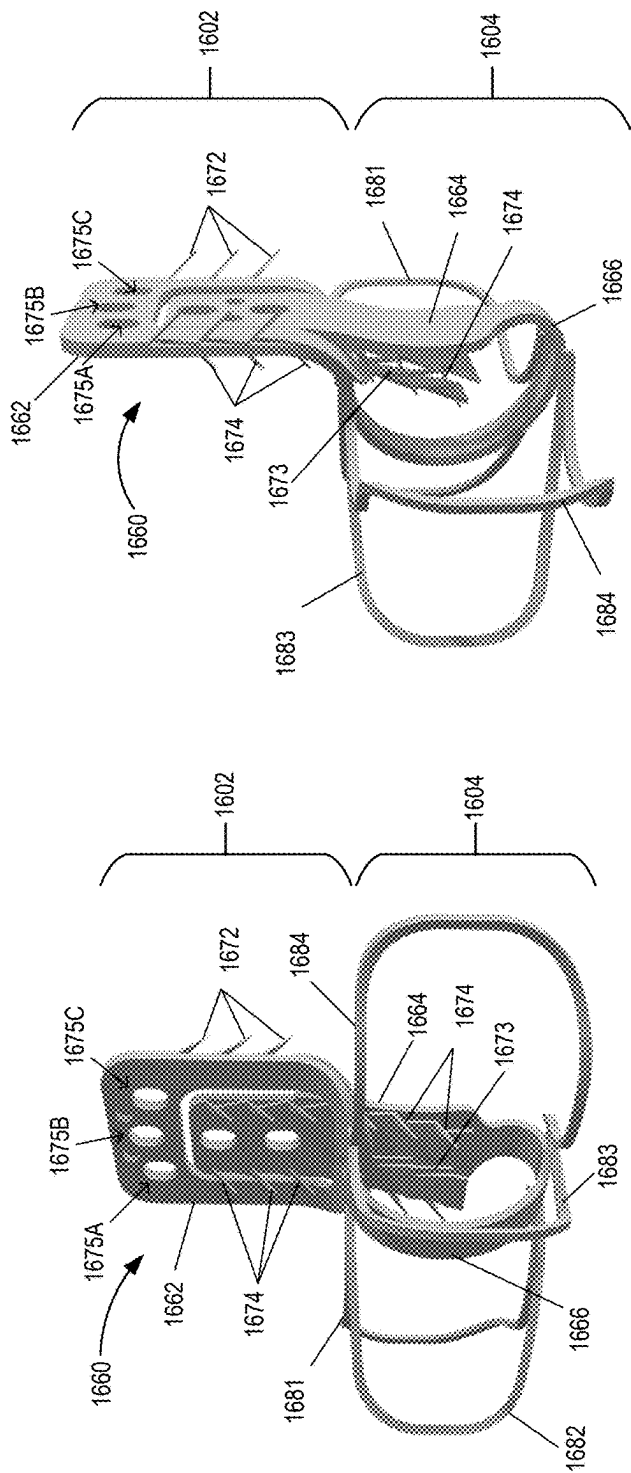


FIG. 16A

FIG. 16B

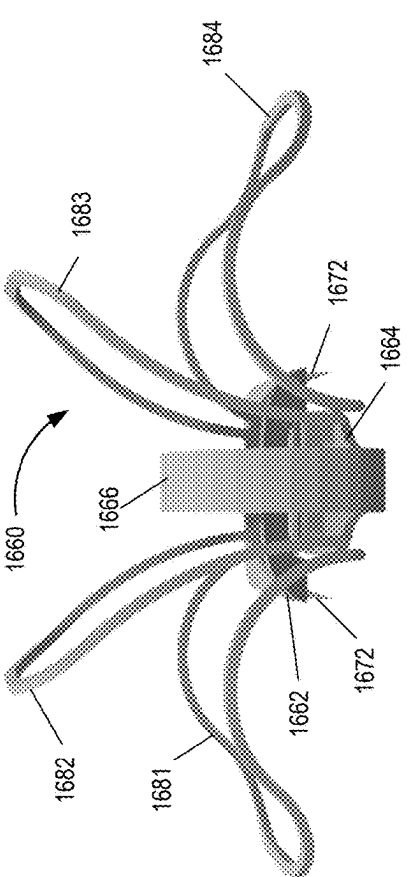


FIG. 16C

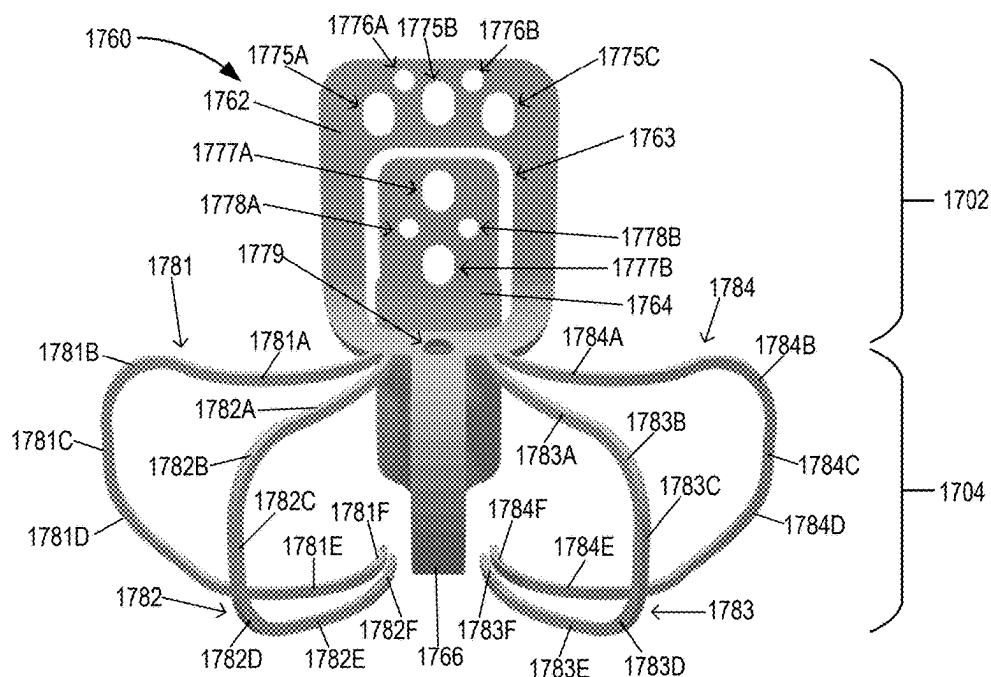


FIG. 17A

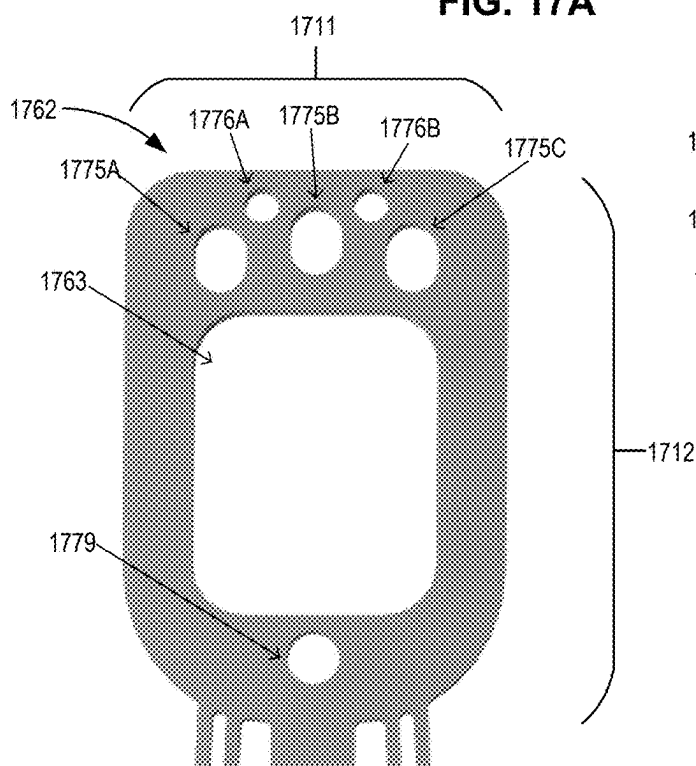


FIG. 17B

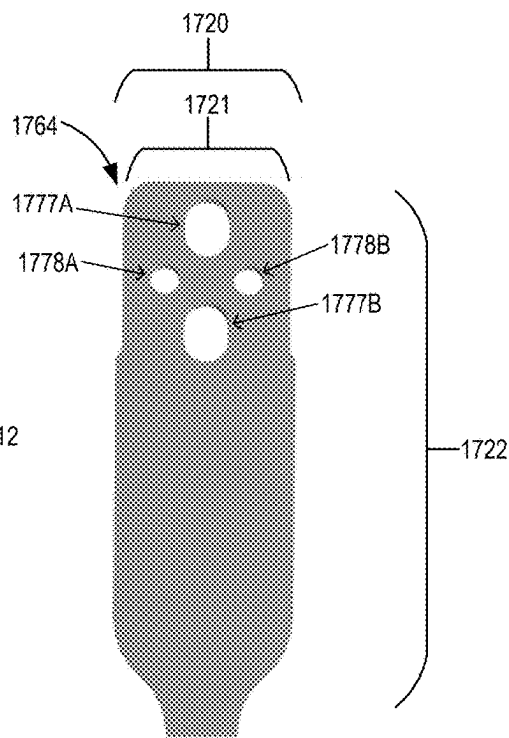


FIG. 17C

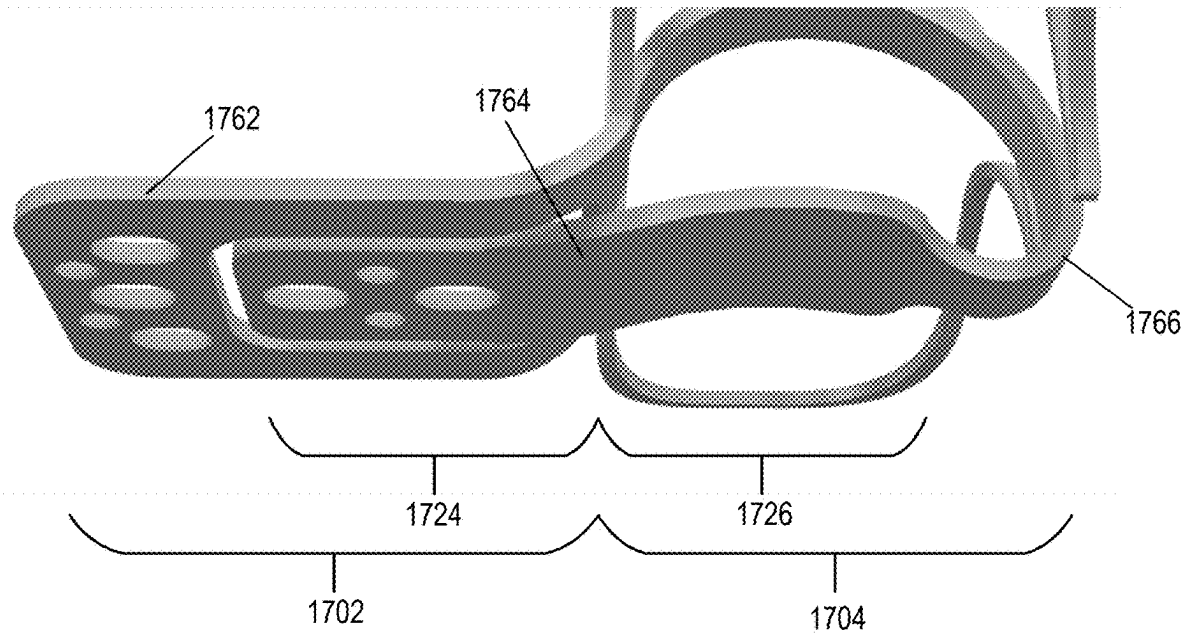


FIG. 17D

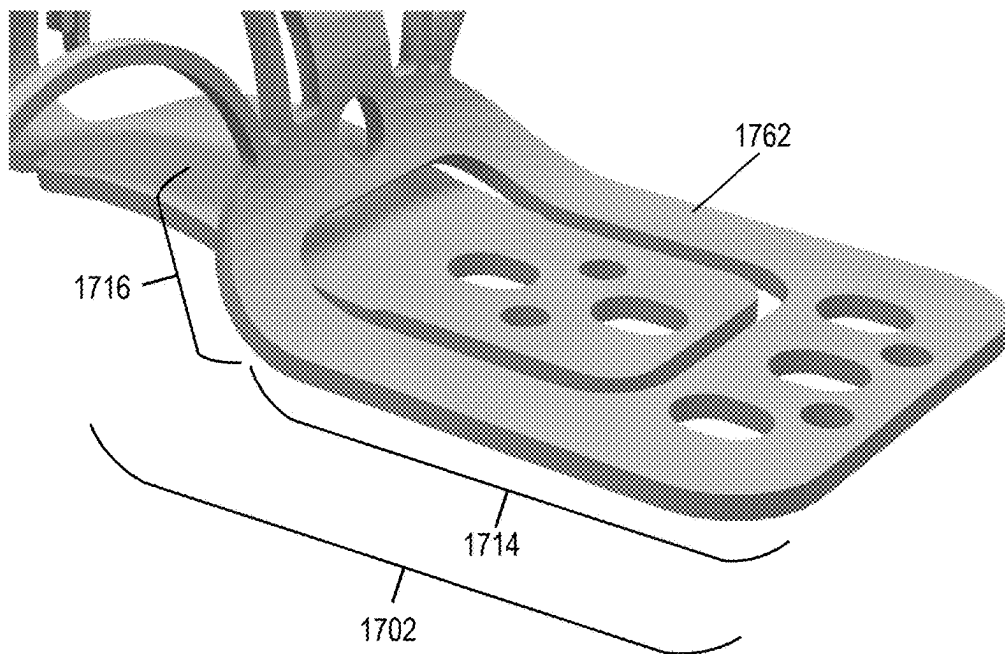


FIG. 17E

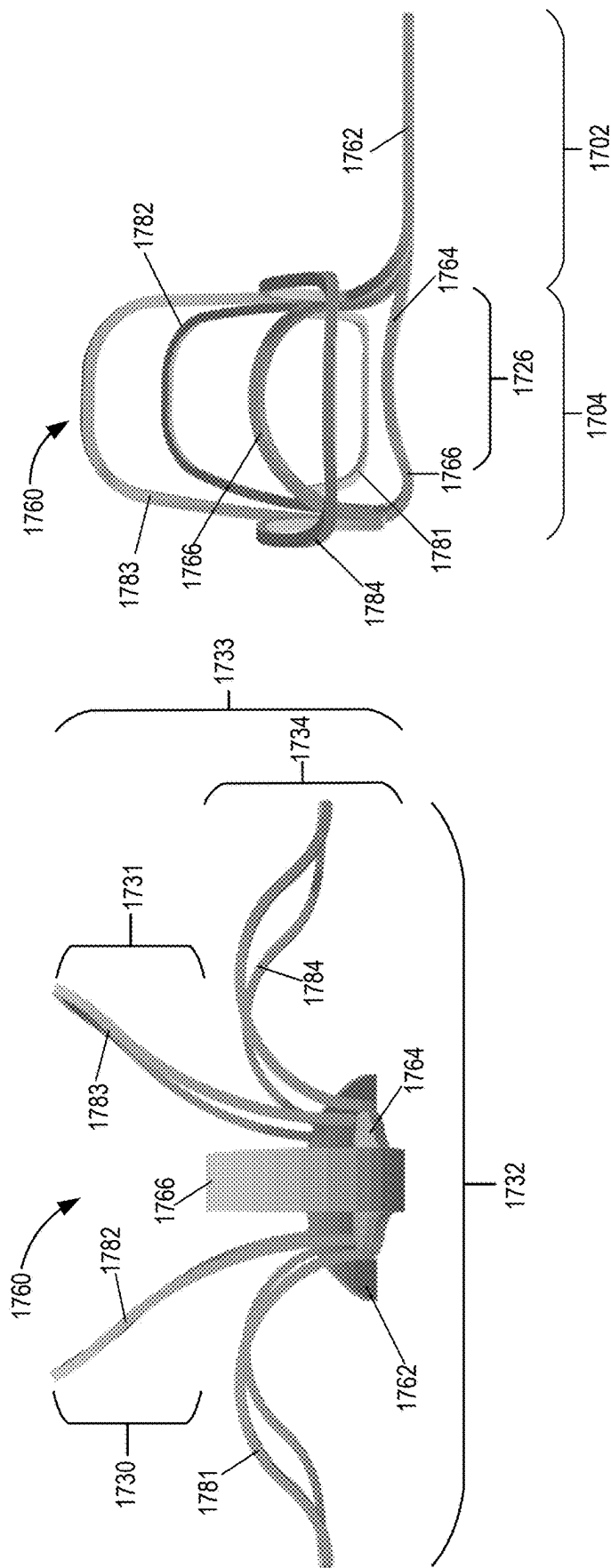


FIG. 17F

FIG. 17G

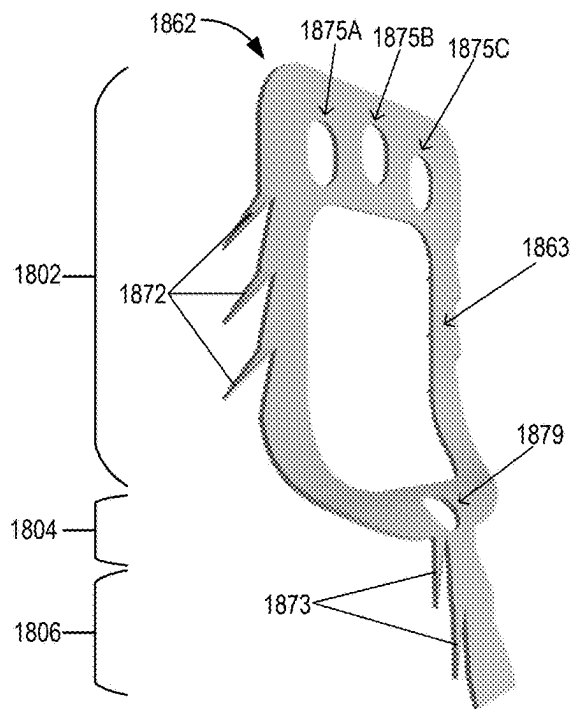


FIG. 18A

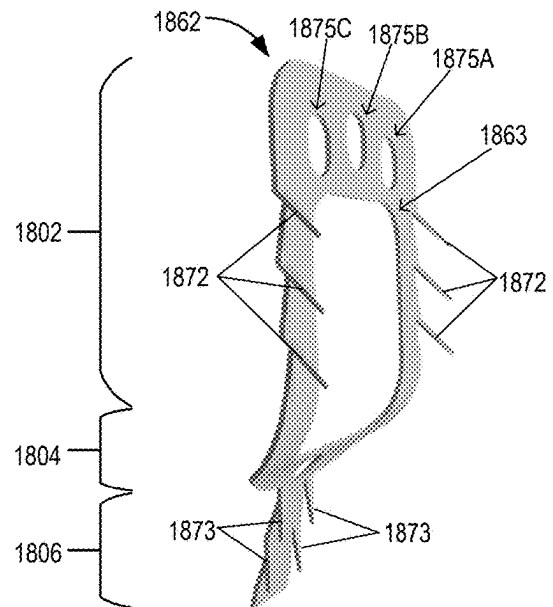


FIG. 18B

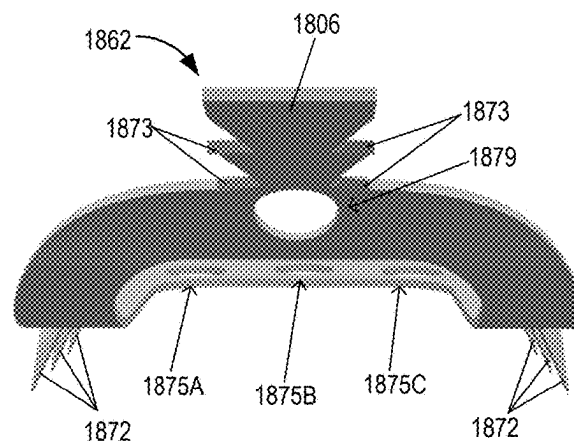


FIG. 18C

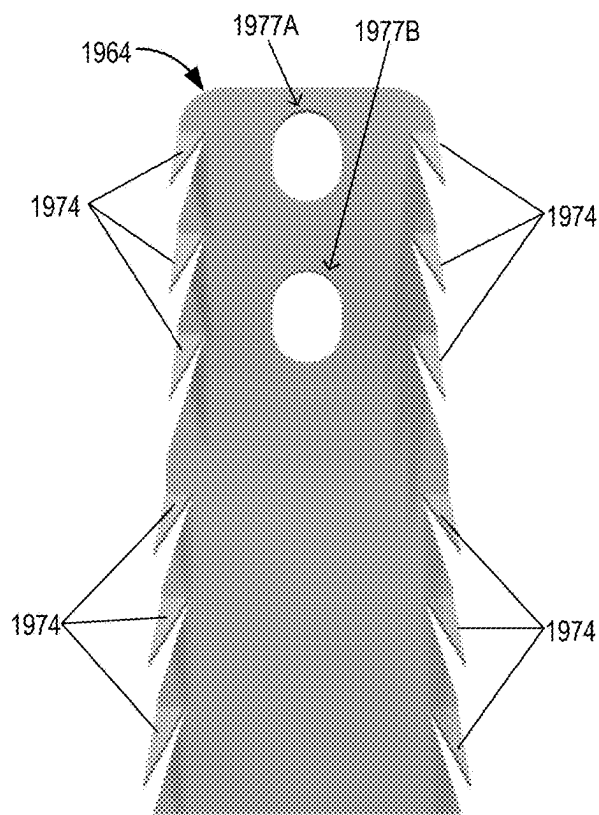


FIG. 19A

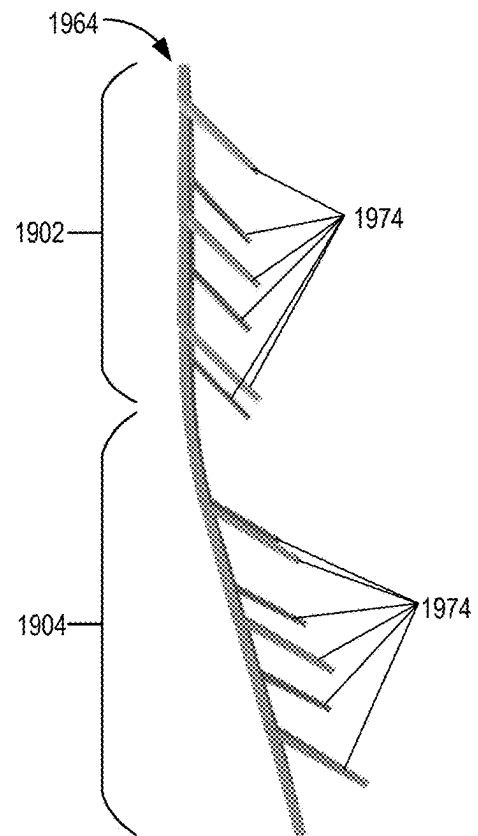


FIG. 19B

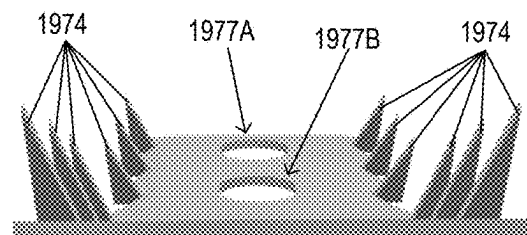


FIG. 19C

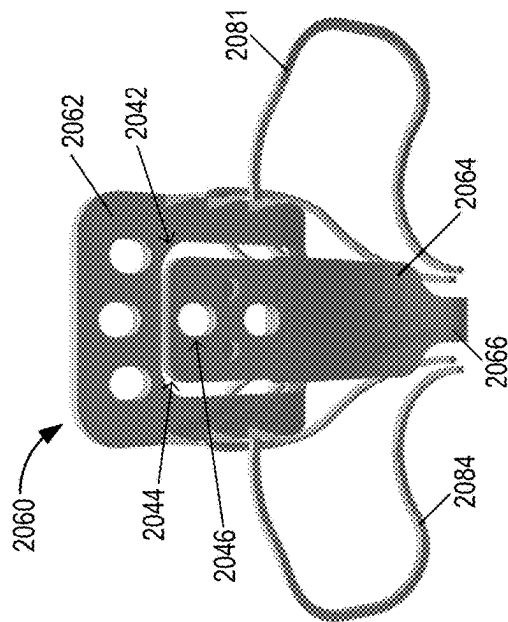


FIG. 20A

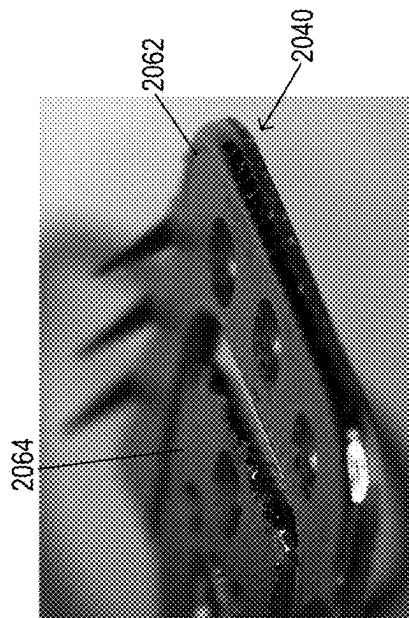


FIG. 20B

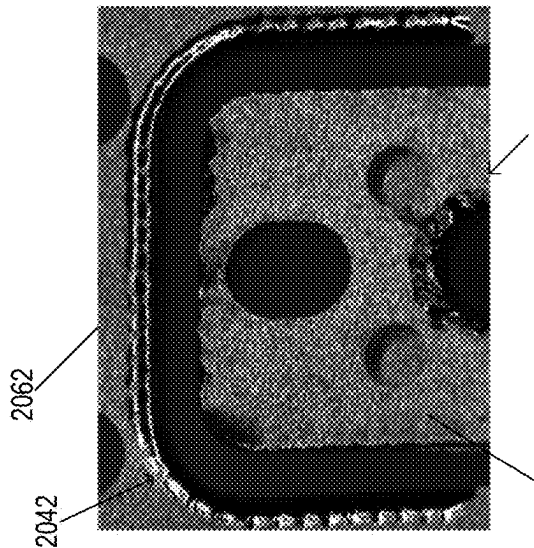


FIG. 20C

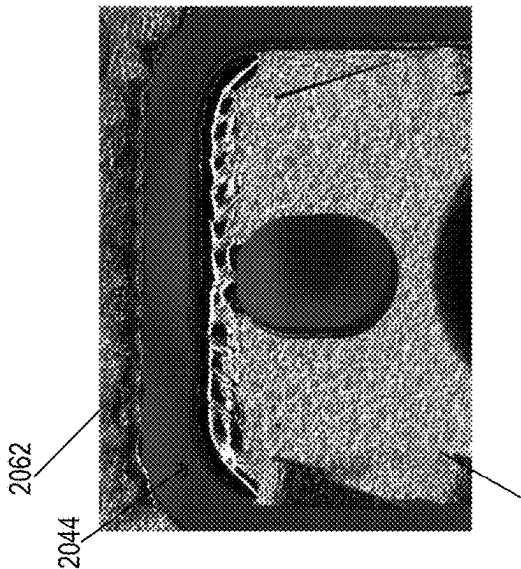


FIG. 20D

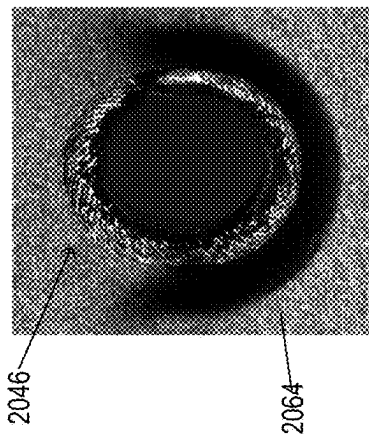


FIG. 20E

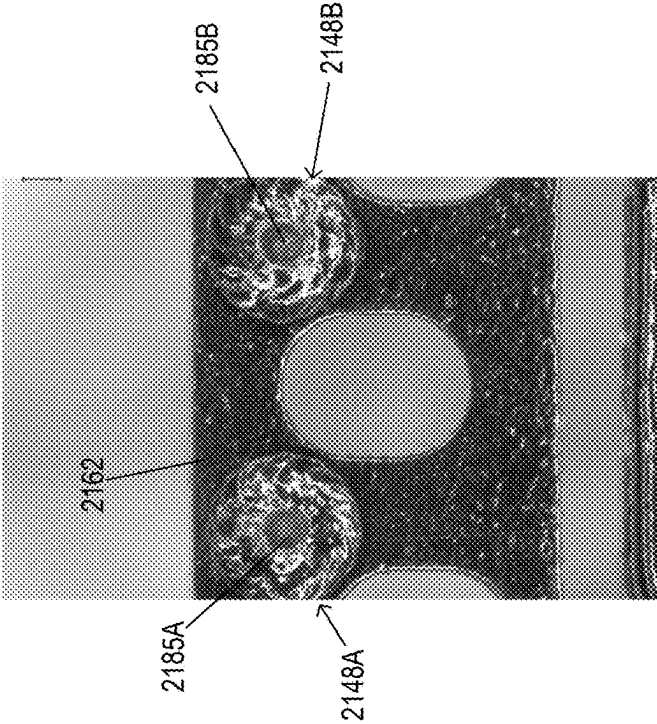


FIG. 21B

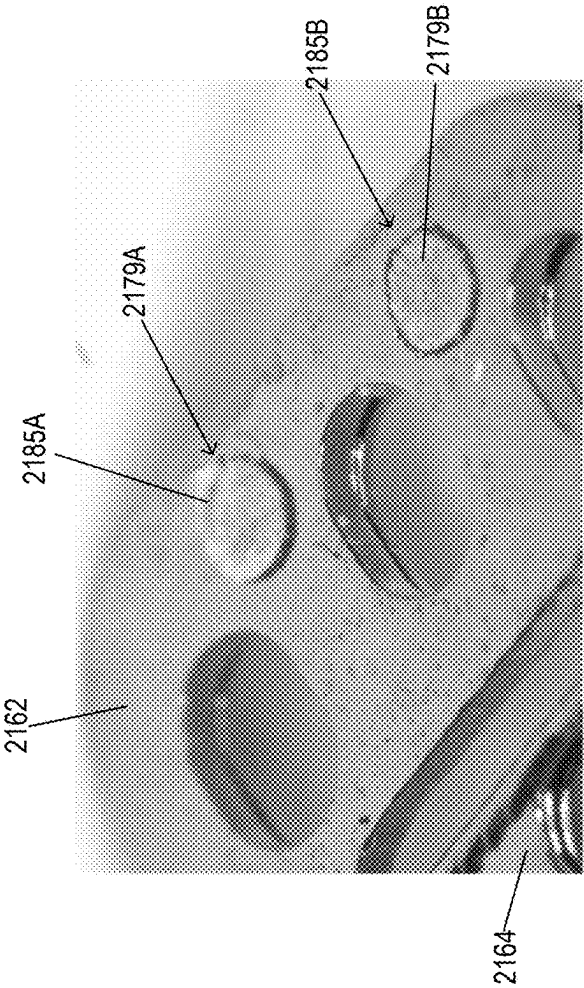


FIG. 21A

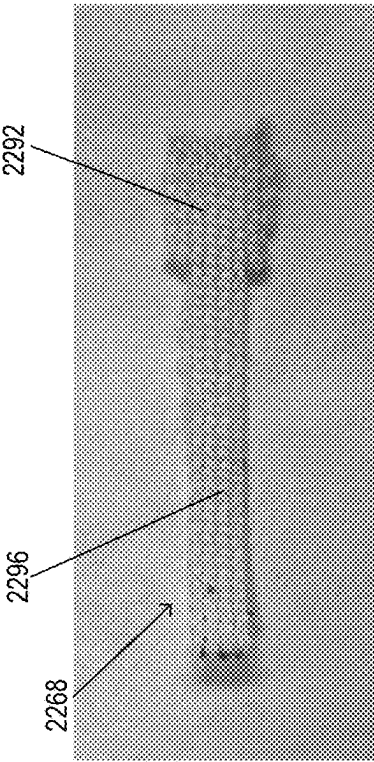


FIG. 22A

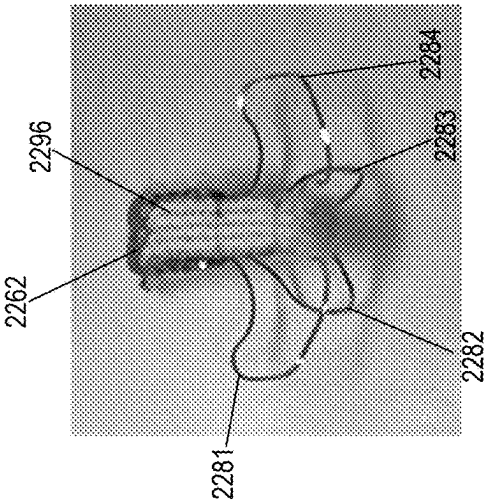


FIG. 22B

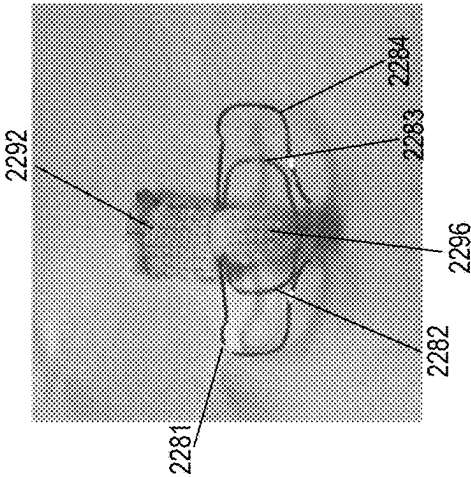
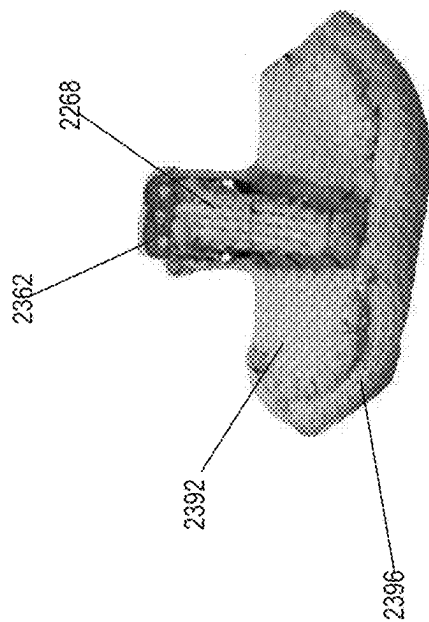
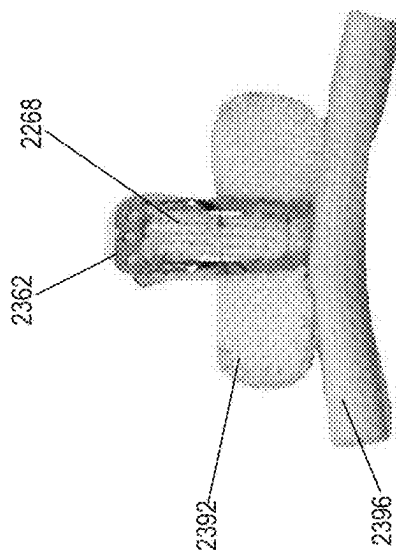
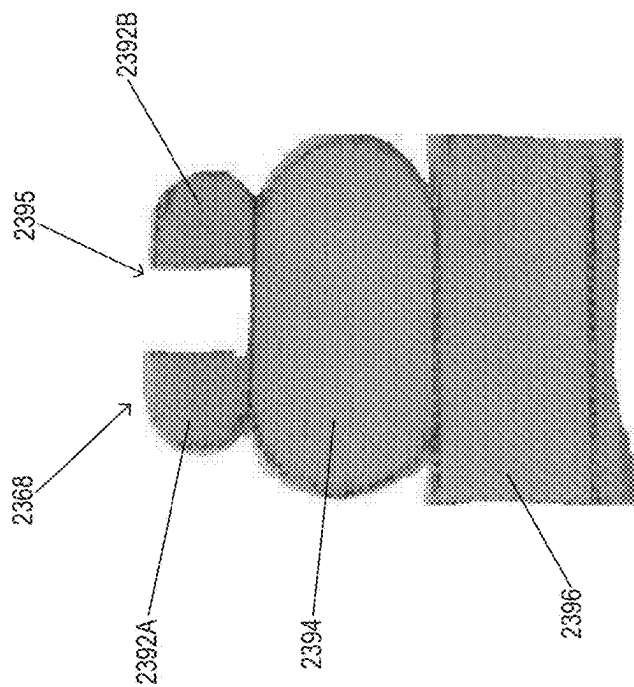


FIG. 22C



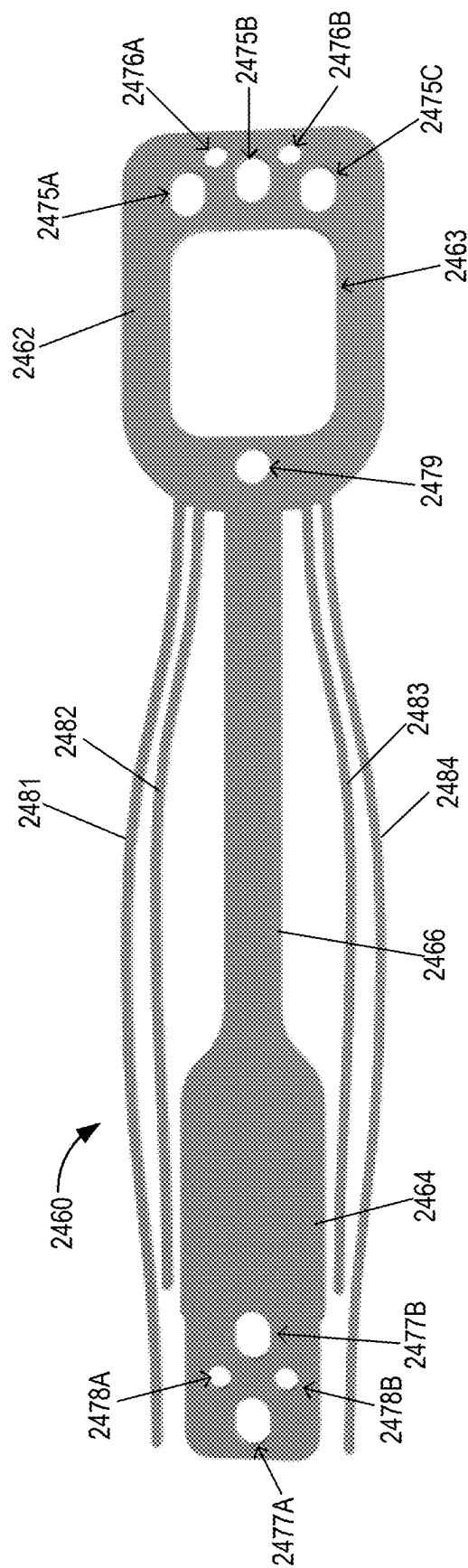


FIG. 24

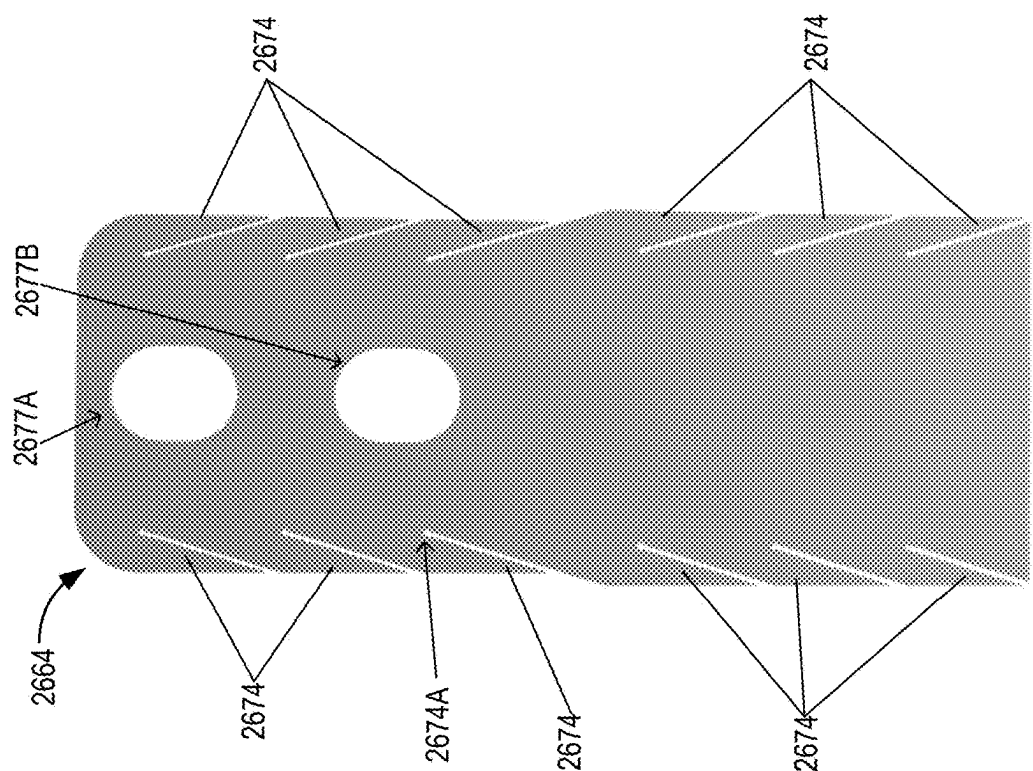


FIG. 25

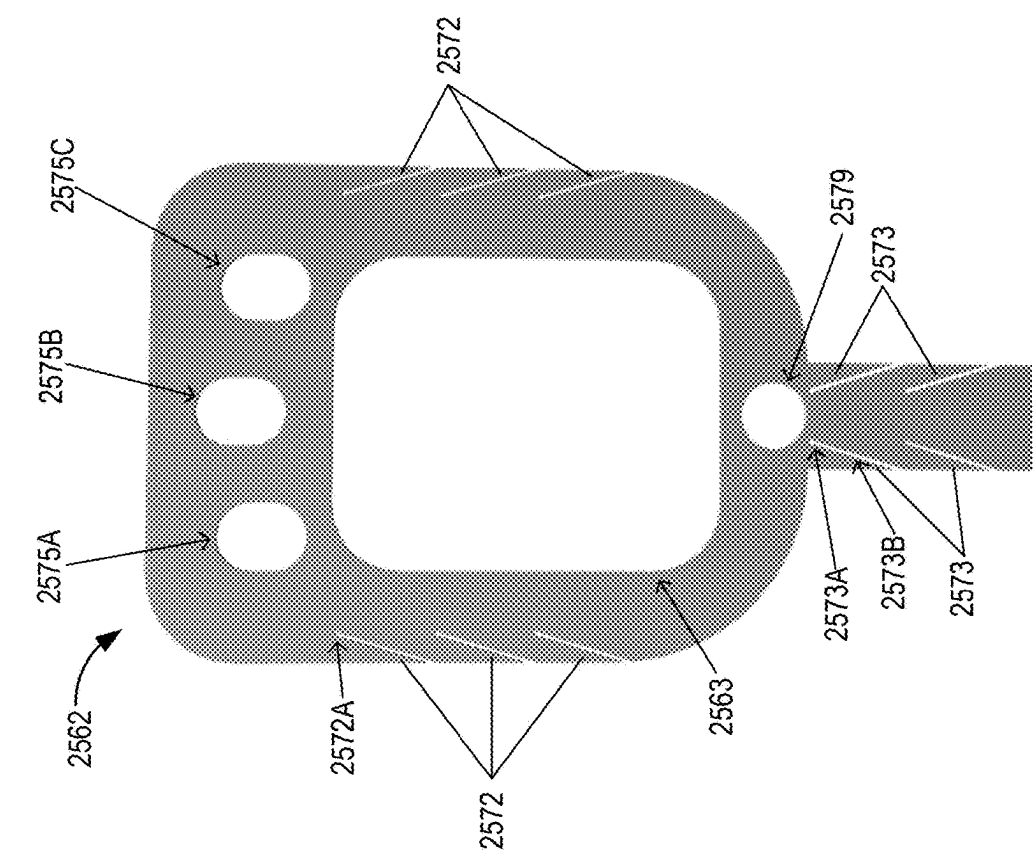


FIG. 26

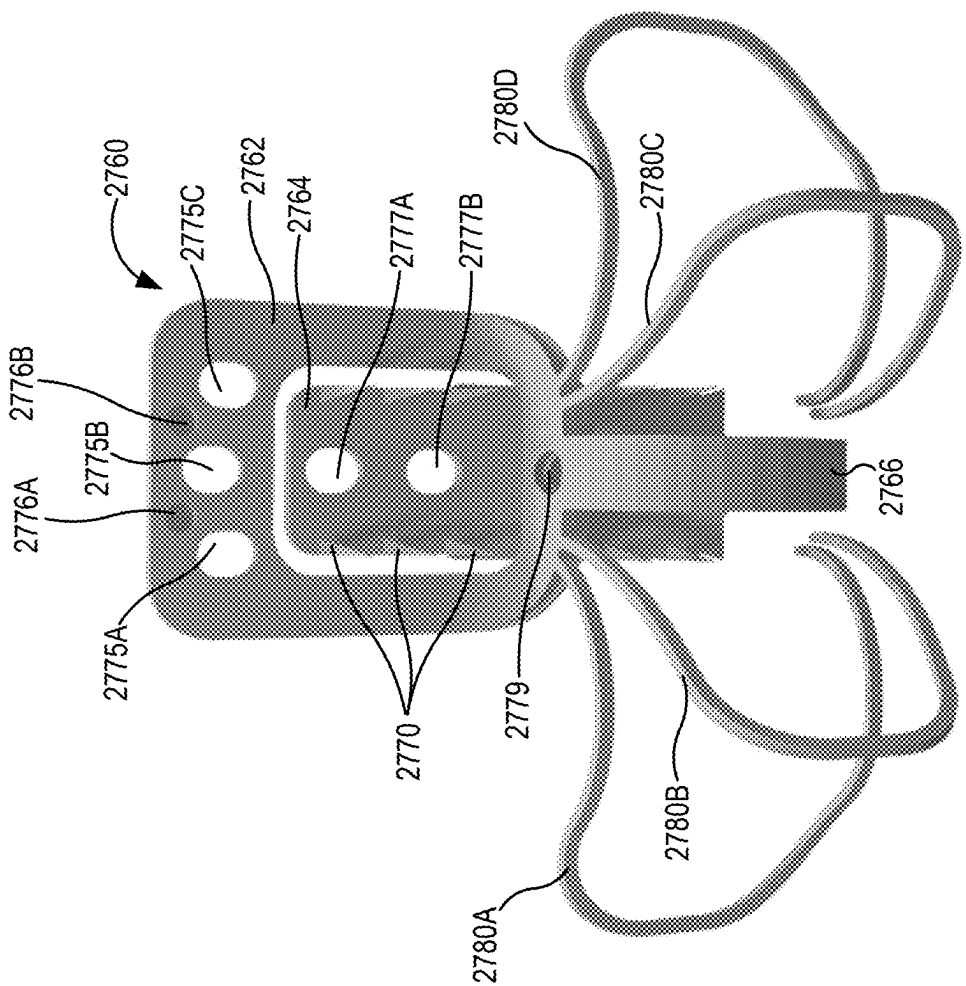


FIG. 27A

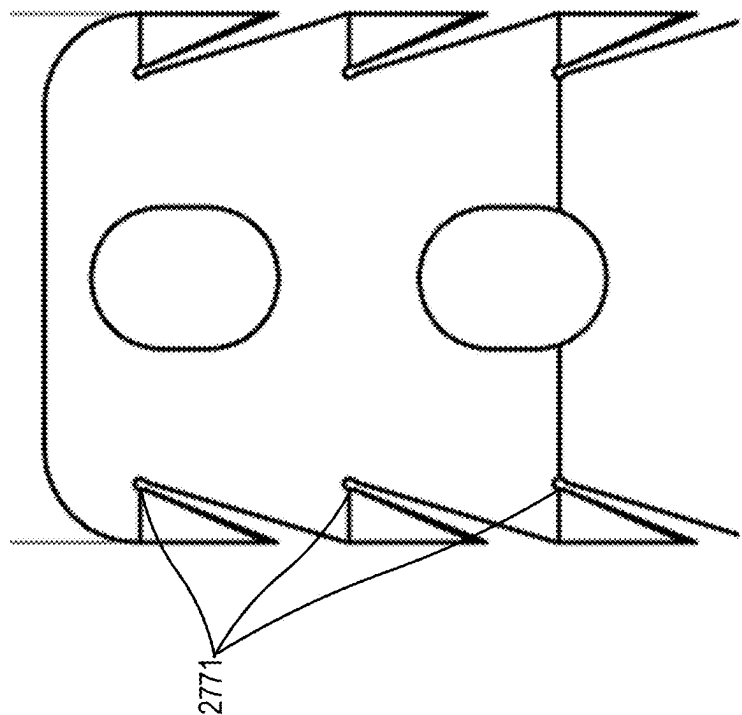


FIG. 27B

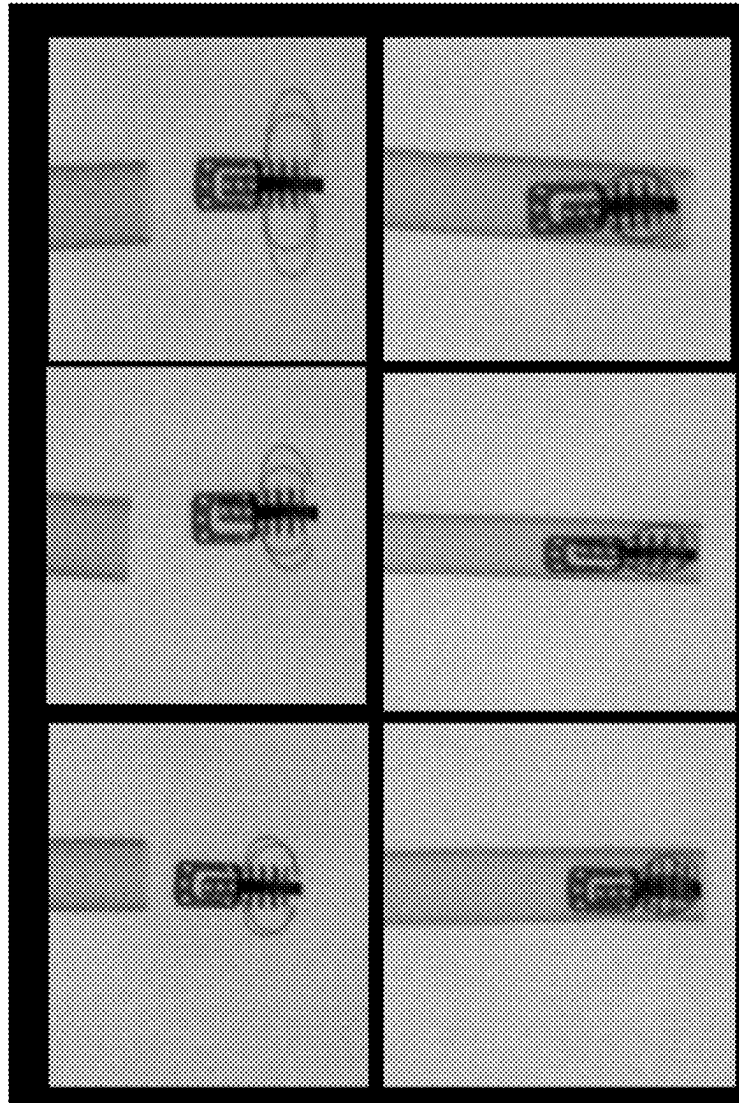


FIG. 28

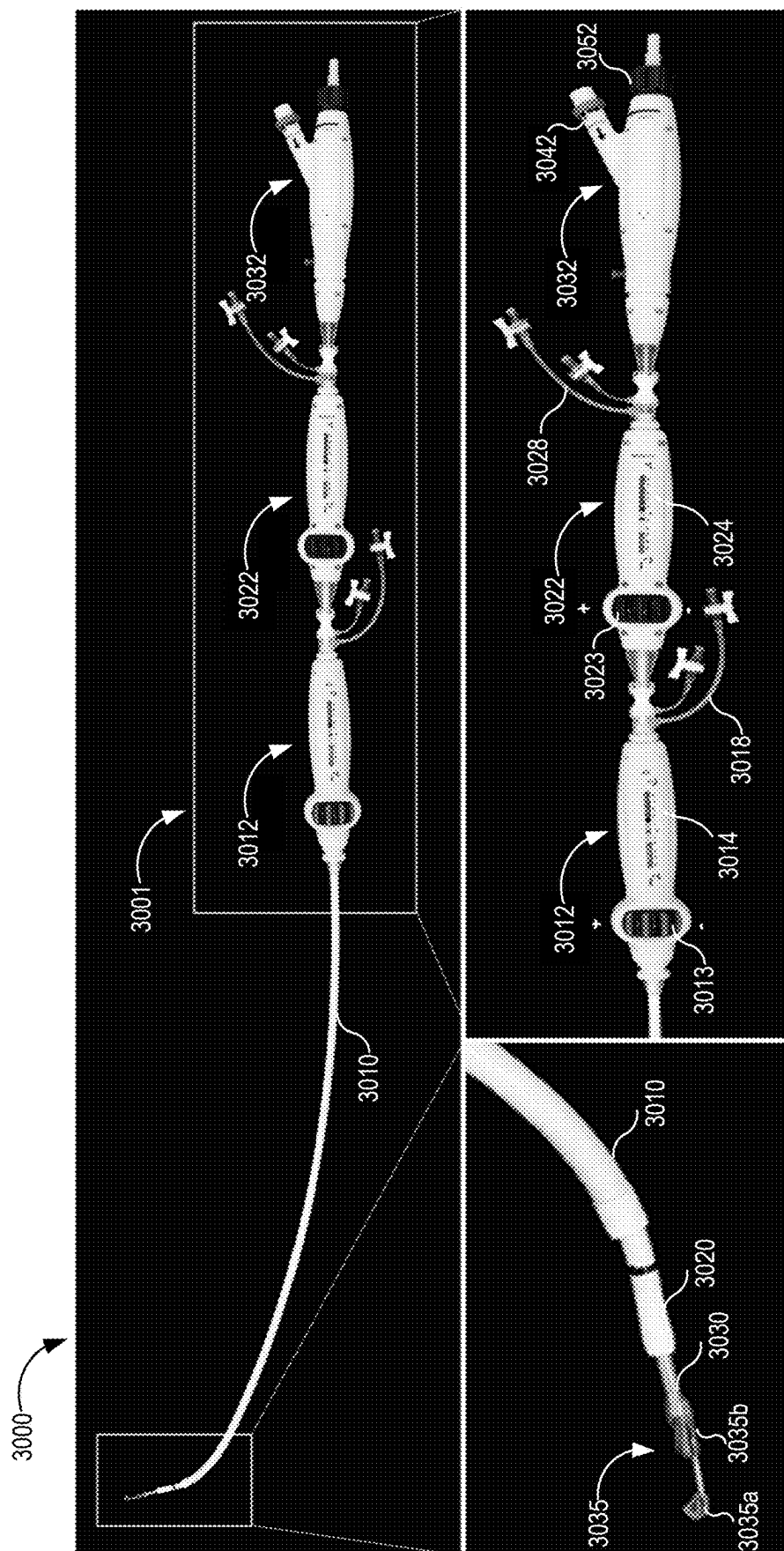


FIG. 29

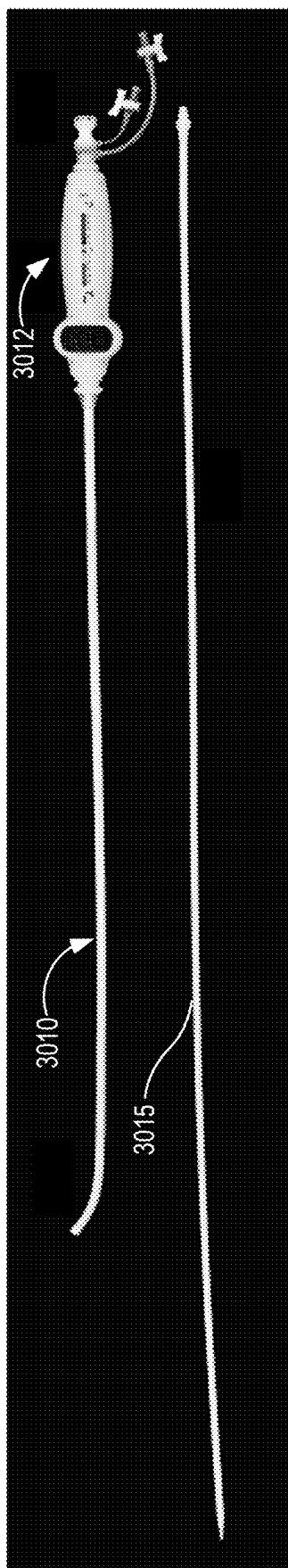


FIG. 30A

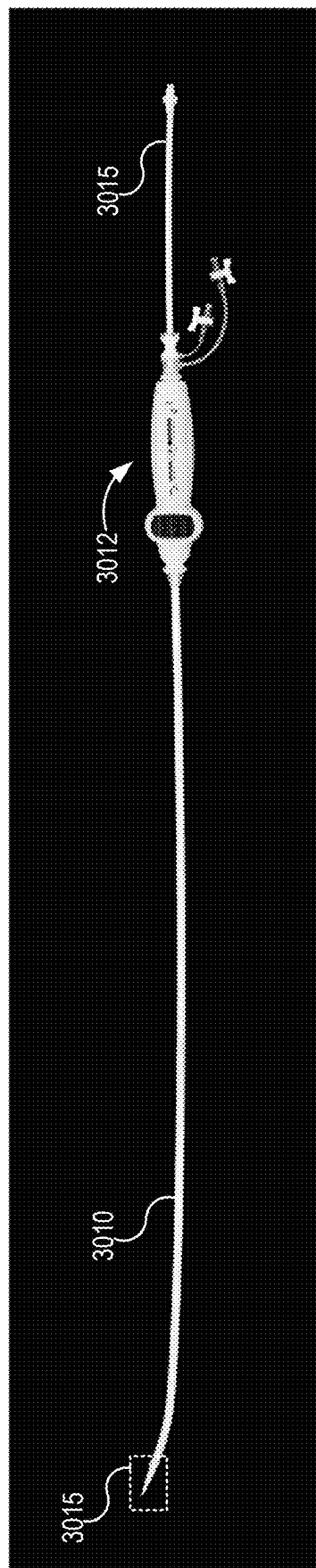


FIG. 30B

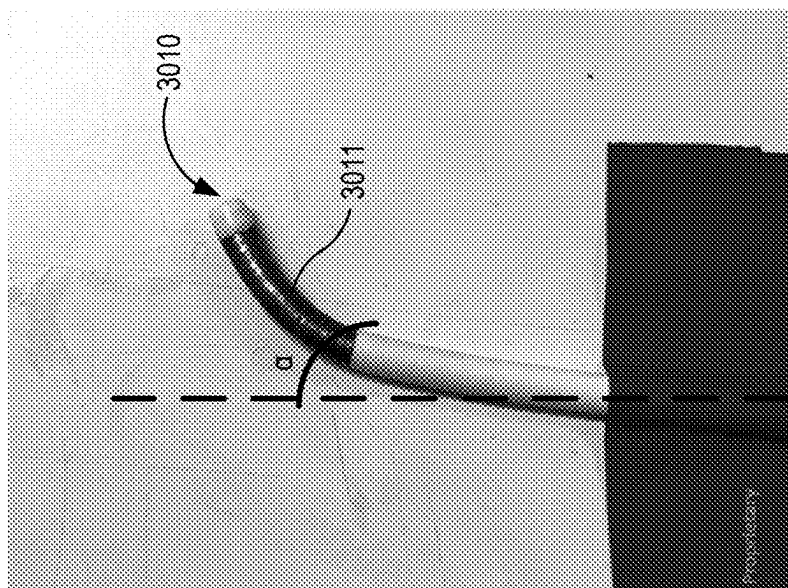


FIG. 31C

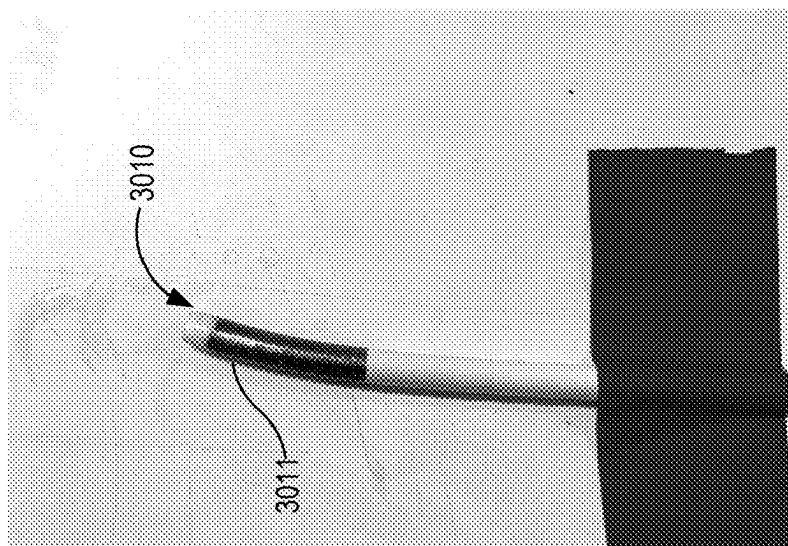


FIG. 31B

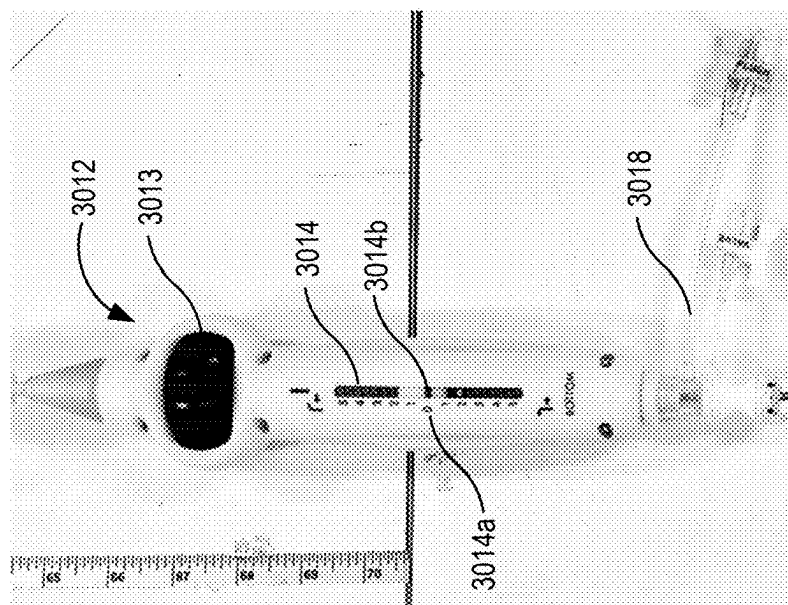


FIG. 31A



FIG. 32A

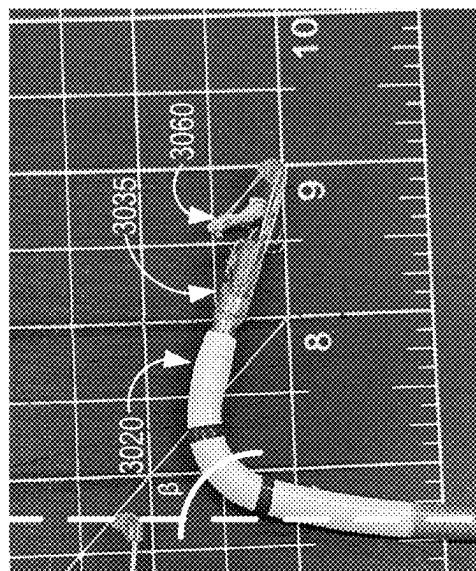


FIG. 32D

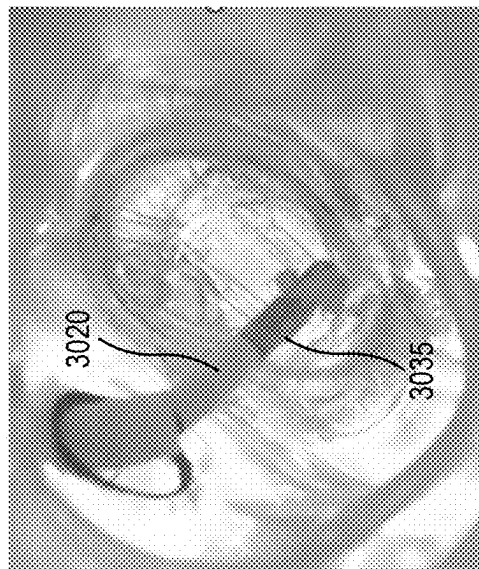


FIG. 32C

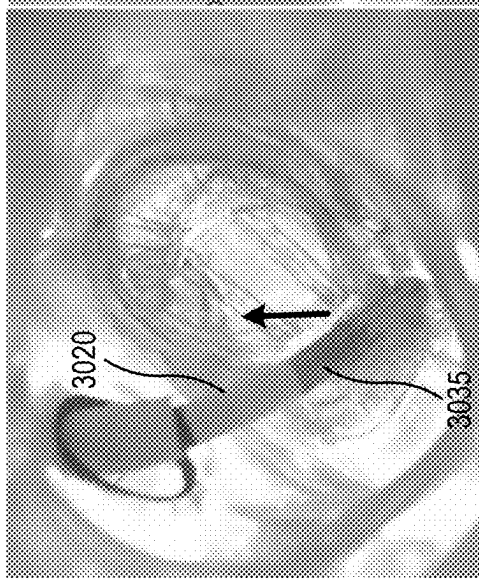


FIG. 32B

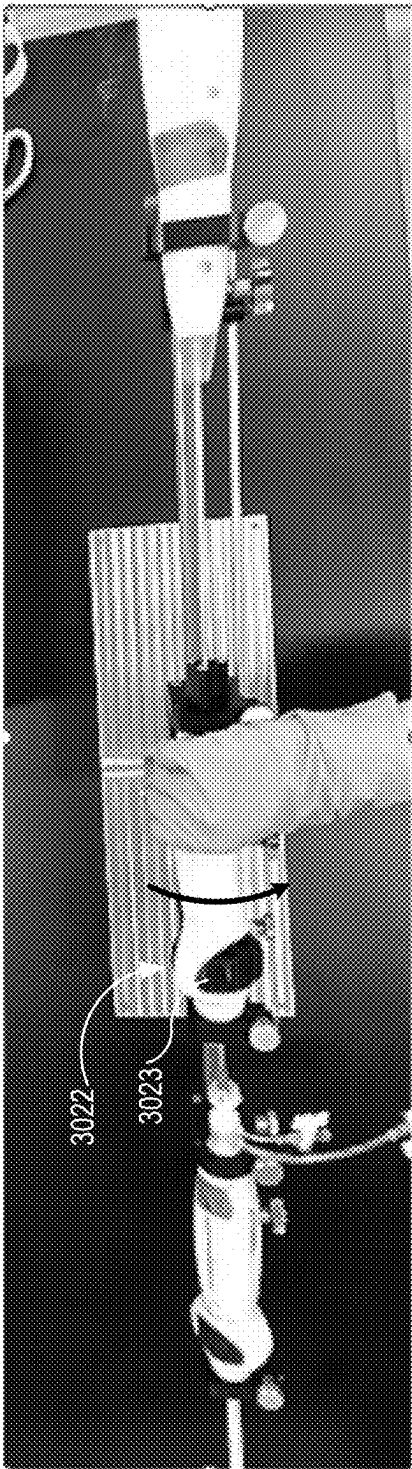


FIG. 33A

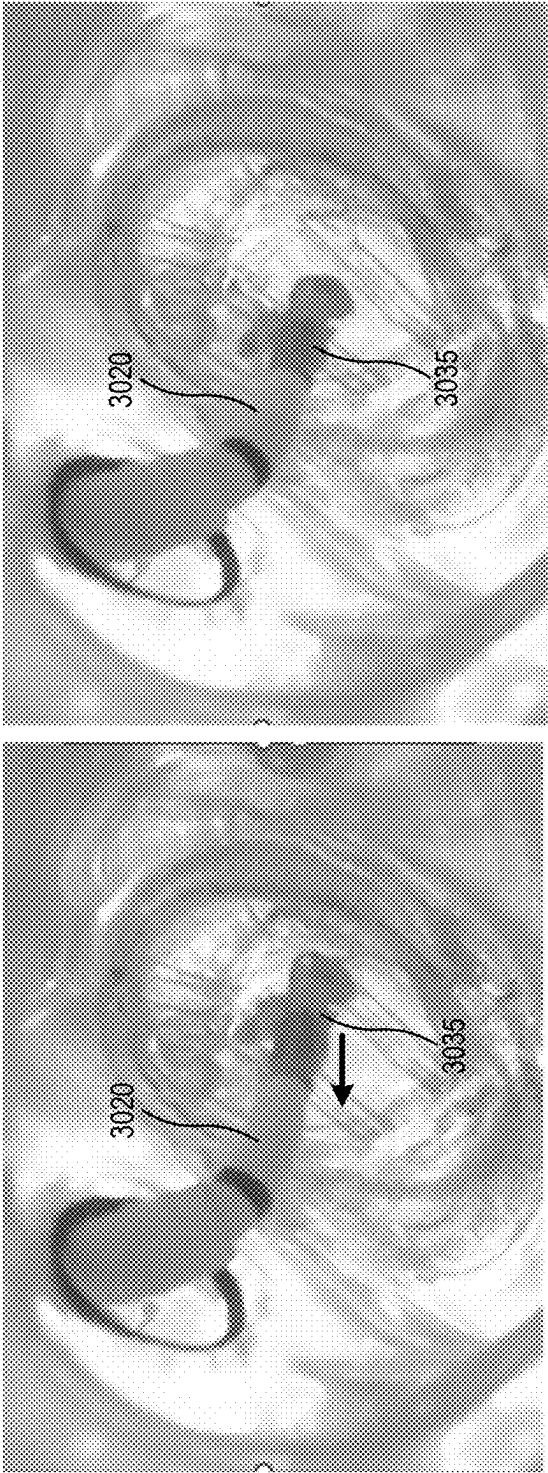
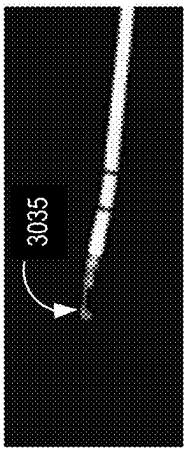
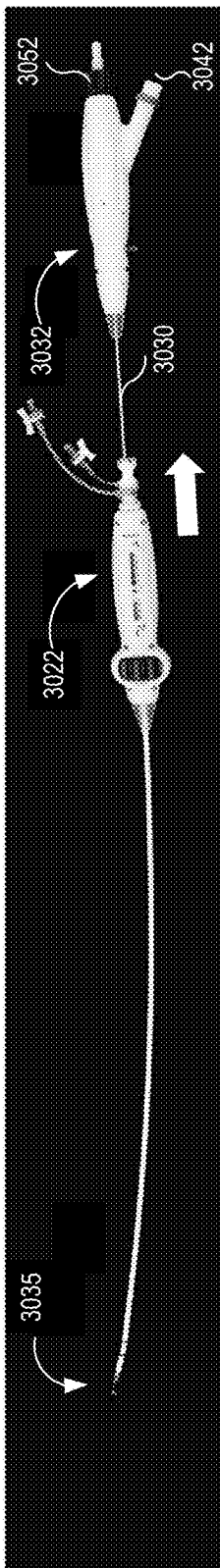
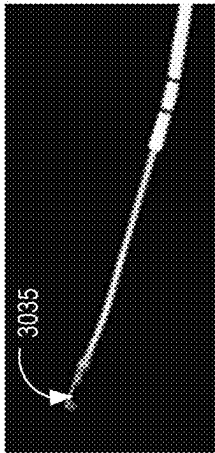
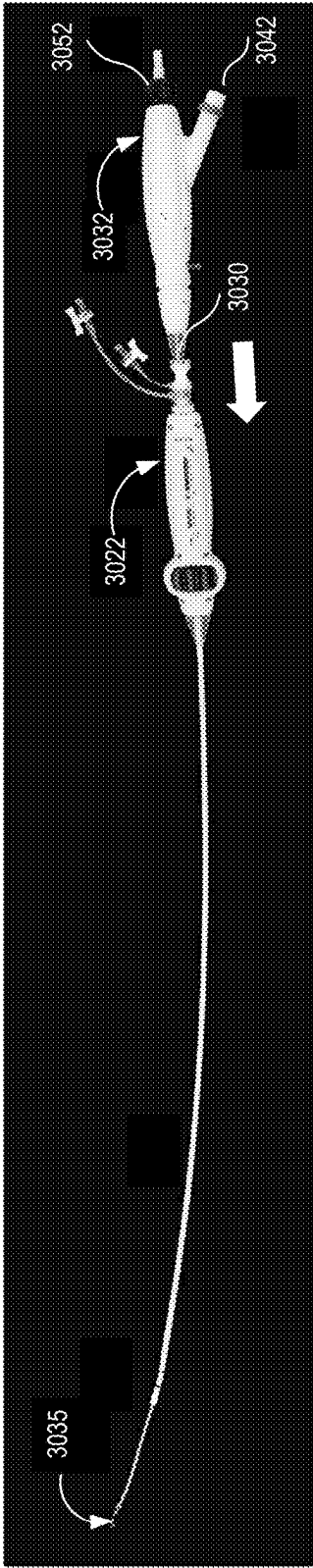


FIG. 33B

FIG. 33C



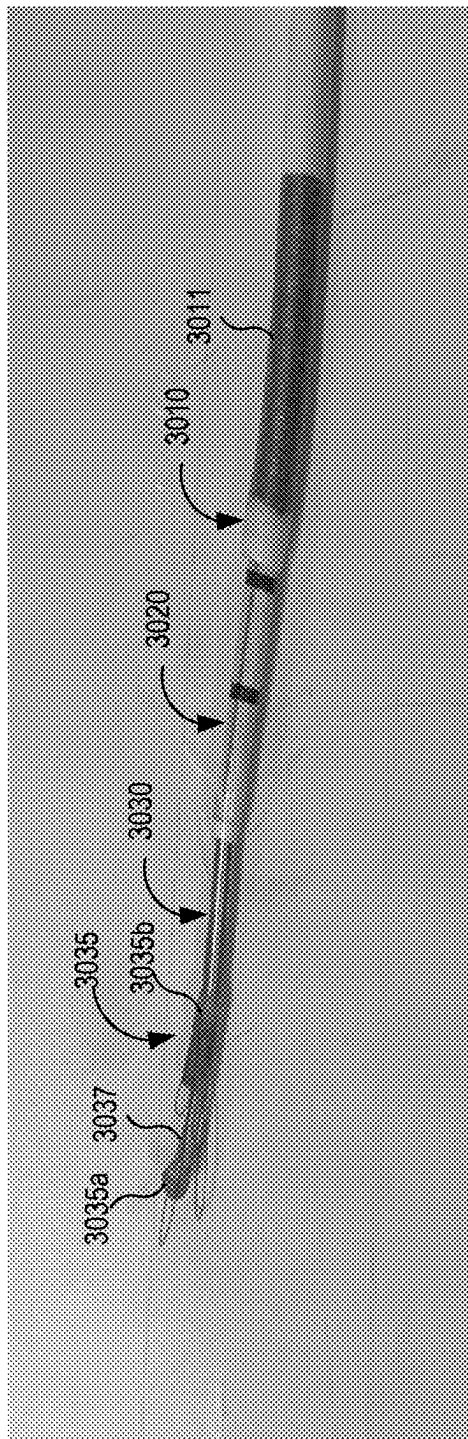


FIG. 35A

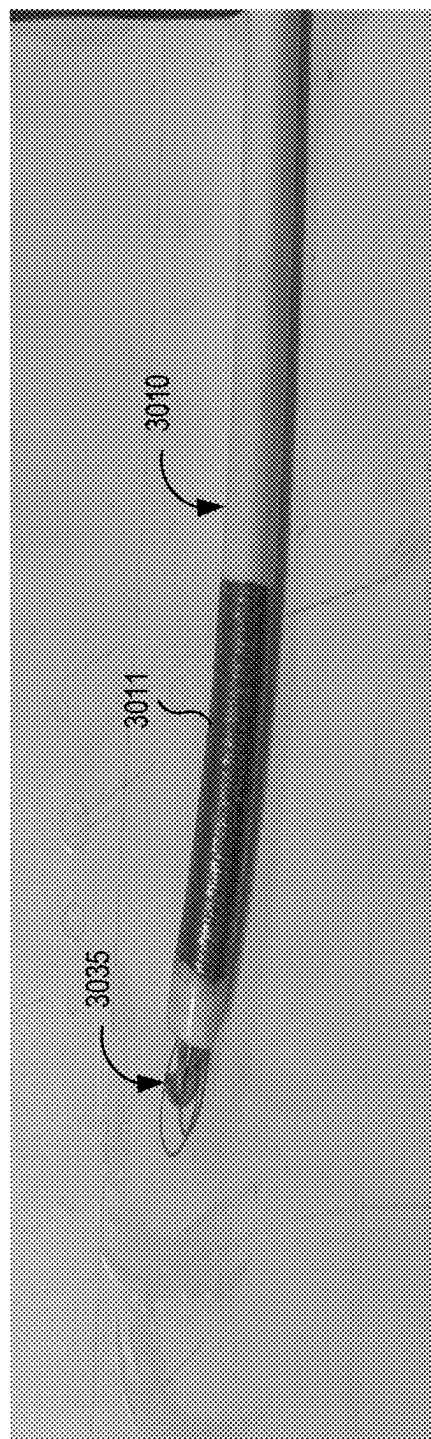


FIG. 35B

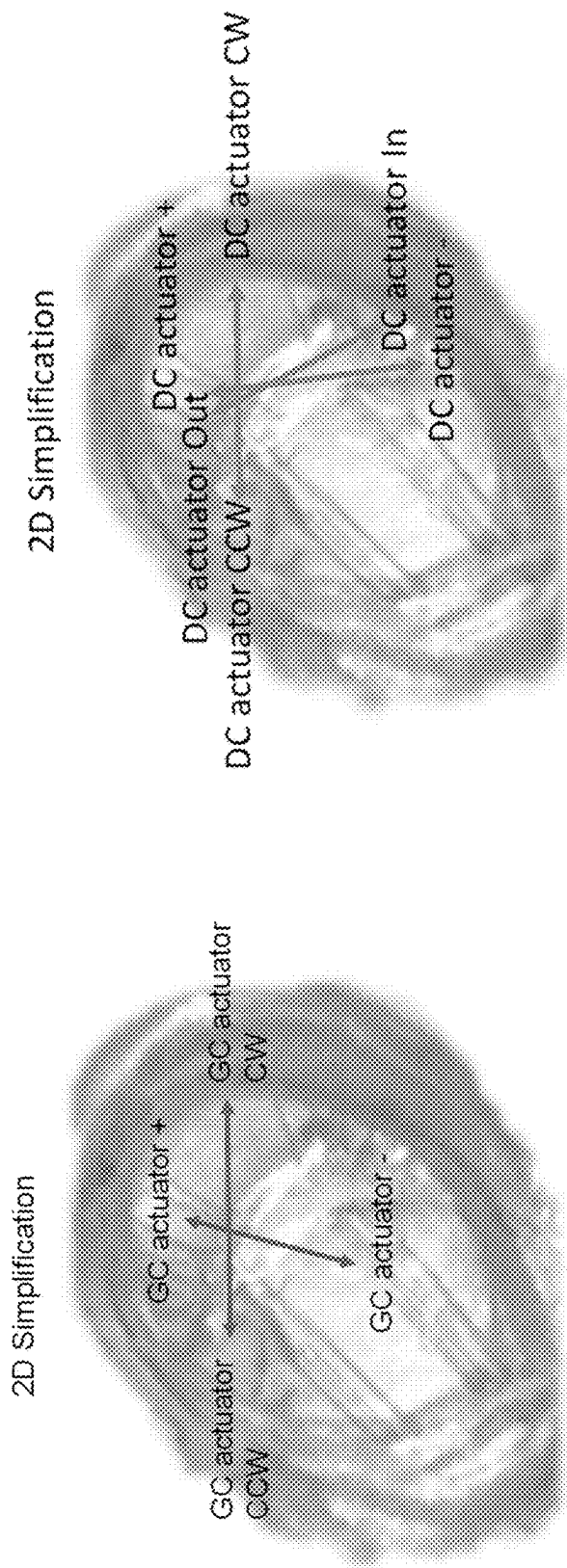
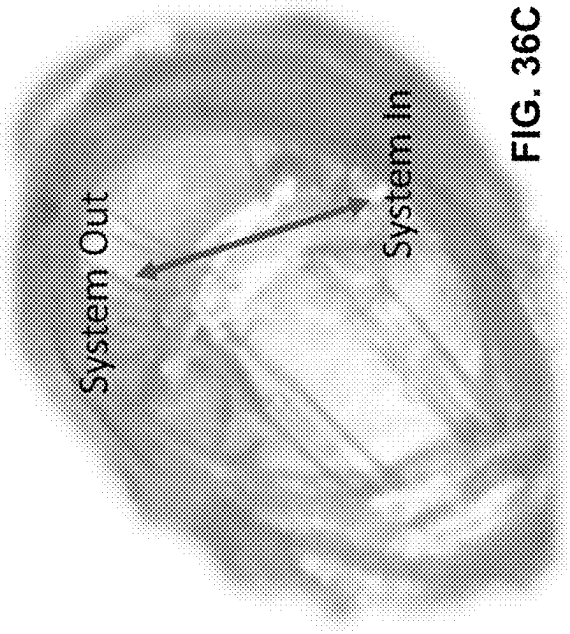


FIG. 36B



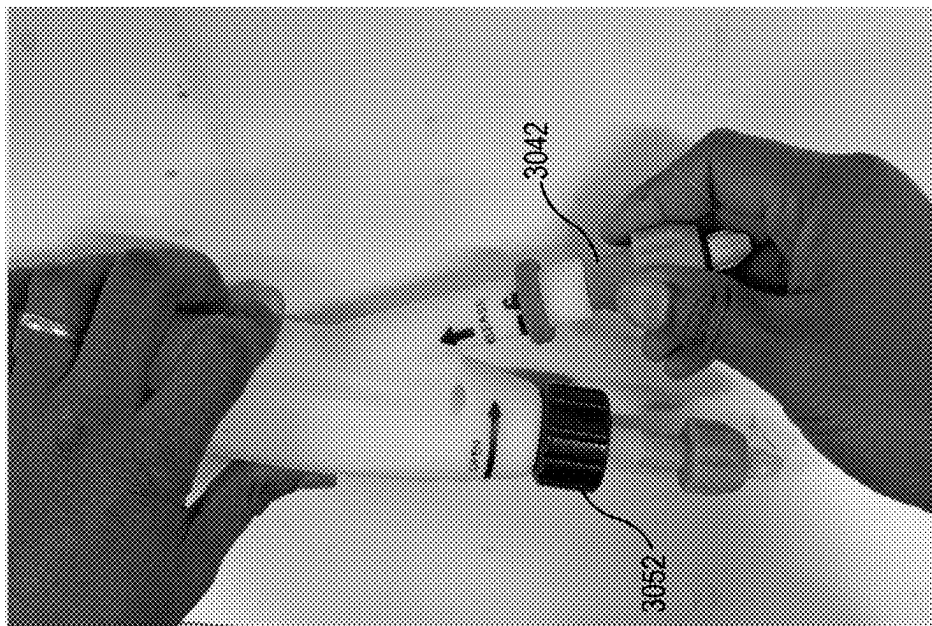


FIG. 37B

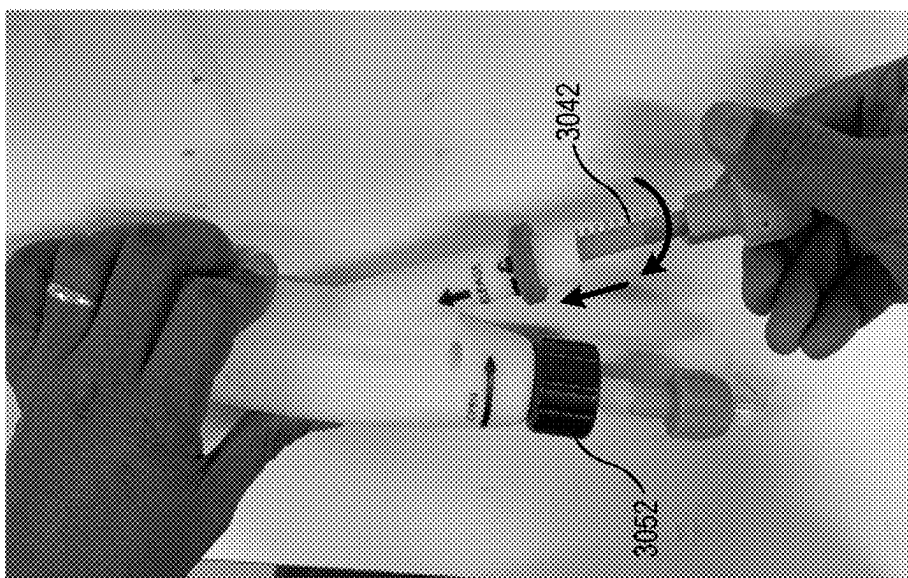


FIG. 37A

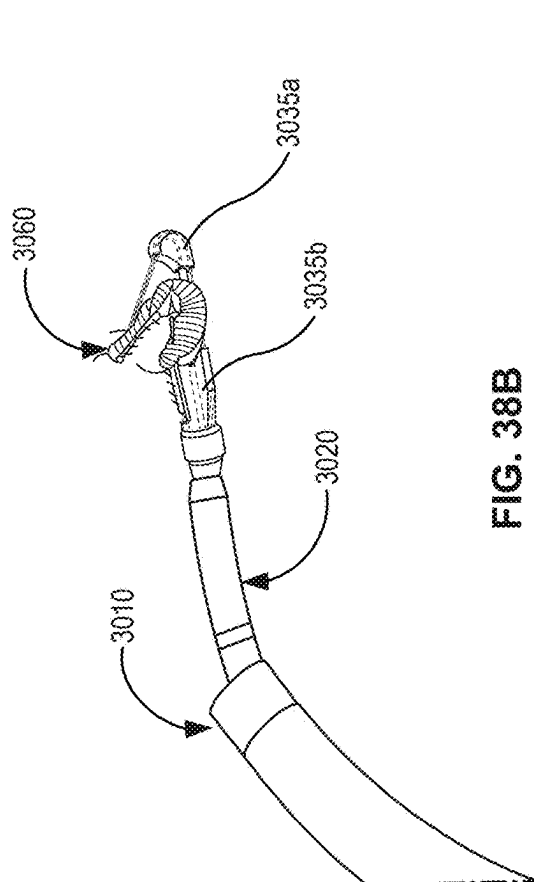


FIG. 38B

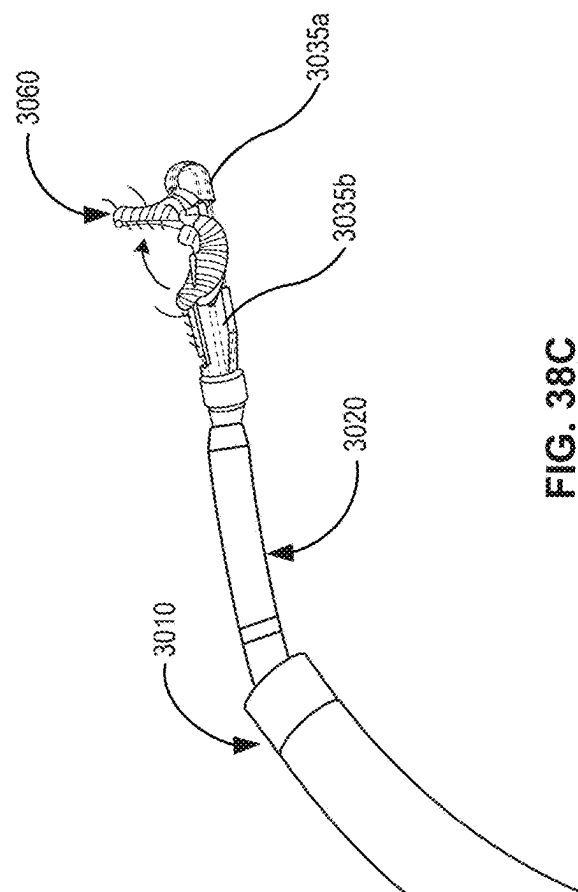


FIG. 38C

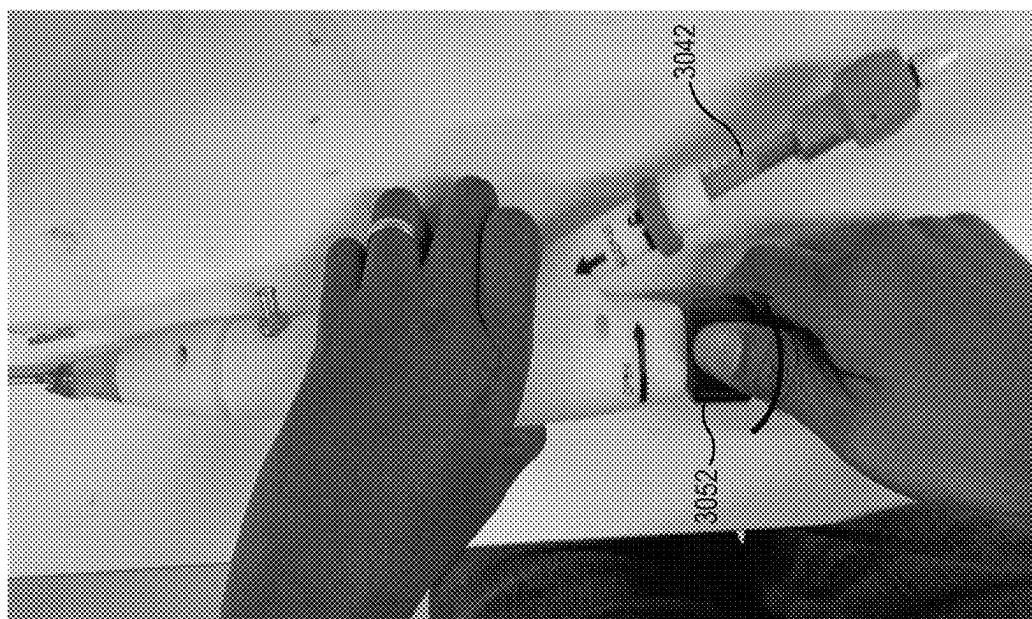


FIG. 38A

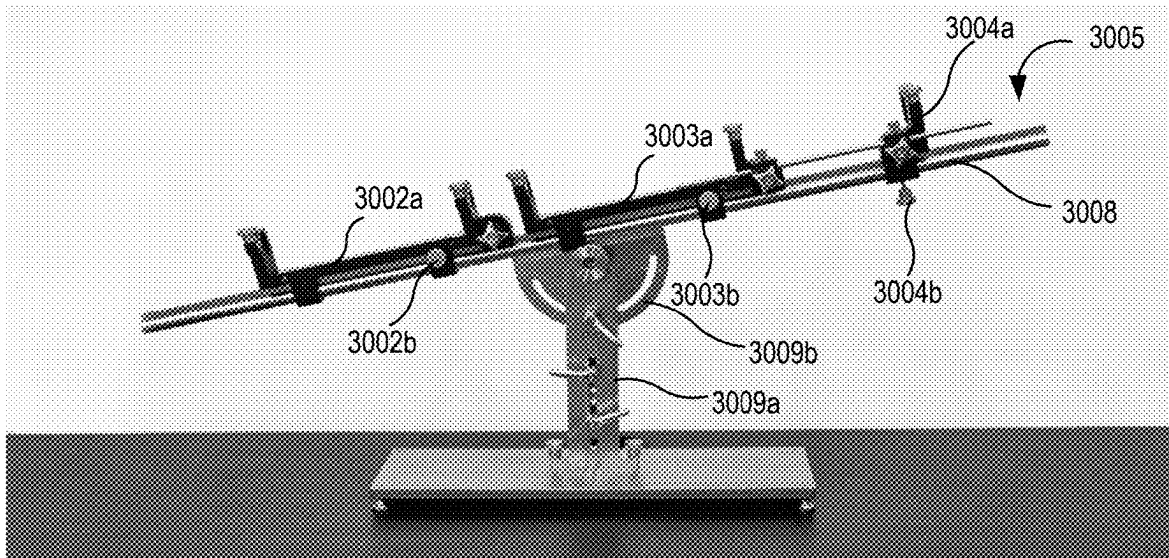


FIG. 39A

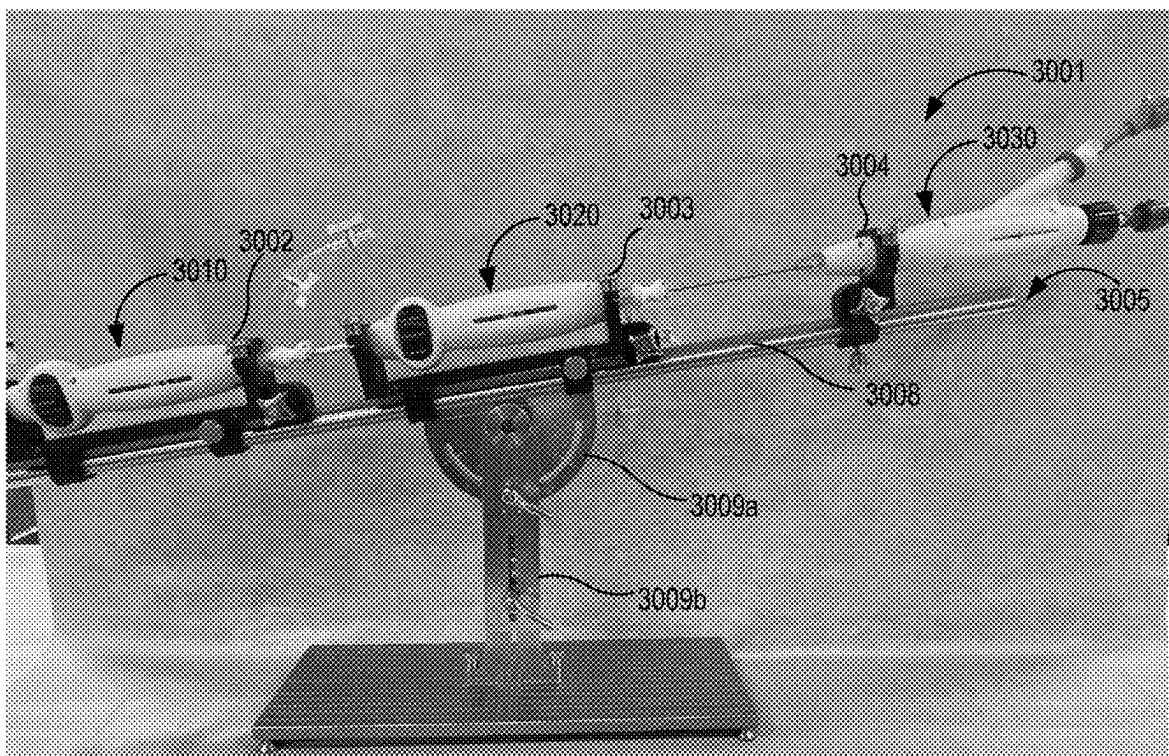


FIG. 39B

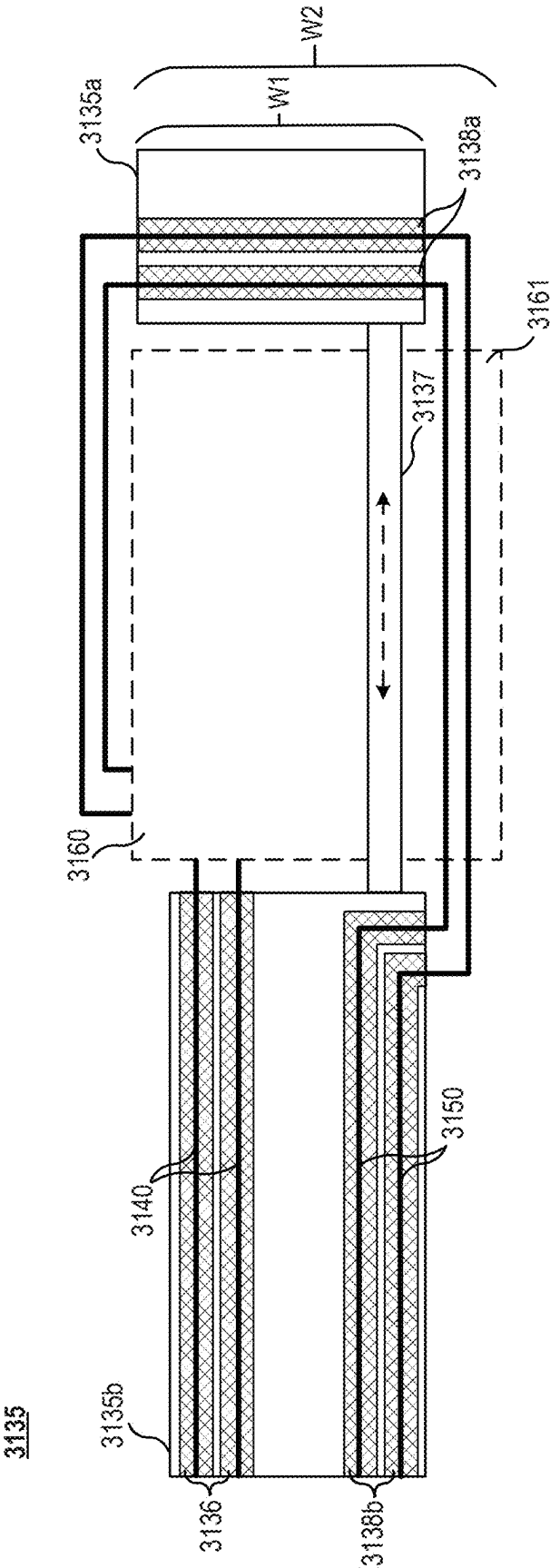


FIG. 40

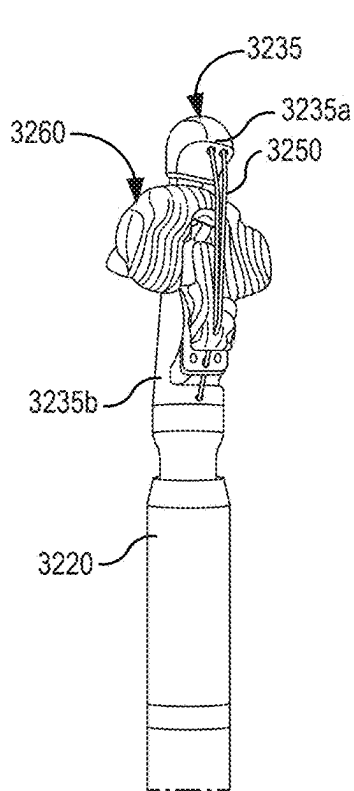


FIG. 41A

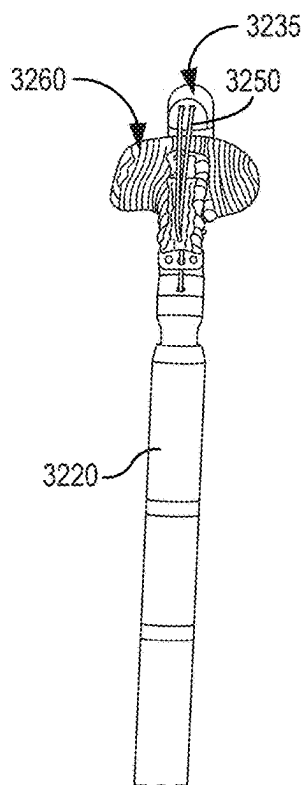


FIG. 41B

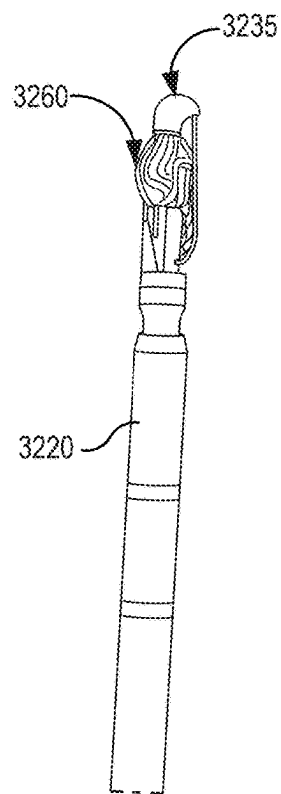


FIG. 41C

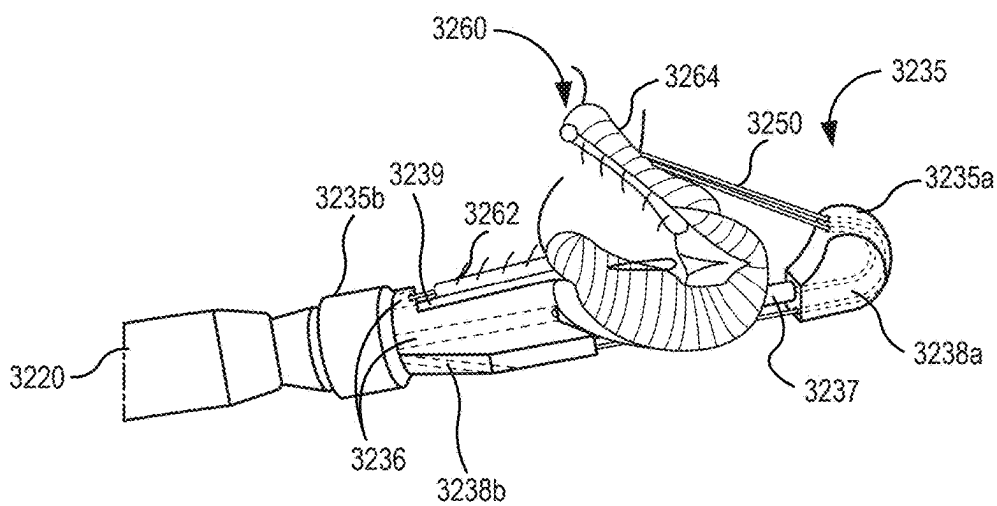


FIG. 41D



FIG. 42A

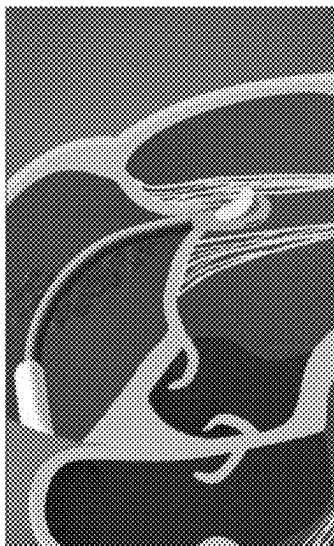


FIG. 42B



FIG. 42C

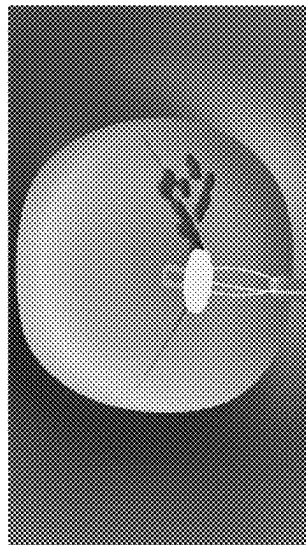


FIG. 42D

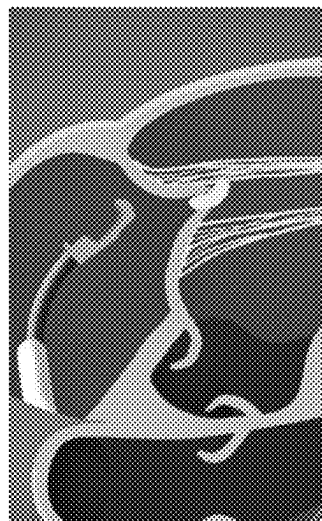


FIG. 42E

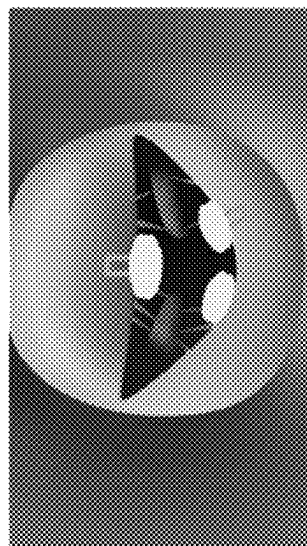


FIG. 42F

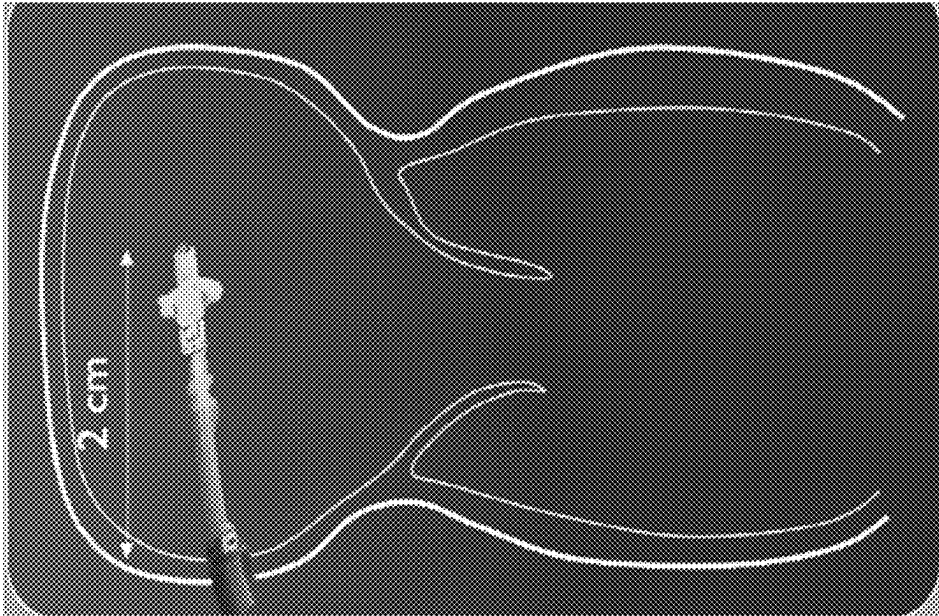


FIG. 43B

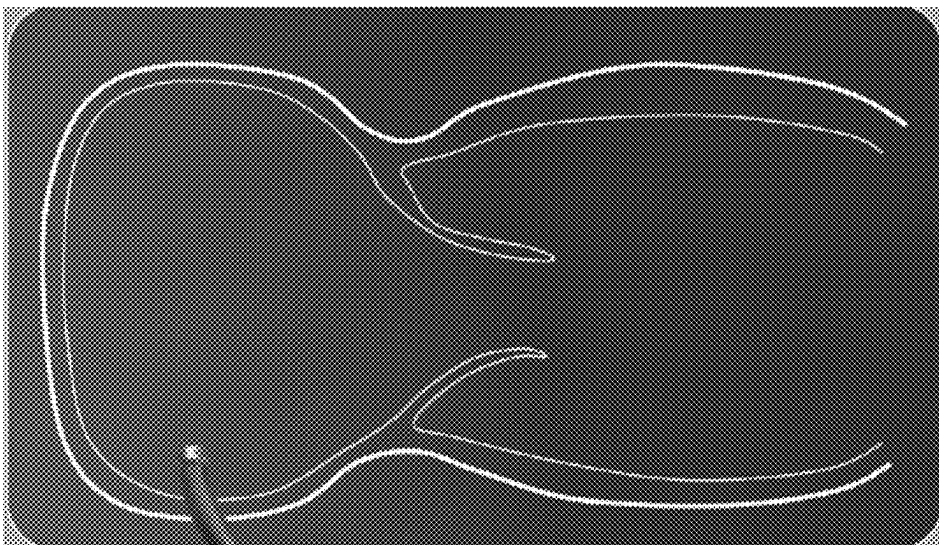


FIG. 43A

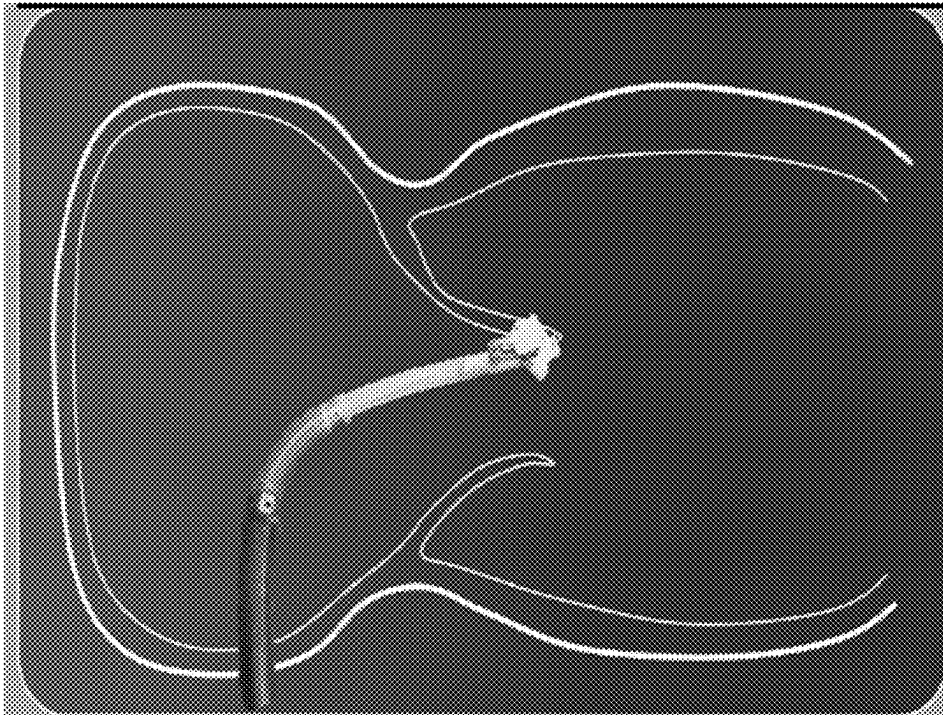


FIG. 43D

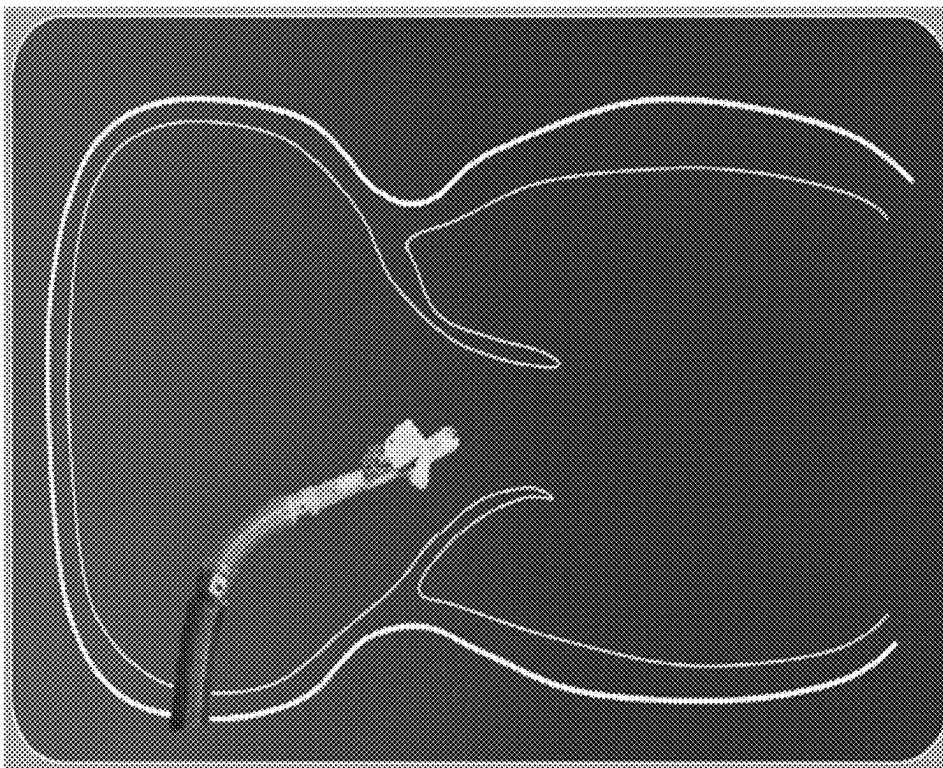
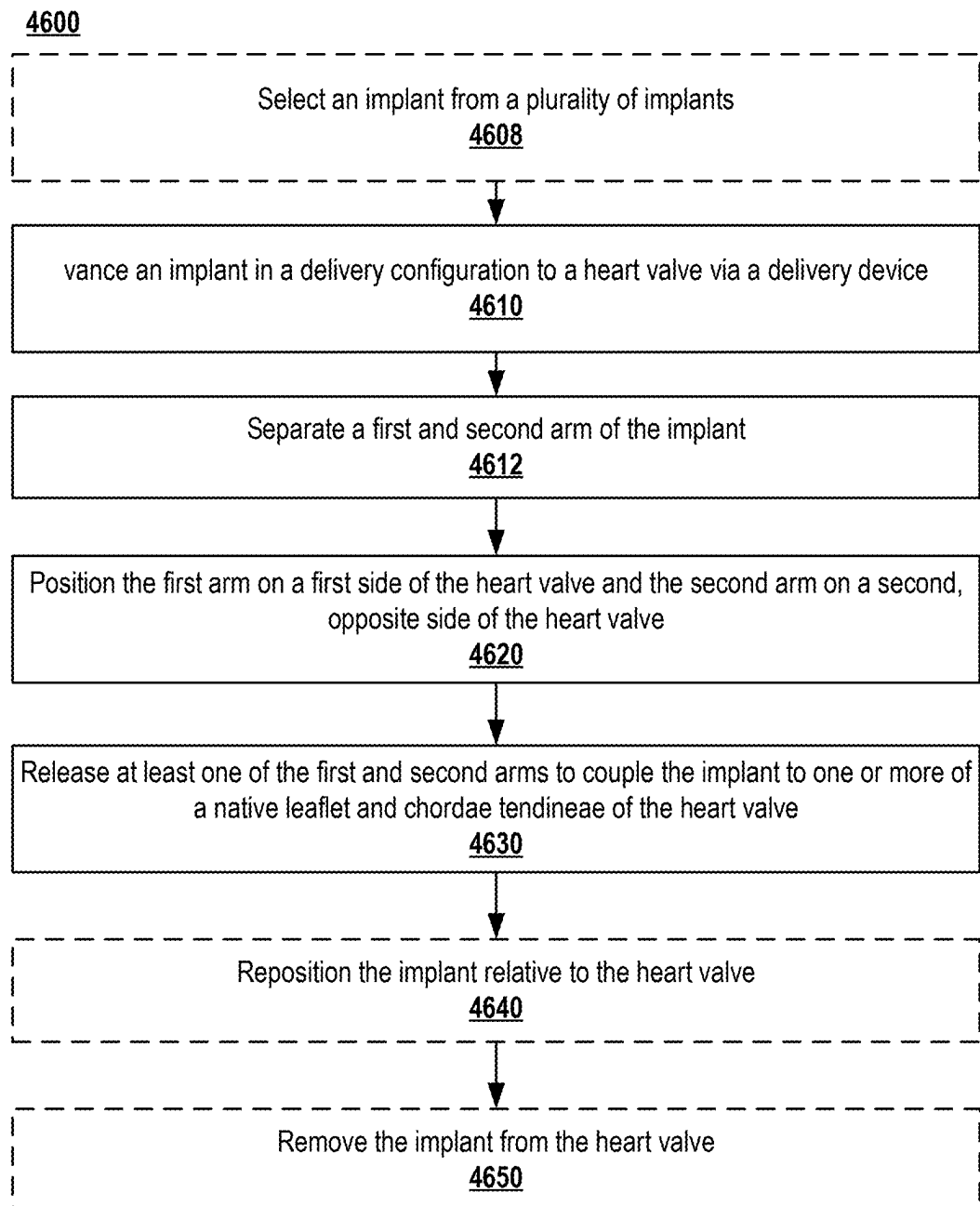


FIG. 43C

**FIG. 44**

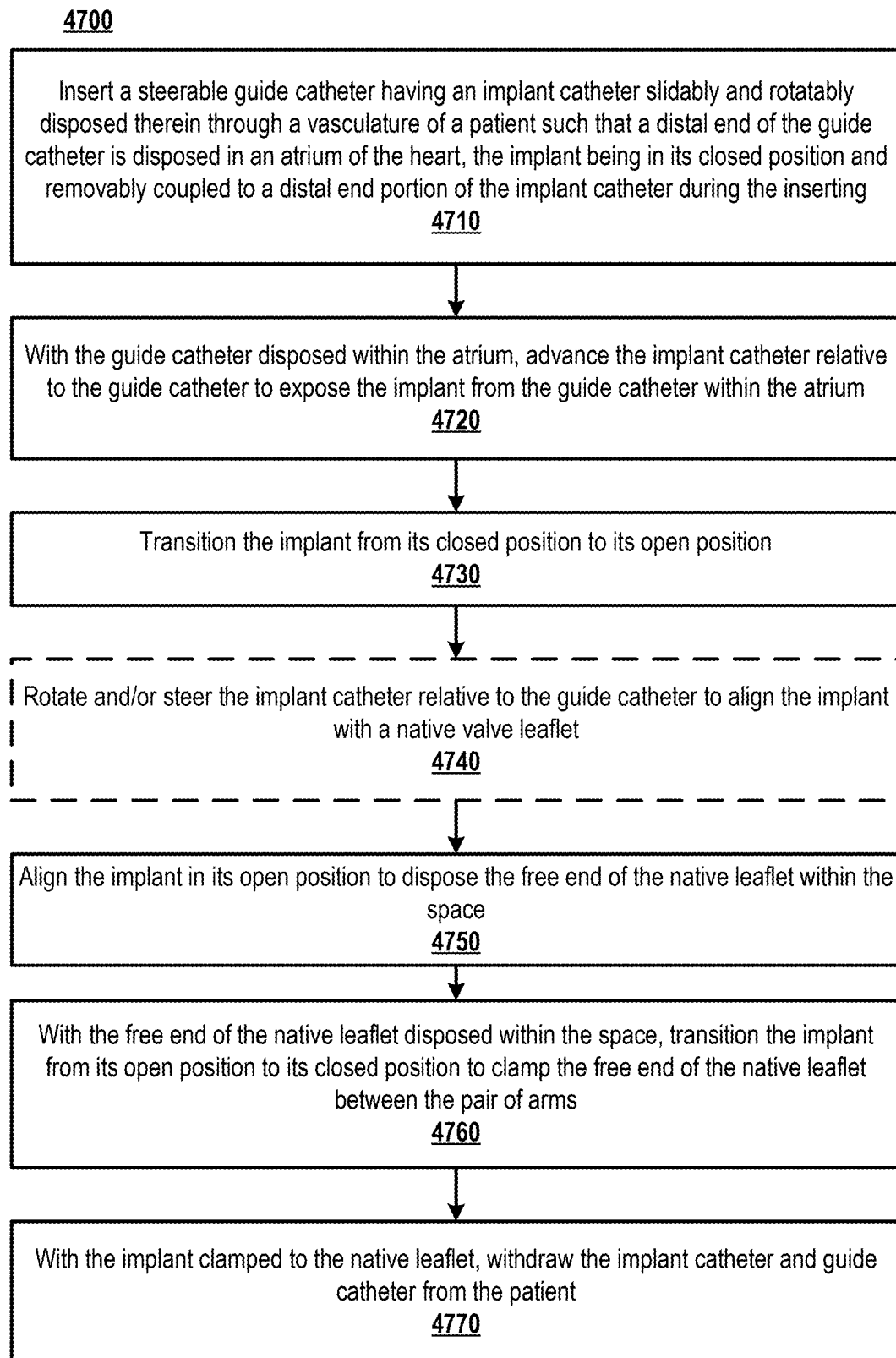
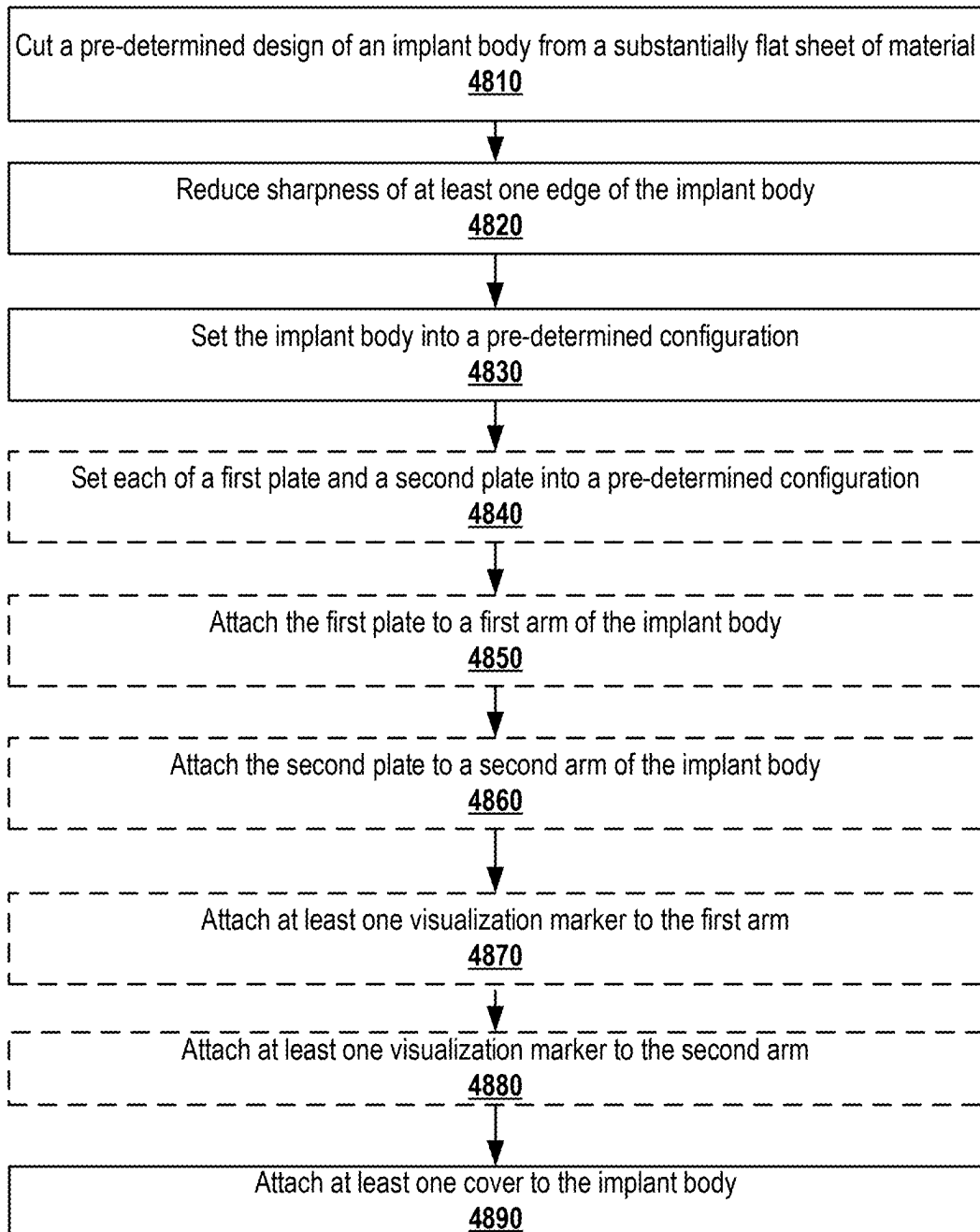


FIG. 45

4800**FIG. 46**

1

SYSTEMS, DEVICES, AND METHODS FOR REDUCING HEART VALVE REGURGITATION

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims priority to and the benefit of U.S. Provisional Application No. 63/675,682, filed Jul. 25, 2024, entitled “Systems, Devices, and Methods for Reducing Heart Valve Regurgitation”, the contents of which are hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

Embodiments described herein relate to an implant and an implant delivery system for reducing heart valve regurgitation. In particular, embodiments described herein relate to a transcatheter delivery system for delivering an implant to the mitral heart valve.

BACKGROUND

Heart valve regurgitation, particularly functional mitral regurgitation (FMR), is a common heart valve lesion seen in heart failure (HF) patients and it has been shown to worsen the progression of heart failure and increase the mortality of patients. Most heart failure patients are too high-risk for surgeries due to their poor health conditions. Minimally invasive transcatheter approaches exist for delivering implants to the heart valves (e.g., the mitral valve); however, current implant technologies fail to adequately treat FMR and can disturb the native valve dynamics resulting in poor long-term durability.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is a diagram illustrating coaptation at an annular plane of a heart valve (left) and dilation of the annulus and the ventricular cavity that deforms the heart valve and that results in heart valve regurgitation.

FIG. 1B is a diagram illustrating mitral regurgitation during systole (left) and treatment of the heart valve regurgitation using an implant (right).

FIGS. 2A-2C are schematic diagrams of an implant delivery system for reducing heart valve regurgitation during different stages of delivery of the implant, according to embodiments.

FIG. 3 is a schematic block diagram of an implant holder of an implant delivery system, according to embodiments.

FIG. 4 is a schematic diagram of an implant for reducing heart valve regurgitation, according to embodiments.

FIG. 5A is an illustration of an implant delivery system including a multi-lumen implant catheter and an implant holder, according to embodiments.

FIG. 5B shows a close up of a distal end of the implant delivery system, according to embodiments.

FIG. 5C is an image of the implant delivery system including a handle assembly, an implant catheter, and an implant holder, according to embodiments.

FIGS. 6A-6E show images of steerability of catheters of an implant delivery system and FIGS. 6F-6J show images of proximal control mechanisms of the implant delivery system configured to steer the catheters, according to embodiments.

FIGS. 7A-7C shows steerability of an implant delivery system including three catheters, according to embodiments.

2

FIG. 8 is a diagram of an implant holder of an implant delivery system, according to embodiments.

FIGS. 9A-9C show images of an implant holder of an implant delivery system including an implant coupled thereto, according to embodiments.

FIGS. 10-15 depict implants for treating heart valve regurgitation, according to various embodiments.

FIGS. 16A-16C illustrate a front perspective view (FIG. 16A), a back perspective view (FIG. 16B), and a bottom view (FIG. 16C) of an implant, according to an embodiment.

FIG. 17A illustrates a front view of an implant, according to an embodiment. FIG. 17B illustrates a front view of a first arm of the implant of FIG. 17A, according to an embodiment. FIG. 17C illustrates a front view of a second arm of the implant body of FIG. 17A, according to an embodiment.

FIG. 17D illustrates a back perspective view of the implant of FIG. 17A, according to an embodiment. FIG. 17E illustrates a front perspective view of the implant of FIG. 17A, according to an embodiment.

FIG. 17F illustrates a bottom view of the implant of FIG. 17A. FIG. 17G illustrates a side view of the implant of FIG. 17A.

FIGS. 18A-18C illustrate a front perspective view, a back perspective view, and a bottom view, respectively, of a first plate of an implant, according to an embodiment.

FIGS. 19A-19C illustrate a front view, a side view, and a bottom view of a second plate of an implant, according to an embodiment.

FIG. 20A illustrate a top perspective view of an implant, according to an embodiment.

FIGS. 20B-20E show edges of the implant where one or more plates are welded to the implant, according to embodiments.

FIGS. 21A-21B illustrate a portion of a first arm showing visualization markers received within an implant, according to embodiments.

FIGS. 22A-22C illustrate a cover for an implant for treating heart valve regurgitation, according to embodiments.

FIG. 23A illustrates a top view of a cover in a flat configuration, according to embodiments. FIGS. 23B-23C illustrate a front view of the cover coupled to the implant, according to embodiments.

FIG. 24 illustrates a front view of the implant shown in FIGS. 17A-17G in a substantially flat configuration, according to embodiments.

FIG. 25 illustrates a front view of a variation of the first plate shown in FIGS. 18A-18C in a substantially flat configuration, according to embodiments.

FIG. 26 illustrates a front view of a variation of the second plate shown in FIGS. 19A-19C in a substantially flat configuration, according to embodiments.

FIGS. 27A-27B are front views of an implant for treating heart valve regurgitation, according to an embodiment.

FIG. 28 are images of different size implants in a delivery configuration (bottom row) and a deployed configuration (top row).

FIG. 29 shows an implant delivery system including a handle assembly coupled to a proximal end of a catheter system and an implant holder coupled to distal end of the catheter system, according to embodiments.

FIGS. 30A-30B show the handle assembly of FIG. 29 including a guide catheter actuator coupled to a guide catheter and a dilator configured to be disposed through a lumen of the guide catheter, according to embodiments.

FIG. 31A shows a close-up view of the guide catheter actuator, according to embodiments.

FIGS. 31B-31C show a distal end of the guide catheter including a bendable section configured to bend in response to actuation of the guide catheter actuator, according to embodiments.

FIG. 32A shows a first actuation of a delivery catheter actuator, according to embodiments.

FIGS. 32B-32D show a distal end of the delivery catheter in response to the first actuation of the delivery catheter actuator, according to embodiments.

FIG. 33A shows a second actuation of the delivery catheter actuator of FIG. 32A, according to embodiments. FIGS. 33B-33C show the distal end of the delivery catheter in response to the second actuation, according to embodiments.

FIGS. 34A and 34C show translation of the delivery catheter actuator relative to an implant catheter actuator, according to embodiments. FIGS. 34B and 34D show the distal end of the implant catheter and the delivery catheter in response to the translation, according to embodiments.

FIGS. 35A-35B show the implant catheter and the delivery catheter in an extended configuration and a retracted configuration, respectively, according to embodiments.

FIGS. 36A-36C show steerability maps of a distal end of the implant delivery system in response to actuation of the guide catheter actuator, the delivery catheter actuator, and the entire handle assembly, respectively, according to embodiments.

FIGS. 37A-37B show actuation of an atrial tether actuator, according to embodiments.

FIGS. 38A-38C show actuation of a ventricular tether actuator to open a ventricular arm of the implant away from an atrial arm of the implant, according to embodiments.

FIGS. 39A-39B show a handle assembly holder to help a user control the handle assembly, according to embodiments.

FIG. 40 is a schematic diagram of an implant holder including a proximal portion and a distal portion joined by a connector therebetween, according to embodiments.

FIGS. 41A-41D shows an implant holder including a proximal portion and a distal portion joined by a connector therebetween, according to embodiments.

FIGS. 42A-42F illustrate a method for delivery an implant to a heart valve using an implant delivery system to treat heart valve regurgitation, according to embodiments.

FIGS. 43A-43D illustrate a method for delivery an implant to a heart valve using an implant delivery system to treat heart valve regurgitation, according to embodiments.

FIG. 44 is a flow chart for an example method of delivering an implant to treat heart valve regurgitation, according to embodiments.

FIG. 45 is a flow chart diagram showing an example method for delivering an implant to reduce heart valve regurgitation, according to embodiments.

FIG. 46 is a flow chart for an example method of manufacture an implant for treating heart valve regurgitation, according to embodiments.

DETAILED DESCRIPTION

Heart valve regurgitation has been shown to worsen the progression of heart failure (HF) and increase the mortality rate of patients. The first line of treatment for HF patients with heart valve regurgitation, such as mitral valve regurgitation, is guideline directed medical therapy (GDMT). Medical therapy targets the hemodynamics or left ventricular (LV) remodeling, rather than the valve itself, in an attempt to improve LV geometry which in turn would have

secondary effect on mitral regurgitation. Cardiac resynchronization therapy (CRT) is also used to treat mitral regurgitation in patients with prolonged QRS. CRT helps with synchronizing the LV contraction thereby increasing its systolic function, which in turn increases the LV closing force. FMR can also often be treated by either surgical repair or replacement of the valve when patients undergo concomitant coronary bypass surgery. However, most HF patients are deemed as too high-risk for surgeries due to their poor health condition. Therefore, minimally invasive transcatheter technologies to repair or replace the valve are in need for such patients. Current technologies fail to adequately treat FMR and disturb the native valve dynamics resulting in poor long-term durability.

In contrast, embodiments described herein augment the valve leaflet at the regurgitant site using a light-weight implant to provide leaflet extension and cover the regurgitant gap in systole; and the implant can move along with the leaflet in diastole to allow unrestricted inflow. In some embodiments, the implant can focally augment the native leaflet when attached. A steerable implant delivery system may be used for transcatheter deployment of the implant under image guidance. The implant described herein overcomes the anatomical constraints for implant interventions for heart valve regurgitation, thereby helping with expanding the patient population who can be treated significantly. The embodiments described herein can use a single leaflet capture approach, so the risk of developing functional stenosis or capturing short leaflets or restricting the possibility of future interventions on the same valve is eliminated.

Disclosed herein are devices, systems, and methods for reducing heart valve regurgitation. Generally, the devices (e.g., implants) described herein may be configured to couple to a heart valve such that the heart valve may be altered to treat a patient's cardiac condition, such as heart valve regurgitation. The devices described herein generally comprise implantable devices configured to couple to cardiac tissue, such as one or more of a valve leaflet, chordae tendineae, and an annulus of a heart valve, to reduce or eliminate heart valve regurgitation. The devices may comprise an attachment section and a coaptation section. The attachment section may be configured to couple the implant to tissue, and the coaptation section may be configured (e.g., alone or in combination with the attachment section) to enhance and/or alter one or more physical characteristics of the heart valve. Physical characteristics may include, but are not limited to, leaflet length, leaflet height, leaflet width, leaflet thickness, leaflet curvature, leaflet stiffness, leaflet shape, leaflet strength, and a combination thereof. In this manner, when implanted, the implant may enhance and/or alter one or more physical characteristics that may affect the movement, deformation, and/or stretch of the leaflet. The altered physical characteristics may affect a portion of the heart valve or, in some variations, the entire heart valve. The altered physical characteristics may be useful in treating (e.g., partially or completely correcting) a cardiac condition, such as heart valve regurgitation (e.g., functional mitral regurgitation (FMR), tricuspid valve regurgitation, aortic valve regurgitation, pulmonary valve regurgitation).

In some variations, the one or more altered physical characteristics of the heart valve may reduce or eliminate a gap between leaflets, thereby reducing or eliminating valve regurgitation. For example, the devices described herein may alter one or more physical characteristics of a valve in a manner that increases a coaptation surface area for the valve leaflets. That is, the devices described herein may couple to a leaflet of a valve such that a length, thickness,

5

and/or width of that leaflet is modified (e.g., increased). The modified length, width, and/or thickness of the leaflet may provide for a modified coaptation surface (e.g., increased coaptation surface area) that decreases or eliminates regurgitation. For instance, the devices described herein may protrude or otherwise extend from a surface of a heart valve leaflet such that another leaflet may coapt onto a portion of the device during systole, which may re-establish unidirectional blood flow through the valve. In some embodiments, the devices described herein may couple to the valve such that leaflet movement is modified in a manner that reduces or eliminates regurgitation.

The devices described herein may provide many benefits. For example, the devices may be configured to capture a single leaflet, which may preserve a valve orifice, as opposed to an edge-to-edge approach. In some embodiments, the single leaflet may be captured while the associated valve may be open (e.g., during diastole). Additionally, the devices may be configured to move with the heart valve during diastole and accordingly diastolic flow may not be impeded and thus the devices may avoid inducing functional stenosis. In a further example, the devices may be used in combination with valve replacement (e.g., transcatheter mitral valve replacement). That is, the devices may be coupled to the same valve that may undergo valve replacement without removing the coupled devices and/or requiring additional techniques or technologies to accommodate the coupled devices, and/or modifying the anatomy of the valve favorably (i.e. increase leaflet dimension) to enable transcatheter mitral valve replacement. Furthermore, the one or more physical characteristics of the heart valve to be altered may be specifically selected based on characteristics (e.g., anatomy, severity of condition to be treated, etc.) of a specific patient. For example, a patient's leaflet thickness may require enhancement so the devices described herein may alter the leaflet thickness without affecting a leaflet width. In a further example, the patient's leaflet thickness and width may require enhancement, such that the devices may facilitate both enhancements. In this way, the devices may be used in a wide range of patients, including patients with relatively small heart valves, relatively older patients, and/or patients with relatively calcified mitral annuluses.

In addition, the devices described herein may be deformable such that the devices may non-destructively (e.g., atraumatically) affect the structure or function of the heart valve and/or may receive a portion of the heart valve (e.g., a leaflet) via coaptation. For example, the devices may releasably attach to heart valve tissue (e.g., native leaflets, chordae tendineae) via a compressive force and/or friction force that may not damage the tissue. The compressive force may be applied by an attachment section comprising a first segment or arm and a second segment or arm of the implant. In some configurations, at least a portion of each of the first and second segments or arms may be coplanar with one another, which may assist in maintaining the position of the device relative to the heart valve. The central member may define a portion of a coaptation section, and the coaptation section may further include one or more support members. The support members may extend from the first segment or arm and may assist in defining a volume within the coaptation section. At least a portion of the coaptation section (e.g., at least a portion of the central member, one or more support members or a portion thereof) may be deflectable and/or deformable upon contact by a native leaflet. For example, one or more support members may deflect and/or deform relative to one or more additional support members. By virtue of the deflection and/or deformation of one or

6

more of the support members (e.g., two, three, four, or more), the coaptation section volume at least partially defined by the support members may change (e.g., may decrease) upon contact by a native leaflet, and may be restored to its pre-contact volume upon loss of contact with the native leaflet.

Moreover, the devices described herein may be adjustable such that a physician may change the configuration and/or position of the devices during the procedure without detrimentally affecting the structure or function of the heart valve. For example, the devices may be configured to transition between a first configuration (e.g., an implanted configuration) and a second configuration (e.g., an open configuration) multiple times, and may avoid damaging tissue while applied to the heart valve (including during reapplication or application several times) during any such transition between the configurations. In this way, the devices may be positioned and repositioned until the devices are placed in an optimal orientation on the heart valve such that the effectiveness (as measured by, for example, a reduction in regurgitation) of the devices is maximized. To further aid in the positioning of the devices, the devices may also be directly or indirectly visualized by a physician during the procedure, such that the physician may determine the position and configuration of specific portions of the devices. Accordingly, the adjustability and visibility of the devices may reduce or eliminate the need for subsequent surgical procedures that may otherwise be required to adjust the position(s) of the devices. In some variations, multiple devices may each be releasably attached to heart valve tissue simultaneously such that the effects of each device may be combined to optimally reduce heart valve regurgitation. The devices may further enable tissue encapsulation such that the devices may be permanently integrated into the heart valve. The tissue encapsulation may provide additional thickness to the coaptation section, which may increase the efficacy of the devices. The efficacy may continue to increase as the thickness of the tissue increases over time. In this way, the efficacy and permanence of the devices described herein may result in patients reducing or eliminating heart valve regurgitation, the use of anti-coagulants, and/or avoiding risks associated with tissue maladaptation.

The systems described herein may include an implant delivery system configured to advance an implant attached thereto within a patient. The implant delivery system may comprise a handle coupled to one or more catheters, with each catheter comprising one or more lumens. One or more elongate members (e.g., wires, sutures, tubes) may be routed through the one or more lumens of the catheter. In some variations, the elongate member may comprise a tube comprising a lumen therethrough, and one or more tubes may be routed through the one or more lumens of the catheter. One or more additional elongate members (e.g., wires, sutures) may be advanced through the one or more tubes. The elongate member(s) may be operatively coupled to the first segment or arm of the implant and/or the second segment or arm of the implant, such that the first and second segments or arms may be moved relative to each other. In some variations, the elongate member may be operatively coupled to the first segment or arm and/or the second segment or arm via a direct attachment between the elongate member and the first segment or arm and/or the second segment or arm, while in other variations the elongate member may be operatively coupled to the first segment or arm and/or the second segment or arm via an indirect connection. For example, in some variations, coupling via an indirect connection may include coupling the elongate member to the

first segment or arm and/or the second segment or arm via a looped elongate member (e.g., suture loop) attached to the first segment or arm or the second segment or arm to which the elongate member couples. Accordingly, the delivery device may facilitate positioning the implant adjacent to a target heart valve location (e.g., native leaflet, annulus, chordae tendineae), release the first and/or second segments or arms to couple the implant to the heart valve, and optionally reapply a force to the first and/or second segment or arm to decouple the implant from the heart valve to facilitate removal, retrieval, and/or repositioning of the implant.

FIG. 1A is a diagram illustrating coaptation at an annular plane of a heart valve (e.g., the mitral valve (MV)) on the left and dilation of the annulus that results in heart valve regurgitation on the right. The mitral valve is the bicuspid atrio-ventricular valve located between left ventricle (LV) 10 and left atrium (LA) of the heart. The valve has two leaflets, the anterior leaflet and the posterior leaflet 12, the base of these leaflets attached to a fibrous structure between LV 10 and LA called annulus 12. The MV annulus is saddle shaped with high points/peaks at anterior and a posterior portion and low points at the lateral and medial commissure. The chordae tendinae (hereinafter, "chords") 14 are string-like structures that connect the leaflets 12 to the papillary muscles and the LV 10. The chords 14 originate from the tip of the papillary muscles and insert into the ventricular side of both anterior leaflet (AL) and posterior leaflet (PL). Mitral regurgitation (MR) occurs when the valve leaflets 12 fail to coapt during systole resulting in backflow of blood through the gap formed between the leaflets. This backflow of blood engorges the pulmonary veins causing shortness of breath, and also causes volume overload in the LV 10 and is one of the significant contributors of heart failure (HF). Functional mitral regurgitation can be broadly classified into atrio-genic or atrial FMR and ventricular FMR based on the underlying pathology.

In FMR, this physiological leaflet motion is perturbed and thus the coaptation seal is inadequate. The dilated annulus draws the leaflets 12 away from each other in diastole, and a larger distance needs to be traversed by the leaflets 12 before they coapt in systole. The dyskinetic and enlarged LV tethers both marginal and strut chordae, and both the anterior and posterior leaflets, resulting in inadequate deformation of the leaflets to enable the verticalization of the leaflet edges to form a coaptation shelf in systole. As shown, in a normal mitral valve, the rough zone 12R of the leaflet edges 12 verticalize and form the coaptation shelf or seal in systole. In the abnormal mitral valve shown on the right, displacement of papillary muscles and dilation of annulus results in an increased chordal tension preventing the leaflet edges 12 to verticalize and forming a coaptation shelf or seal.

FIG. 1B is a diagram illustrating mitral regurgitation during systole (left) and treatment of the heart valve regurgitation using an implant 160 (right). As shown on the right, an implant 160 may be atraumatically attached to the edge of one of the leaflets 12 at the regurgitant site to provide the vertical shelf needed for coaptation. This implant 160 protrudes off the leaflet 12 surface to focally increase the length and width at the leaflet edge 12, which can potentially provide enough leaflet 12 reserve and also reduce the extent of motion of opposing leaflet needed to get into coaptation. The protrusion of the implant can be appropriately sized based on the regurgitant jet dimensions and characteristics such that this focal augmentation using the implant fits the regurgitant gap in systole but would not interfere with the opening of the valve during diastole to avoid the risk of

stenosis. The protrusion of the implant can be appropriately oversized based on the regurgitant jet dimensions and characteristics such that this focal augmentation using the implant may be larger than the regurgitant gap in systole to achieve a large reserve of coaptation, but does not interfere with the opening of the valve during diastole to avoid the risk of stenosis.

FIGS. 2A-2C are schematic diagrams of an implant delivery system 100 for reducing heart valve regurgitation during different stages of delivering an implant 160, according to embodiments. The implant delivery system 100 may include one or more catheters and a handle assembly coupled to a proximal end of the one or more catheters. In some embodiments, the implant delivery system 100 may include an implant catheter (IC) 130 extending alongside or through an inner lumen of a delivery catheter (DC) 120. The delivery catheter 120 may extend alongside or through an inner lumen of a guide catheter (GC) 110. A distal end of the implant catheter 130 may be coupled to an implant holder 135 including a cavity, space, or opening configured to receive the implant 160.

At least one of the implant catheter 130, the delivery catheter 120, or the guide catheter 110 may be steerable (e.g., may articulate in one or more planes) to navigate the implant delivery system 100 to a leaflet of the heart valve such that the implant 160 can be secured to the leaflet. In some embodiments, the delivery catheter 120 and the guide catheter 110 may be steerable to position the implant 160 near a heart valve of the patient. In some embodiments, the implant catheter 130 may not be steerable. For example, a stiffness of the implant catheter 130 may be less than that of the delivery catheter 120 such that a portion of the implant catheter 130 disposed in the delivery catheter 120 bends or flexes with the delivery catheter 120. As shown in FIG. 2B, the implant catheter 130, the delivery catheter 120, and the guide catheter 110 may be configured to navigate the implant holder 135 through a left atrium LA of the patient, past the valve leaflet L, and at least partially into the left ventricle LV such that the implant 160 can be secured to the leaflet of the mitral valve.

In some embodiments, the guide catheter 110 may be capable of bending to an angle from the longitudinal axis of up to 90 degrees, 100 degrees, 120 degrees, 135 degrees, 145 degrees, 150 degrees along one plane (e.g., a first plane). In some embodiments, the delivery catheter 120 may be configured to bend along one or more planes (e.g., along the first plane and a second plane perpendicular to the first plane). In some embodiments, the delivery catheter 120 may be configured to bend to an angle from the longitudinal axis of up to about 110 degrees, 120 degrees, 130 degrees, 140 degrees, 150 degrees, 160 degrees, 170 degrees, 180 degrees, 190 degrees along the first plane. In some embodiments, the delivery catheter 120 may be configured to bend to an angle from the longitudinal axis of up to about 30 degrees, 40 degrees, 45 degrees, 50 degrees, 55 degrees, 60 degrees along the second plane. In some embodiments, the guide catheter 110 can be bent to an angle of up to about 135 degrees, to access the left atrium. In some embodiments, the delivery catheter 120 can be bent to an angle of up to about 120 degrees, to help to point the implant 160 towards the valve and get access to different valve segments. In some embodiments, the implant catheter may not be steerable. In some embodiments, the implant catheter 130 may be configured to translate linearly (e.g., proximally and distally) relative to the delivery catheter 120 for positioning of the implant 160 near or on the leaflet. In some embodiments, the implant catheter 130 may be configured to translate linearly

a distance of up to about 3 cm, 4 cm, 5 cm, 6 cm, 7 cm, 8 cm, 9 cm, 10 cm, inclusive of all ranges and subranges therebetween.

In some embodiments, the delivery catheter **120** may be configured to rotate (e.g., between about 180 degrees to about 360 degrees) about its own longitudinal axis. In some embodiments, the implant catheter **130** may be configured to rotate about its own longitudinal axis. This rotation enables deployment on either the anterior or posterior leaflet of the heart valve. In some embodiments, the catheters **110**, **120**, **130** (or a portion thereof) may be formed from any suitable material such as, Nylon, Polyethylene, Nylon/Polyethylene, Pebax®, etc. In some embodiments, the catheters **110**, **120**, **130** may be extruded. In some embodiments, the catheters (e.g., the steerable catheters **120**, **130**) may be extruded with a metal ring along a length of the catheter (e.g., at a distal end) and one or more pull wires coupled to the ring. In some embodiments, the catheters (e.g., the steerable catheters **120**, **130**) may include two pull wires coupled to the ring with a predetermined degree of separation therebetween (e.g., about 180 degrees) such that tension to each of the pull wires articulates the catheter along one plane. In some embodiments, the catheters **110**, **120**, **130** may include any number of pull wires spaced at any interval around the circumference of the ring. In some embodiments, the catheters **110**, **120**, **130** may include 1 pull wire, 2 pull wires, 3 pull wires, 4 pull wires, 5 pull wires, 6 pull wires, 7 pull wires, 8 pull wires. The pull wires may run along a length of the respective catheter (e.g., the steerable catheters **120**, **130**) and be anchored to an actuator on the proximal end of the implant delivery system **100** (e.g., a GC actuator **112**, a DC actuator **122**, and/or an IC actuator **132** described in further detail below). In some embodiments, the actuator may be any suitable actuator such as, for example, rack and pinion, gear system, knob, slider etc. In some embodiments, the actuators may allow the user to steer the catheters **120**, **130**.

In some embodiments, at least one of the catheters **110**, **120**, **130** may include a port (e.g., a flushing port) connected to the inner lumen of the catheters **110**, **120**, **130**. The port may introduce fluid to reduce frictional contact between the catheters **110**, **120**, **130** and/or to remove air or prevent air from entering the patients' circulation. In some embodiments, a valve (e.g., a hemostatic valve) may be included in the handle assembly proximal to the port to close the proximal end of the catheters **110**, **120**, **130** and prevent leakage of blood.

In some embodiments, the catheters **110**, **120**, **130** have varying stiffness or hardness along their length. For example, the catheters **110**, **120**, **130** may include harder material at the proximal end transitioning to softer material at the distal end to allow bending of the softer portion while keeping the harder portion in place. In some embodiments, the stiffness or hardness may be measured using the Shore durometer scale with "D" referring to an index of the Shore durometer scale. In some embodiments, a stiffness of the guide catheter **130** may transition from about 45D to about 80D at a proximal end to about 15D to about 35 D at a distal end. In some embodiments, a stiffness of the delivery catheter **120** may transition from about 45D to about 80D at a proximal end to about 15D to about 35 D at a distal end. In some embodiments, a stiffness of the implant catheter may be less than 35 D. In some embodiments, a stiffness of the implant catheter may be less than 25 D. In some embodiments, the catheters **110**, **120**, **130** may have a column strength to be introduced and advanced into the vasculature, while also having flexibility to be articulated through the geometry of the heart. In some embodiments, a

bending length (e.g., a length of a bending portion of the catheter configured to bend or a length of a steerable portion) of the guide catheter **110** may be in a range of about 10 mm to about 140 mm, inclusive of all ranges and subranges therebetween. In some embodiments, the bending length of the guide catheter **110** may be in a range of about 30 mm to about 110 mm, inclusive of all ranges and subranges therebetween. In some embodiments, the bending length of the guide catheter **110** may be about 100 mm. In some embodiments, a bending length of the delivery catheter **120** may be in a range of about 10 mm to about 90 mm, inclusive of all ranges and subranges therebetween. In some embodiments, the bending length of the delivery catheter **120** may be about 55 mm. In some embodiments, the bending length of the delivery catheter may be about 37 mm.

In some embodiments, the guide catheter **110** and/or the delivery catheter **120** may include a bendable portion (e.g., a shape-set section) configured to bend a predetermined angle from the longitudinal axis. In some embodiments, the bendable portion may include a different material than the rest of the guide catheter **110** and/or delivery catheter **120**. In some embodiments, the guide catheter **110** and/or the delivery catheter **120** may have a constant hardness aside from the bendable portion. In some embodiments, the bendable portion may bend to an angle of about 10 degrees, about 20 degrees, about 30 degrees, about 40 degrees, about 50 degrees, inclusive of all ranges and subranges therebetween. In some embodiments, the predetermined angle may correspond to shape of the heart. For example, the bendable portion may be configured to bend to a 20 degree angle to maintain the distal end of the implant delivery system in the left atrium (e.g., by hooking the atrium) and stabilize the distal end of the implant delivery system. In some embodiments, the guide catheter **110** may include or be coupled to one or more valves. For example, the guide catheter may be coupled to a hemostatic valve for flushing bodily fluid (e.g., blood) from the lumen of the guide catheter **110**. In some embodiments, the delivery catheter. In some embodiments, the implant delivery system **100** may be configured to receive a dilator. The dilator may define a central lumen configured to accommodate a guide wire. The dilator may have a tapered distal end that forms a point. In some embodiments, a proximal portion of the tapered distal end may have a diameter corresponding to the diameter of the guide catheter **110** such that the dilator may expand tissue in preparation for the guide catheter **110** to be disposed there-through.

In some embodiments, the implant delivery system may only include two steerable catheters, and the implant holder may be coupled directly to one of the steerable catheters (e.g., a two-catheter delivery system). The guide catheter **610** may include a proximal portion that is of a hardness between about 60D to 65D (e.g., 63D), a middle portion of hardness between about 50D to 60 D (e.g., 55D), and a distal portion of hardness between about 20D to 30D (e.g., 25D). In some embodiments, the guide catheter **610** may gradually transition between hardness levels. The delivery catheter **620** may include a proximal portion having a hardness between 60D to 65D (e.g., 63D) and a distal portion having a hardness between 30D to 40D (e.g., 35D).

In some embodiments, the catheters **110**, **120**, **130** may be visible on fluoroscopy and echocardiography to aid in navigation of the implant delivery system **100**. In some embodiments, a distal tip of the guide catheter **110** may include a radio-opaque marker such as a ring including radio-opaque material. In some embodiments, a sleeve of the delivery catheter **120** may include a radio-opaque marker disposed at

11

or near a distal tip of the delivery catheter **120**. In some embodiments, the implant holder **135** may include a radio-opaque marker at its distal tip. In some embodiments, one or more radio-opaque markers may be disposed at a predetermined distance from a distal end of the guide catheter **110**. For example, the predetermined distance from the distal end of the guide catheter may be about 2.5 mm, about 5 mm, about 7.5 mm, about 9 mm, about 10.5 mm, about 13 mm, about 14.5 mm, about 17 mm, about 18.5 mm, about 20 mm. In some embodiments, one or more radio-opaque markers may be disposed at a predetermined distance from a distal end of the delivery system. For example, the predetermined distance from the distal end of the delivery catheter **120** may be about 2.5 mm, about 5 mm, about 7.5 mm, about 9 mm, about 10.5 mm, about 13 mm, about 14.5 mm, about 17 mm, about 18.5 mm, about 20 mm. In some embodiments, the delivery catheter **110** may include two radio-opaque markers disposed thereon: a first radio-opaque marker disposed about 5 mm from the distal end and a second radio-opaque marker disposed about 20 mm from the distal end.

In some embodiments, the proximal end of the implant delivery system **100** may include a handle assembly. The handle assembly may include one or more actuators (e.g., a button, a lever, a trigger, a linear slider, a knob, a wheel, or the like) configured to control a configuration of the implant **160** and/or an articulation of the catheters **110**, **120**, **130**. In some embodiments, the handle assembly may include a GC actuator **112** coupled to the guide catheter **110** (e.g., and one or more pull wires) and configured to articulate the guide catheter **110** during navigation of the implant delivery system **100** and/or delivery of the implant **160**. In some embodiments, the handle assembly may include a DC actuator **122** coupled to the delivery catheter **110** (e.g., and one or more pull wires) and configured to articulate the delivery catheter **120** during navigation of the implant delivery system **100** and/or delivery of the implant **160**. In some embodiments, the handle assembly may include an IC actuator **132** coupled to the implant catheter **130** and configured to articulate the implant catheter **130** during navigation of the implant delivery system **100** and/or to move the implant catheter **130** linearly (e.g., proximally and/or distally) relative to the steerable catheters **120**, **130**. In some embodiments, the three catheters **110**, **120**, **130** may work together to accommodate the geometry of the vasculature and the heart to deliver the implant **160** to a target heart valve.

In some embodiments, the implant **160** may include a central member coupled to a first segment or arm and a second segment or arm. In some embodiments, the implant **160** may further include one or more support members, one or more positioning openings, and/or one or more friction elements. The implant **160** may be configured to move between one or more configurations including a delivery configuration, an open configuration, a closed configuration, and an implanted configuration. For example, the implant **160** may move from the delivery configuration in which the first arm and the second arm are substantially coplanar and the one or more support members are in a partially or completely extended shape (e.g., an elongated state) and/or in a partially or completely collapsed shape (e.g., a compressed state), to an open configuration in which the first arm and the second arm are separated from one another. The implant **160** may move between the open configuration and an implanted configuration in which the first arm and the second arm are disposed around the leaflet and apply a clamping force to the leaflet. In some embodiments, the implant **160** may be in a closed configuration in which each

12

of the first arm and second arm are substantially coplanar with one another. In this way, the implant **160** in the open configuration can be positioned around a portion of a leaflet of the heart valve, and the implant **160** in the implanted configuration can apply the clamping force on the leaflet to secure the implant **160** to the leaflet. In some embodiments, the implant **160** may include a cover disposed around the central member to prevent damage to the leaflet tissue and/or to promote tissue growth. The implant **160** is described in further detail with respect to FIG. 4 and FIGS. 10-29F.

In some embodiments, the implant catheter **130** may be coupled to the implant holder **160** such that there is no rotational movement therebetween. For example, the implant catheter **130** may be welded to the implant holder **160**. In some embodiments, the implant catheter **130** may define one or more lumens extending therethrough. In some embodiments, one or more elongate members (e.g., wires, sutures, tethers, braids, etc.) may extend through the one or more lumens of the implant catheter **130** from a proximal end to a distal end of the implant catheter **130**.

In some embodiments, the implant catheter **130** may define a main lumen, and include a pattern cut on one side thereof (e.g., via laser cutting) to improve flexibility while maintaining torque-ability (e.g., maintaining a column strength of the implant catheter **130** for rotation). In some embodiments, the main lumen of the implant catheter **130** may be formed from a metal (e.g., stainless steel), alloy (e.g., Nitinol), or a combination thereof. In some embodiments, the main lumen may include one or more hypotubes extending therethrough. In some embodiments, four hypotubes may be disposed inside the main lumen of the implant catheter **130**. Each hypotube may define a lumen in which a portion of an elongate member may be disposed. In some embodiments, the hypotubes may be formed from a material to prevent deformation of or friction with the one or more elongate members. In some embodiments, the hypotubes may include a metal or metal alloy such as stainless steel, Nitinol or a polymer such as Nylon, Polyethylene, Polyurethane, Polyimide, etc. In some embodiments, the one or more hypotubes may include a polymer such as polyamide. In some embodiments, the one or more hypotubes may be braided polyamide tubes. In some embodiments, the one or more hypotubes may be configured to move within the main lumen of the implant catheter **130**. In some embodiments, the one or more elongate members may be formed from electropolished Nitinol or a polymer such as Teflon, Polyethylene, Kevlar etc. Additionally or alternatively, the elongate members (e.g., the atrial **142** and ventricular tethers **152**) may include a braided suture including a polymer. In some embodiments, the elongate members may be configured to have a stretchability below a predetermined threshold. Therefore, when a force is applied to the elongate member (e.g., the elongate member is actuated), the elongate member may impart a substantially equal force on the implant (e.g., without absorbing force by stretching). In some embodiments, the one or more elongate members may include a coating configured to reduce or limit friction between the elongate member and the lumen of the hypotubes and/or delivery catheter **130**. In some embodiments, the hypotubes may have a diameter corresponding to that of the elongate members. For example, in some embodiments, the hypotubes may each have an inner diameter in a range between about 0.010 inches and about 0.015 inches, inclusive of all ranges and subranges therebetween. In some embodiments, the hypotubes may have an outer diameter in a range between about 0.015 inches to about 0.021 inches, inclusive of all ranges and subranges therebetween. In some

13

embodiments, the hypotubes may have an inner diameter of 0.012 inches. In some embodiments, the hypotube may have an outer diameter of 0.018 inches.

In some embodiments, the one or more elongate members may include an atrial tether (AT) **140** and a ventricular tether (VT) **150**. In some embodiments, the atrial tether **140** and the ventricular tether **150** may each form a loop removably coupleable to a portion of the implant **160** (e.g., be disposed through one or more openings defined by the implant **160**). In some embodiments, the implant (e.g., the atrial arm and the ventricular arm) may include suture loops coupled thereto. In some embodiments, the atrial tether **140** and ventricular tether **150** may be coupleable to the one or more suture loops on the implant **160**. In some embodiments, the atrial tether **140** and the ventricular tether **150** may extend through at least a portion of the implant holder **135** and may secure the implant **160** in the cavity of the implant holder **135**. Each of the atrial tether **140** and the ventricular tether **150** may include a first end and a second end coupled to the proximal end of the implant delivery system **100** such that each form a loop (e.g., about 0.005 inches in diameter) at the distal end of the implant delivery system **100**. For example, a portion of the atrial tether **140** and the ventricular tether **150** may run along the length of the implant catheter **130** inside each hypotube, with their first and second ends anchored on the handle. In some embodiments, the second end of the atrial tether **140** and/or the second end of the ventricular tether **150** may not be coupled to the handle and may instead be coupled to a portion of the tether **140**, **150** distal to the handle.

In some embodiments, the handle assembly may further include an AT actuator **142** coupled to the atrial tether **140** and a VT actuator **152** coupled to the ventricular tether **150**. The AT actuator **142** and the VT actuator **152** may be configured to control a configuration of the atrial tether **140** and the ventricular tether **150** (e.g., apply/release tension the tethers, decouple the tethers from the implant **160**, etc.). In some embodiments, the AT actuator **142** and the VT actuator **152** may each include linear sliders coupled to the atrial tether **140** and the ventricular tether **150**, respectively, such that when each of the AT actuator **142** and the VT actuator **152** are moved proximally, tension is applied to the atrial tether **140** and the ventricular tether **150**, respectively.

In some embodiments, the atrial tether **140** and the ventricular tether **150** may be configured to manipulate the implant **160** into different configurations when tension is applied to the atrial tether **140** and/or the ventricular tether **150** (e.g., via the AT actuator **142** and/or the VT actuator). For example, the atrial tether **140** may be operatively coupled to the first segment or arm of the implant **160** (e.g., the atrial arm) and the ventricular tether **150** may be operatively coupled to the second segment or arm of the implant **160** (e.g., the ventricular arm). In this way, the physician may manipulate the AT actuator **142** and/or the VT actuator **152** (e.g., pull a trigger, move the linear slider proximally, rotate a knob in a first direction) to retract the atrial tether **140** and/or ventricular tether **150**, which may pull the first and second segments of the implant **160** apart from one another. In some embodiments, the AT actuator **142** may be actuated or maintained in position such that the first segment or arm (e.g., the atrial arm) of the implant **160** is stabilized. When the atrial arm of the implant **160** is stabilized, the VT actuator **152** may be actuated to tension the ventricular tether **150** such that the second segment or arm of the implant **160** (e.g., the ventricular arm) is moved away from the first segment or arm. This may be performed by the physician when the implant **160** is within the patient and

14

adjacent the target heart valve location (e.g., native leaflet, annulus, chordae tendineae), such that the implant **160** may surround the valve tissue. The physician may release the VT actuator (e.g., stop pulling the trigger, move the linear slider distally, rotate a knob in a second direction) to release the ventricular tether **150**, which may cause the first and second segments or arms of the implant **160** to move towards one another and couple the first and/or second segments or arms to the heart valve tissue.

In some embodiments, a distance an actuator (e.g., the AT actuator **142** and/or the VT actuator **152**) travels or moves may correspond to a degree the implant **160** opens (e.g., an angle between the first segment or arm and the second segment or arm). In embodiments where the actuator is a linear slider, a distance a linear slider moves corresponds to the degree the implant **160** opens. In some embodiments, a maximum length the linear slider travels may be in a range between about 5 mm to about 50 mm, inclusive of all ranges and subranges therebetween. In some embodiments, the maximum length the linear slider travels may be about 15 mm. In some embodiments, about 5 mm of linear motion may correspond to about a 45 degree increment of opening between the segments or arms of the implant **160**.

In some embodiments, the actuator (e.g., the AT actuator **142** and/or the VT actuator **152**) may include a rotatable actuator (e.g., a knob) to control tension applied to the atrial tether(s) **140** and/or ventricular tether(s) **150**, and therefore control a degree the implant **160** opens. In some embodiments, the AT actuator **142** may include a linear slider and a locking mechanism. In some embodiments, the AT actuator **142** may be tensioned to secure the atrial tether **140** in a desired position, and once the atrial arm is in the desired position, the locking mechanism may be activated (e.g., pulled, pushed, rotated, etc.) to lock the atrial tether **140** with the tension corresponding to the desired position of the atrial arm. In some embodiments, the VT actuator **152** may include a knob, and a degree of rotation of the knob may control an amount of tension applied to the ventricular tether **150**, and therefore, a degree to which the implant **160** opens. In embodiments where the actuator is a knob, a degree of rotation of a knob may correspond to the degree the implant **160** opens. In some embodiments, 180 degrees of rotation of the VT actuator **152** may correspond to about a 45 degree increment of opening between the segments or arms of the implant **160**. In some embodiments the actuator may be hybrid, translating a rotary knob motion to a linear translational motion.

In some embodiments, the proximal end of the implant delivery system **100** may include one or more stopping members for precise control over articulation of the implant **160**. In some embodiments, the stopping members may be disposed at a predetermined position along a length of the sliding path of the AT actuator **142** and/or the VT actuator **152** such that the implant **160** may only be opened or closed a predetermined amount and/or to indicate to the user (e.g., tactically) a degree the implant **160** has opened and/or closed. The stopping members may be protrusions, grooves, indents, detents, openings, slots, or the like configured to engage the actuators to prevent movement past the predetermined amount. For example, in some embodiments, a portion of the AT actuator **142** and/or a portion of the VT actuator **152** may have a shape complimentary to the sliding path on the proximal end of the implant delivery system **100** to indicate to the user a degree with which the implant **160** has been opened or closed. In some embodiments, the stopping members may at least temporarily stop or prevent movement of the AT actuator **142** and/or VT actuator **152**. In

15

some embodiments, the stopping members may be disposed along the length of the sliding path to indicate the inter-arm angle of the implant **160** is 0 degrees, 45 degrees, 90 degrees, and/or 120 degrees.

In some embodiments, the distal end of the implant delivery system **100** may be navigated near the heart valve such that the implant catheter **130** is oriented substantially perpendicular to the annular plane of the heart valve (e.g., mitral valve). In some embodiments, the second arm of the implant (e.g., the ventricular arm) may be opened (e.g., by tensioning the ventricular tether **150**), and the distal end of the delivery system may be visualized (e.g., via echocardiography). In some embodiments, the implant holder **135** may be oriented (e.g., rotated) to align the ventricular arm with the desired leaflet capture location. For example, the ventricular arm may be positioned near a portion of the leaflet that is not achieving coaptation. In some embodiments, after visualization, one or more locations of the implant **160** may be demarcated (e.g., by the user) on the echocardiogram. For example, the following markers may be annotated and/or considered: (1) a first marker can correspond to a proximal end of the atrial arm, (2) a second marker can correspond to a proximal end of the ventricular arm, and (3) third marker can correspond to a distal end of the ventricular arm. In some embodiments, the implant catheter **130** may be advanced towards the heart valve until the proximal end of the atrial arm aligns with an edge of the leaflet. In some embodiments, once the proximal end of the atrial arm aligns with the edge of the leaflet, a location at which the implant catheter **130** protrudes from the delivery catheter **120** can be demarcated on the implant catheter **130**. At this location, the implant is entirely below the leaflet edge (e.g., and positioned in the ventricle V).

Once the implant **160** is positioned below the leaflet edge, the ventricular arm can be opened away from the atrial arm. For example, tension can be applied to the ventricular tether **150** to open the ventricular arm (e.g., approximately to 90 degrees). For example, the VT actuator **152** may be actuated to open the ventricular arm. In some embodiments, while the ventricular arm is opened, the implant catheter **130** may be biased towards the leaflet. With the ventricular arm open, the implant catheter **130** may be retracted proximally. In some embodiments, the implant catheter **130** may be retracted proximally until the second marker location (e.g., to assess a degree to which the implant **160** covers the opening of the heart valve). The implant catheter **130** may be retracted further proximally to the third marker on the implant catheter **130**. Once retracted to the third marker, the ventricular arm may be closed a predetermined amount. For example, the VT actuator **152** may be actuated to loosen the ventricular tether **150** such that the ventricular arm closes to 45 degrees (i.e., 45 degrees between the atrial and ventricular arm). Closing the arm to 45 degrees can restrain the leaflet from opening beyond the ventricular arm. In some embodiments, the atrial arm can then be loosened from the implant holder **135** to move the atrial arm towards the leaflet. For example, the AT actuator **142** may be actuated to loosen the atrial tether **140** such that the atrial arm moves towards the leaflet. Moving the atrial arm towards the leaflet can cause the implant **160** to move with the leaflet such that the position of the protrusion of the implant **160** off of the leaflet (e.g., the location of augmentation) becomes clear. If the position of the implant **160** is not satisfactory (e.g., does not prevent valve regurgitation), tension to the atrial arm can be applied, and then tension onto the ventricular arm can be applied to open the implant **160** and repeat the previously described steps after steering to a new location.

16

Once the implant **160** is in the desired position, tension may be applied to the atrial arm to move the atrial arm towards the implant holder **135**, and then the ventricular tether **150** may be decoupled from the ventricular arm. The atrial arm tether can be decoupled after the ventricular arm tether is decoupled to release the implant **160** away from the implant holder **135**.

As shown in FIG. 2C, once the implant **160** is clamped onto a portion of the valve leaflet L, the atrial tether **140** and/or the ventricular tether **150** can be manipulated (e.g., loosened, de-tensioned) to move the implant **160** away from the cavity of the implant holder **135** such that the physician can monitor the placement of the implant **160** without detaching the implant **160** from the implant delivery system **100**. This allows for real time assessment of regurgitation correction; it allows the implant to go into coaptation freely without complete disengagement of the implant **160** from the implant delivery system **100**. If adjustments need to be made, the tethers can be tensioned to transition the implant **160** towards its open configuration sufficient to release the implant **160** from the valve leaflet L and pull the implant back into the cavity of the implant holder **135**, and the implant **160** can be repositioned using the tethers **140**, **150**. Once the placement of the implant **160** is confirmed by determining sufficient coaptation of the heart valve with the implant **160**, the atrial tether **140** and the ventricular tether **150** can be decoupled from the implant **160**, and the implant delivery system **100** can be navigated out of the heart, leaving the implant **160** attached to the valve leaflet L. For complete disengagement of the implant **160** from the implant catheter **130**, anchors can be removed or otherwise disengaged to release the tethers **140**, **150**, and the wire loops can be pulled out by the user. For example, one end of the tether can be released from the proximal end of the delivery system, and second end of the tether can be pulled proximally until the first end is pulled through the arm of the implant.

In some embodiments, the implant delivery system **100** may include a snare in the handle assembly and/or positioned along the catheters **110**, **120**, **130** (e.g., the implant catheter **130**). In some embodiments, a portion of the atrial tether(s) **140** and/or a portion of the ventricular tether(s) **150** may be coupled to the snare. For example, a first end of each of the atrial tether(s) **140** and ventricular tether(s) **150** may be coupled to the snare. Once the implant **160** has been attached to the leaflet and positioning of the implant **160** has been confirmed, the snare may be configured to cut, sever, clip, etc. the first end of each of the atrial tether(s) **140** and the ventricular tether(s) **150** so that the tethers **140**, **150** can be pulled out of the proximal end of the system **100**. In some embodiments, a cutting portion of the snare may be positioned near a distal end of the implant delivery system **100** such that a distance to pull the tethers **140**, **150** to remove them is reduced. The snare positioned near a distal end of the implant delivery system **100** may reduce likelihood that the tethers **140**, **150** get stuck in the lumen of the implant catheter **130**.

FIG. 3 is a schematic block diagram of an implant holder **335** of an implant delivery system, according to embodiments. In some embodiments, the implant holder **335** may include a proximal portion, a central portion, and a distal portion. The proximal portion may include a first cross-sectional area, the central portion may include a second cross-sectional area, and the distal portion may include a third cross-sectional area. The second cross-sectional area may be smaller than the first and second cross-sectional areas such that the central portion forms a cavity **337**. The

implant holder 335 may define the cavity 337 in the central portion configured to receive at least a portion of an implant 360 during navigation and delivery of the implant 360 onto the heart valve. In some embodiments, the distal portion of the implant holder 335 may form an atraumatic tip (e.g., blunt, rounded, etc.) to prevent damage to the tissue during delivery of the implant 360. In some embodiments, the implant holder 335 may have a straight distal tip to increase the moment length. In some embodiments, the implant holder 335 may define one or more atrial channels 336 and one or more ventricular channels 338. The atrial channel(s) 336 and the ventricular channel(s) 338 may extend along a portion of a length of the implant holder 335. In some embodiments, the implant holder 335 may include two atrial channels (e.g., a first atrial channel and a second atrial channel) and two ventricular channels 338 (e.g., a first ventricular channel and a second ventricular channel). In some embodiments, the atrial channel(s) 336 each define an opening near a proximal end of the implant holder 335 and the ventricular channel(s) 338 define an opening near a distal end of the implant holder 335. One or more atrial tethers 340 may extend through the atrial channel(s) 336, and one or more ventricular tethers 350 may extend through the ventricular channel(s) 338. In some embodiments, a first portion of an atrial tether 340 may extend through the first atrial channel and a second portion of the atrial tether 340 may extend through the second atrial channel such that a loop is formed between an opening of the first atrial channel and an opening of the second atrial channel. In some embodiments, a first portion of a ventricular tether 350 may extend through the first ventricular channel and a second portion of the ventricular tether 350 may extend through the second ventricular channel such that a loop is formed between an opening of the first ventricular channel and an opening of the second ventricular channel.

In some embodiments, a loop of the atrial tether 340 may be disposed through a first portion of the implant 360 (e.g., an opening defined by the atrial arm of the implant 360) to couple the atrial arm to the atrial tether 340. For example, a first end of the atrial tether 340 may be fixed to the proximal end, and the second end of the atrial tether may be disposed through the opening of the implant 360 and fed through the implant catheter to the proximal end of the implant catheter. The atrial tether 340 may be configured to stabilize a position of the implant 360 relative to the implant holder 335. The first atrial channel and the second atrial channel may have unequal lengths such that each atrial channel 336 defines an opening at different locations of the implant holder 335. The length of each atrial channel 336 may correspond to a respective predetermined region of the atrial arm of the implant 360 such that the atrial tether 340 couples to the atrial arm of the implant 360 at its respective predetermined region. The atrial tether 340 may couple to the atrial arm at the predetermined regions may stabilize the implant 360 relative to the implant holder 335 (e.g., prevent sliding or displacement of the implant 360). For example, the first atrial channel may define a first opening and the second atrial channel may define a second opening distal to the first opening such that the atrial tether 340 extends from the second opening, through the atrial arm of the implant 360, and into the first opening or vice versa. The first opening and the second opening may correspond to locations along the atrial arm of the implant 360.

In some embodiments, the ventricular tether 350 may be disposed through a second portion of the implant 360 (e.g., an opening defined by the ventricular arm of the implant) to couple the ventricular arm of the implant 360 to the ven-

tricular tether 350. The ventricular tether 350 may be coupled to the implant 360 in a similar manner as the atrial tether 340. In some embodiments, the ventricular channels 338 may be configured such that the ventricular tether(s) 350 extend partially outside of the implant holder 335. The implant holder 335 may include two sets of ventricular channel(s) 338. A first set of ventricular channel(s) 338 in the proximal portion of the implant holder 335 and a second set of ventricular channel(s) 338 in the distal portion of the implant holder. The first set of ventricular channels may define a first set of openings in the proximal portion of the implant holder 335 and the second set of ventricular channels may define a second set of openings at the distal portion of the implant holder 335. The second set of openings may include two openings on a backside of the implant holder 335 and two openings on a frontside of the implant holder 335. The ventricular tether(s) 350 may be configured to (i) extend through a ventricular channel from the first set of ventricular channel(s) 338, (ii) exit the back side of the implant holder 335 and extend outside of the implant holder 335, (iii) extend back into a backside of the implant holder 335 at the distal end, (iv) exit a front side of the implant holder 335 at the distal end, and (v) loop through the ventricular arm of the implant 360. The implant 360 may be transitioned from a closed configuration to an open configuration in response to tension applied to the ventricular tether 350 (e.g., by actuating the ends of the ventricular tether proximally), thereby pulling the ventricular arm away from the atrial arm of the implant 360. The implant 360 may be transitioned from the open configuration back to the closed configuration by releasing tension from (e.g., introducing slack to) the ventricular tether 350 (e.g., actuating the ends of the ventricular tether distally). In some embodiments, the implant 360 may be loosened from the cavity 337 by releasing tension from (e.g., introducing slack to) the atrial tether 340 (e.g., actuating the ends of the at least one atrial tether distally). In some embodiments, tension may also be released from the ventricular tether 350 after tension is released from the atrial tether 340 to loosen (e.g., further loosen) the implant 360 from the cavity 337. The implant catheter may be withdrawn proximally such that effectiveness of the implant 360 clamped onto the leaflet of the heart valve in reducing heart valve regurgitation can be visualized (e.g., via imaging). In some embodiments, tensioning the atrial tether 340 can tighten or pull the implant 360 against the implant holder 335, releasing tension on the atrial tether 340 allows the distal end of the delivery system to be moved away from the implant 360, and re-tensioning the atrial tether 340 can re-tighten or pull the implant 360 against the implant holder 335.

In some embodiments, a first atrial channel may define an opening (e.g., the first opening) on the implant holder 335 configured to align with a proximal edge of the first arm or segment of the implant 360 when the implant 360 is coupled to the implant holder. In some embodiments, a second atrial channel may define an opening (e.g., the second opening) on the implant holder 335 configured to align with a location of a distal edge of the first arm or segment (e.g., where the central member begins) when the implant 360 is coupled to the implant holder 335. In this way, the first arm or segment of the implant 360 may be secured to the implant holder 335 at two points along its length. In some embodiments, a first atrial tether may extend through the first atrial channel and a second atrial tether may extend through the second atrial channel. In some embodiments, the first atrial tether may couple to the first segment or arm of the implant 360 at a first location, and the second atrial tether may couple to the first

segment or arm of the implant **360** at a second location distal to the first location. In some embodiments, one atrial tether may extend through both the first and second atrial channels, and a first portion of the atrial tether may couple to the implant **360** at the first location and a second portion of the atrial tether may couple to the implant **360** at the second location.

In some embodiments, a total length of the implant holder **335** (e.g., a distance between a proximal end and a distal tip of the implant holder **335**) may be in a range of about 15 mm to about 45 mm, inclusive of all ranges and subranges therebetween. In some embodiments, the total length of the implant holder **335** may be about 21 mm. In some embodiments, a distal portion of the implant holder **335** from a distal end of the implant **360** in the closed configuration to a distal tip of the implant holder **335** may have a length corresponding to a moment arm for articulating the implant **360**. In some embodiments, the distal portion of the implant holder **335** may be straight. In some embodiments, the distal portion of the implant holder **335** may form a curve. In some embodiments, the distal portion of the implant holder **335** may form a dome shape. The curved or dome shape distal portion may act as a fulcrum when the ventricular tether(s) **350** is tensioned to open the arm. For example, at least a portion of the second set of ventricular channel(s) **338** in the distal end of the implant holder **335** may act as a fulcrum for the ventricular tether(s). In some embodiments, increased length of the distal portion may increase the moment arm to enable the implant **360** to transition between the open and closed configurations more easily (e.g., with less tensile force acting on the tethers). In some embodiments, the length of the distal portion of the implant holder **335** may be about 3 mm, 4 mm, 5 mm, 6 mm, 7 mm, 8 mm, 9 mm, or about 10 mm, inclusive of all ranges and subranges therebetween. In some embodiments, a length of the implant holder **335** may be at least 2 mm longer, at least 3 mm longer, at least 4 mm longer, at least 5 mm longer, at least 6 mm longer, at least 7 mm longer than a length of the implant **360** to ensure a desired moment arm to facilitate articulation and/or to open the ventricular arm of the implant **360** to an angle that is greater than 90 degrees without the distal portion of the implant holder **335** obstructing the ventricular arm from such motion. In some embodiments, the implant holder **335** may be at least 5 mm longer than a length of the implant **360**. The distal tip should be longer than the implant to increase the moment arm for easy articulation. However, the distal tip should not be so long that it is incapable of navigating through curved paths of the delivery system. Therefore, the distal tip should have a length shorter than a bend length of the guide catheter (as described with respect to FIGS. 1A-1B).

In some embodiments, a shape of the cavity **337** may correspond to dimensions of the implant **360**. For example, the cavity **337** may define a radius of curvature at least that of a radius of curvature of the implant **360** including the cover. In some embodiments, the implant **360** may sit in the cavity **337** to reduce catching of the implant **360** on nearby tissue during navigation of the implant holder **335** through the vasculature and the heart. In some embodiments, a profile of the implant holder **335** may be in a range of about 5 mm to about 6.5 mm, inclusive of all ranges and subranges therebetween. In some embodiments, the implant holder **335** may be substantially cylindrical. In some embodiments, diameter of the implant holder **335** may be in a range of about 4 mm to about 10 mm, inclusive of all ranges and subranges therebetween.

In some embodiments, the implant holder **335** may include a proximal engagement surface (e.g., a flat portion) on which the first segment or arm (e.g., the atrial arm) of the implant **360** may be secured. For example, when the atrial tether(s) **340** are tensioned, at least a portion of the atrial arm of the implant **360** may be configured to abut at least a portion of the proximal engagement surface of the implant holder **335** to help secure a position of the implant **360** relative to the implant holder **335**. In some embodiments, the proximal engagement surface may be in a plane parallel to a plane through a longitudinal axis of the implant holder **335**. In some embodiments, the proximal engagement surface may be slanted (e.g., not parallel to a plane through the longitudinal axis of the implant holder **335**). In some embodiments, a length of the proximal engagement surface may correspond to a length of the first segment or arm of the implant **360**. For example, the length of the engagement surface may be in a range of about 4 mm to about 10 mm, inclusive of all ranges and subranges therebetween. In some embodiments, the slanted proximal portion may reduce the overall profile of the implant holder **335**. In some embodiments, the implant holder **335** may be formed from a rigid biocompatible material (e.g., a material similar to that of the catheters **110**, **120**, **130**).

FIG. 4 is a schematic diagram of an implant **460** configured to be implanted on a leaflet of a heart valve to reduce heart valve regurgitation, according to embodiments. In some embodiments, the implant **460** may have a coaptation section comprising a 3-dimensional shape and an attachment section configured to couple the device to a heart valve (e.g., mitral, tricuspid, pulmonic, aortic) at a treatment site thereof. The treatment site may correspond to a portion of the heart valve that contributes to a cardiac condition, such as heart valve regurgitation during systole. For example, the treatment site may be located along the anterior leaflet of the mitral valve (e.g., A1, A2, A3), posterior leaflet of the mitral valve (e.g., P1, P2, P3), chordae tendineae of the mitral valve, anterior leaflet (i.e., cusp) of the tricuspid valve, posterior leaflet (i.e., cusp) of the tricuspid valve, septal leaflet (i.e., cusp) of the tricuspid valve, chordae tendineae of the tricuspid valve, left coronary leaflet (i.e., cusp) of the aortic valve, right coronary leaflet (i.e., cusp) of the aortic valve, non-coronary leaflet (i.e., cusp) of the aortic valve, left leaflet (i.e., cusp) of the pulmonary valve, right leaflet (i.e., cusp) of the pulmonary valve, or anterior leaflet (i.e., cusp) of the pulmonary valve.

The implant **460** may be coupled to heart valve tissue at the treatment site (e.g., a first native leaflet) in a manner that reduces or eliminates regurgitation. For example, the implant **460** may be coupled to the heart valve tissue such that a gap between the two native leaflets of the heart valve is reduced or eliminated. In one variation, the implant **460** may be coupled to the heart valve tissue such that the coaptation section of the implant **460** may be coaptated by a second native leaflet. For example, in some variations in which the coaptation section of the implant **460** provides a coaptation surface for the second native leaflet, the treatment site may be referred to as P3-A3, in that the device may be coupled to the P3 portion of the posterior leaflet of the mitral valve and may be coaptated by the A3 region of the anterior leaflet of the mitral valve. In a further example, the implant **460** may be coupled to the A3 portion of the posterior leaflet of the mitral valve and may be coaptated by the P3 region of the anterior leaflet of the mitral valve. In some variations, the implant **460** may be coupled to a leaflet having a gap or cleft, such that the coaptation section of the device may cover the gap or cleft of the leaflet while enabling coaptation

of an opposing leaflet onto the coaptation section. In some variations, multiple implants **460** may be coupled to the leaflet having a gap, such that they are adjacent to each other on the same leaflet, opposing each other on different leaflets, or a combination thereof.

The treatment site may correspond to portions of the heart valve associated with injury, disease, geometric distortion, and/or poor leaflet mobility and may be determined by a physician using clinical data, such as blood pressure, heart rate, audio signals, images generated by non-invasive techniques (e.g., MRI, CT, ultrasound, fluoroscopy), images generated by invasive techniques (e.g., intracardiac echocardiography, transesophageal echocardiography), or a combination thereof. For example, a physician may use a measurement of leaflet mobility to determine the treatment site, such as excursion angle. Excursion angle may correspond to the magnitude of leaflet movement between diastole and systole. In some variations, the excursion angle may be between about 5 degrees to about 75 degrees, about 5 degrees to about 60 degrees, about 5 degrees to about 45 degrees, or about 5 degrees to about 25 degrees. For example, in some variations, the excursion angle may be about 5 degrees, about 15 degrees, about 25 degrees, or about 45 degrees, or less than about 25 degrees. In this way, the implant **460** may be advanced to the heart valve and coupled to the treatment site to reduce or eliminate heart valve regurgitation (e.g. reduce or close a gap between leaflets), despite the reduced excursion angles of the leaflets.

The implant **460** may include one or more sections configured to couple to one or more portions of a heart valve, such that the implant may reduce or eliminate heart valve regurgitation. In some embodiments, the attachment section may be configured to releasably engage heart valve tissue and may advantageously enable the implant to be securely coupled to the heart valve tissue and, if so desired by a physician, decoupled from the heart valve tissue, either for removal or for repositioning at another treatment site of the heart valve. In some variations, the length of the leaflet coupled (e.g., grasped) by the implant **460** may be adjusted, such that the attachment section may grasp more or less of the leaflet. In this way, the physician may optimize the effectiveness of the implant by repositioning (e.g., one or more times) at a treatment site associated with the greatest reduction in heart valve regurgitation (e.g., reducing a gap between the posterior and anterior leaflets of the mitral valve).

The attachment section of the implant **460** may include two segments (e.g., planar segments) or arms configured to couple to portions of a heart valve thereby coupling the implant **460** to one or more of a native leaflet, and chordae tendineae. For example, the attachment section may comprise a first segment or arm **462** and at least a portion of a second segment or arm **464**. The first and second segments or arms **462**, **464** may be biased towards one another such that the first and second segments or arms **462**, **464** hold (e.g., compress) heart valve tissue (e.g., posterior or anterior leaflet of the mitral valve) received therebetween. In some variations, the first and second segments or arms **462**, **464** may be biased towards one another such that the first and second segments or arms **462**, **464** are at least partially, and in some variations, fully coplanar in a delivery configuration or a closed configuration (e.g., a resting position). The first and second segments or arms **462**, **464** may be configured to securely attach to the heart valve, such that the implant **460** may not come loose after implantation. The attachment section may define a distal portion of the device (relative to the physician during implantation).

The attachment section may further comprise one or more friction elements **470** configured to apply a friction force to heart valve tissue. The one or more friction elements **470** may extend from at least one of the first and second segments or arms **462**, **464** to assist in preventing unintentional movement and/or removal of the implant **460** during and/or after implantation. The friction force resulting from the friction elements **470** may be mitigated by a physician during the implantation process (e.g., during removal and/or repositioning if so desired) by, for example, separating the first and second segments or arms **462**, **464** and/or pulling the first and second segments or arms **462**, **464** upwards first and then away from heart valve tissue to disengage the friction elements **470** from the heart valve tissue. In this way, the implant **460** may remain securely coupled to heart valve tissue during normal cardiac events (e.g., heart beating under nominal conditions, which may include strenuous exercise associated with high heart rates) while providing flexibility to the physician to position and reposition the implant **460** one or more times (e.g., two, three, four or more times) during implantation.

The coaptation section of the implant **460** may be configured (alone or in combination with the attachment section) to enhance and/or alter a physical characteristic of the native leaflet (e.g., a first native leaflet). In some variations, the coaptation section may be configured to provide a coaptation surface for another native leaflet of the heart valve (e.g., a second native leaflet). For example, the coaptation section may be configured to provide a contact surface for non-destructively receiving the second native leaflet during systole, which may reduce or eliminate heart valve regurgitation (e.g. reduce or close a gap between leaflets). At least a portion of the coaptation section may deflect upon coaptation by the native leaflet. In some embodiments, the coaptation section may be configured to receive a portion of the first native leaflet therein (i.e., the leaflet to which the attachment section is coupled). Put differently, the coaptation section may define a volume, and a portion of the first native leaflet may be positioned within the volume when the device is coupled to the first native leaflet.

The coaptation section may comprise a central member **466** from which the first and second segments or arms **462**, **464** may each extend. The first segment or arm **462** may extend from a first portion of the central member **466** and the second segment or arm **464** may extend from a second, different portion of the central member **466**. In some embodiments, the first and second segments or arms **462**, **464** may extend from opposite ends of the central member **466**. The central member **466** may be shaped (e.g., curved) to bias or assist in biasing the first and second segments or arms **462**, **464** towards one another. The biasing force provided by the central member **466** may be sufficient to maintain the first and second segments or arms **462**, **464** in place when coupled to heart valve tissue. The biasing force may be overcome by applying a force to at least one of the segments or arms **462**, **464**, such as by the elongate members or tethers of an implant delivery system (e.g., the atrial tether **140** and/or ventricular tether **150**), such that the first and second segments or arms **462**, **464** may be separated from one another. In this way, the implant **460** may be opened to receive heart valve tissue and/or may be removed from heart valve tissue (e.g., for repositioning). Additionally, the change in a curvature and/or shape of the central member **466** may enable the first and second segments or arms **462**, **464** to be separated from one another.

The coaptation section may further comprise one or more support members **480** configured to provide a surface for

coaptation by a native leaflet, such as by supporting a cover **468**. The one or more support members **480** supporting the cover **468** may be referred to as a first portion of the coaptation section. The coaptation section may further comprise a second portion extending from a proximal end of the first portion. The second portion may be devoid of support members **480**. In some embodiments, the support members **480** may extend from the central member **466**. In some embodiments, the support members **480** may extend from the first segment or arm **462** and may be configured to elastically deform with each interaction with the opposing leaflet, which may contribute to the durability and effectiveness of the implant **460**. For example, mechanical stress concentrations may be reduced by extending the support members **480** from the first segment or arm **462**, which may increase the durability of the implant **460**. The support members **480** may extend from the first segment or arm **462** to define a larger volume of the coaptation section relative to a variation in which the support members **480** extend from the central member **466**. For example, a radius of curvature of the support members **480** extending from the first segment or arm **462** may be greater than a radius of curvature of support members **480** extending from the central member **466**. The larger radius of curvature may allow the implant **460** absorb a force (e.g., via coaptation) from the native leaflet without being damaged, loosening from the first native leaflet, and/or damaging any part of the native leaflet.

The support members **480** may comprise a free end such that the support members **480** may move (e.g., deflect, extend, contract) relative to other portions of the implant **460**. For example, the support members **480** may deflect when a leaflet applies a force to the coaptation section (e.g., during systole) and elastically return to their previous configuration when the leaflet is not applying the force to the coaptation section (e.g., during diastole). The movement facilitated by the free ends may advantageously minimize mechanical stresses at the connection point between the support members **480** and the first segment or arm **462**. Minimizing such mechanical stresses may allow the support members to deflect, bend, or otherwise move without breaking upon coaptation by another portion of the heart valve, which may facilitate complete closure of a heart valve and/or reduce the size of a fluid flow path (e.g., by reducing a gap between two native leaflets). Implants partially attached to chordae tendineae may behave similarly.

The coaptation section may define a volume configured to receive a portion of a native leaflet upon coaptation and/or surround a portion of a native leaflet (e.g., the native leaflet coupled by the attachment section). In some variations, the volume of the coaptation section may be determined by a total width and a total height of the outer portions of two or more support members **480** and a coaptation section length associated with a portion of the second segment or arm **464** and/or a portion of the central member **466**. Accordingly, the total width, total height, and coaptation section length may be used to determine the coaptation section volume. The coaptation section may form a bulbous shape. In some embodiments, a maximum length of the coaptation section (e.g., a maximum length along a longitudinal axis of the coaptation section) may be shorter than a maximum width of the coaptation section (e.g., a maximum length along a transverse axis of the coaptation section). A height of the coaptation section may be defined as a distance from a first surface of the cover **468** contacting the leaflet to a second surface of the cover **468** facing away from the leaflet. The maximum length and maximum width of the coaptation section and/or a ratio of the maximum length to maximum

width may each be modified by modifying the shape of the support members **480**. The coaptation section may taper (e.g., a cross-sectional area may decrease) along the longitudinal axis and/or along the transverse axis. For example, the length, the width, and/or the height of the coaptation section may decrease toward the outer edges of the implant **460** such that the coaptation section forms an oval shape, a football shape, an almond shape, or the like. Along the longitudinal axis, the coaptation section may be flush with the leaflet surface, reach a peak (e.g., increase in height from the leaflet surface), and then taper back toward the leaflet surface. In some embodiments, a volume captured between the native leaflet surface and the surface defined by the cover **468** disposed over the support member **480** may be symmetric (or substantially symmetric) along the longitudinal axis.

The implant **460** may be at least partially covered by the cover **468**. In some embodiments, any portion of the implant **460** may include one or more materials configured to elastically deflect. For example, the implant **460**, including the first and/or second segments or arms **462**, **464**, may include any suitable material such as, for example, a metal (e.g., nitinol, titanium, aluminum, gold, silver, alloys thereof) or a polymer (e.g., polypropylene, polyvinyl chloride, polyethylene, polyurethane). In some embodiments, the implant **460** may include Nitinol with austenite finish temperature below room temperature. This means the implant **460** may remain in its set form at room temperature and/or body temperature, and the implant **460** may be restored to its set shape after deformation when the deformation force is released. The implant **460** may be configured to withstand strain up to about 3%, about 4%, about 5%, about 6%, about 7%, about 8%, about 9%, about 10%, about 11%, about 12%, about 13%, about 14%, about 15%, about 16%, about 17%, about 18%, about 19%, about 20%. In some embodiments, the implant **460** may be configured to withstand strain up to about 10%.

In some embodiments, the cover **468** may include any suitable material such as, for example, a fabric (e.g., textile) or a polymer (e.g., polyester, polytetrafluoroethylene (PTFE)). In some embodiments, the cover **468** may include, for example, a polyester fabric, a knitted polyester, a woven polyester, PTFE, or a combination thereof. In some embodiments, the cover **468** may include two different materials. The fabric may be further reinforced with a metallic wire along its boundary or part of the boundary, so as to provide it a shape that is distinct from the shape that the implant provides. The attachment section and the coaptation section of the implant may be covered in distinct materials and using distinct methods. The attachment section may be covered to enhance visibility of clinical imaging modalities during implantation and be optimized to have a thickness that enables partial protrusion of the teeth/friction elements from the fabric. The coaptation section may be covered to provide a smooth surface for coaptation of the opposing native leaflet, to avoid wear and tear with repeated coaptation with opposing leaflet, and/or to be hemocompatible and allow tissue growth and endothelialization. In some embodiments, the material of the first and/or second arms **462**, **464** and/or the material of the cover **468** may be selected to optimize (e.g., minimize) the weight of the implant **460**, which may prevent the implant **460** from pulling on portions of the heart valve due to the force of gravity, which in turn could reduce the effectiveness of the implant in reducing heart valve regurgitation. Accordingly, in some embodiments, the implant **460** may have a weight of about 50 mg to about 500 mg, about 100 mg to about 150 mg, or about 130 mg to about

140 mg, including about 100 mg, about 110 mg, about 120 mg, about 130 mg, about 140 mg, or about 150 mg. In some embodiments, the body of the implant **460** (e.g., the arms **462**, **464** and central member **466**) may be formed from a single sheet of metal or metal alloy. Therefore, no tiny parts or complex assemblies may be required to assemble the implant **460**. In some embodiments, the body of the implant may be formed from a first sheet of metal or metal alloy, and the friction elements **470** may be formed separately from a second sheet of metal or metal alloy thinner than the first sheet and welded onto the body. In some embodiments, the friction elements **470** may be welded onto the body with weld seams along specific regions of the implant **460**. Therefore, a thickness of the body may be larger than a thickness of the friction elements **470**. For example, the first sheet (and therefore the body of the implant **460**) may have a thickness of about 250 μ m and the second sheet (and therefore the friction elements **470**) may have a thickness of about 100 μ m. In some embodiments, a total length of the implant (e.g., along a longitudinal axis of the implant **460**) may be about 8 mm, about 9 mm, about 10 mm, about 11 mm, about 12 mm, about 13 mm, about 14 mm, about 15 mm, about 16 mm, about 17 mm, about 18 mm, inclusive of all ranges and subranges therebetween. In some embodiments, the implant **460** may be manufactured in a 10 mm, a 12 mm, and a 16 mm size.

In some embodiments, the cover **468** (e.g., the material of the cover **468**) may promote tissue encapsulation of the implant **460**. For example, the cover **468** may enable permanent integration of the implant **460** into the native leaflet within 30-60 days of implantation. Patients thus can be weaned off anticoagulants and survive a life free of bleeding. In some embodiments, a gap between yarns of the fabric of the cover **468** (e.g., a knitted polyester) may impact the incorporation rate. In some embodiments, a diameter of the yarn may impact the incorporation rate. So, in some implementations, the cover **468** may define gaps within fabric configured to increase and/or optimize the incorporation rate, and/or a diameter of the fabric or yarn may be selected or configured to increase and/or optimize the incorporation rate.

a. Segments or Arms

As described herein, the implant **460** includes first and second segments or arms **462**, **464** configured to facilitate coupling to heart valve tissue by applying a force thereto (e.g., compressing tissue therebetween). Accordingly, the shape of the first and/or second segments or arms **462**, **464** may be configured to provide a distributed compressive force to heart valve tissue while minimizing stress concentrations during deployment, implantation, and/or removal methods described herein with respect to FIGS. 29A-29F, 30A-30D, and 31-33. The first and second segments or arms **462**, **464** may be configured to couple different sides of the same native leaflet. For example, the first segment or arm **462** may be configured to couple a first side of a first native leaflet (e.g., an atrial side) and the second segment or arm **464** may be configured to couple a second side of the first native leaflet (e.g., a ventricular side).

In some embodiments, the first and second segments or arms **462**, **464** may each include one or more sections configured to non-destructively couple heart valve tissue. For example, the first and second segments or arms **462**, **464** may each include a planar section and a curved section. More specifically, the first segment or arm **462** may include a first planar section and a first curved section and the second segment or arm **464** may include a second planar section and a second curved section. The planar sections of the first

and/or second segments or arms **462**, **464** may be configured to contact heart valve tissue without causing damage to the heart valve tissue. In some variations, the first and second planar sections may be substantially coplanar when the implant **460** is in the delivery configuration or closed configuration. For example, the first segment or arm **462** may have a greater maximum width than the second segment or arm **464** and may define an opening sufficient to receive the second segment or arm **464** such that the first and second segments or arms **462**, **464** may be coplanar. In some embodiments, the second segment or arm **464** may be disposed at least partially through the opening defined by the first segment or arm **462** such that the leaflet is compressed at multiple areas to provide a better grip on the leaflet. For example, the first segment or arm **462** may push the leaflet toward the second segment or arm **464** such that when the leaflet is disposed the first and segment arms or segments **462**, **464**, the leaflet forms a wave-like shape, which can prevent displacement or sliding of the implant **460**.

The first and second curved sections may extend from the first and second planar sections of the first and second segments or arms **462**, **464**, respectively. The first and second curved sections may each be configured to provide a biasing force to the first and second planar sections, respectively. More specifically, the first curved section may provide a biasing force to the first planar section when the first planar section may be in contact with heart valve tissue, such that the first planar section may not unintentionally separate from the heart valve tissue. Similarly, the second curved section may provide a biasing force to the second planar section when the second planar section is in contact with heart valve tissue, such that the second planar section may not unintentionally separate from the heart valve tissue. While described above as each comprising a planar and a curved section, it should be appreciated that the first and/or second segment or arm may each be entirely planar or entirely curved.

The size (e.g., length, width) of the first and second segments or arms **462**, **464** may be determined based on a variety of factors, including but not limited to a patient's height, age, gender, valve to be treated, valve anatomy, and severity of the cardiac condition. Accordingly, the implant **460** may be manufactured in a variety of sizes, such that a physician may pre-select the implant **460** based on patient specific factors, such as those mentioned above. Pre-selecting the implant **460** may increase the effectiveness of the implant **460** by more closely matching the size of the implant **460** to the particular patient's characteristics. In some variations, the first segment or arm **462** may comprise a first length of about 1 mm to about 10 mm, about 2 mm to about 9 mm, about 3 mm to about 8 mm, or about 4 mm to about 7 mm, including about 1 mm, about 2 mm, about 3 mm, about 4 mm, about 5 mm, about 6 mm, about 7 mm, about 8 mm, about 9 mm, or about 10 mm. The second segment or arm may comprise a second length of about 1 mm to about 10 mm, about 2 mm to about 9 mm, about 3 mm to about 8 mm, or about 4 mm to about 7 mm, including about 1 mm, about 2 mm, about 3 mm, about 4 mm, about 5 mm, about 6 mm, about 7 mm, about 8 mm, about 9 mm, or about 10 mm.

In some embodiments, the implant **460** may occupy about 4%, about 5%, about 6%, about 7%, about 8%, about 9%, about 10%, about 11%, about 12%, about 13%, about 14%, about 15% of the total leaflet surface area of the heart valve. In some embodiments, the implant **460** may occupy about 5% to about 10% of the total leaflet surface area. In some embodiments, when attached to the leaflet, the implant **460**

occupies 8% of the total leaflet surface area and is not expected to restrict flow across the valve in diastole.

In some variations, the first segment or arm **462** may comprise a length different than a length of the second segment or arm **464**. The different lengths may correspond to the surface area of the native leaflet surfaces to which the respective segments or arms may couple. Accordingly, the length of the first segment or arm **462** may be about 1% to about 75%, about 5% to about 60%, about 5% to about 50%, or about 5% to about 25% greater than the length of the second segment or arm **464**. For example, in some variations, the length of the first segment or arm **462** may be about 1%, about 5%, about 25%, or about 50% greater than the length of the second segment or arm **464**. The relative lengths of the first segment or arm **462** to the second segment or arm **464** may define a ratio. In some variations, the ratio of lengths may be about 1.01 to about 1.75, about 1.05 to about 1.6, about 1.05 to about 1.5, or about 1.05 to about 1.25. For example, in some variations, the ratio of lengths may be about 1.01, about 1.05, about 1.25, or about 1.5.

In addition to the lengths of the first and second segments or arms **462**, **464**, the widths of each segment or arm **462**, **464** may be configured to distribute the compressive force. In some variations, the width of the first and/or second segment or arm **462**, **464** may vary along its length. In some variations, varying the width of the first and/or second segment or arm **462**, **464** may reduce mechanical stresses and/or avoid stress concentration “hot-spots” during or after implanting the device. For example, each of the first arm and/or second arm **462**, **464** may transition from a first width to a second width at any point along the longitudinal dimension first arm and/or the second arm **462**, **464**, such as about $\frac{1}{4}$, about $\frac{1}{3}$, about $\frac{1}{2}$, about $\frac{2}{3}$, or about $\frac{3}{4}$ from the distal edge of the first arm and/or the second segment or arm **462**, **464**. In some embodiments, a maximum width of the first segment or arm **462** may be about 1 mm to about 10 mm, about 2 mm to about 8 mm, about 3 mm to about 7 mm, and about 4 mm to about 6 mm, including about 1 mm, about 2 mm, about 3 mm, about 4 mm, about 5 mm, about 6 mm, about 7 mm, and about 8 mm. In some variations, a maximum width of the second segment or arm **464** may be about 1 mm to about 8 mm, about 2 mm to about 8 mm, or about 3 mm to about 5 mm, including about 1 mm, about 2 mm, about 3 mm, about 4 mm, or about 5 mm. In some embodiments, the first segment or arm **462** may have a greater maximum width than the second segment or arm **464** such that the first segment or arm **462** may define an opening sufficient to receive the second segment or arm **464**. Accordingly, in some variations, a maximum width of the first segment or arm **462** may be about 1.01 to about 2, about 1.05 to about 1.75, or about 1.05 to about 1.5 times greater than a maximum width of the second segment or arm **464**. For example, in some variations, the maximum width of the first segment or arm **462** may be about 1.05, about 1.25, about 1.5, or about 1.75 times greater than the maximum width of the second segment or arm **464**.

The first and second segments or arms may each include any shape suitable for attaching to heart valve tissue and retaining the position of the implant relative to the anatomy. For example, the shape of the first and/or section segment or arm **462**, **464** may be square, rectangular, circular, trapezoidal, oval, triangular, combinations thereof, or the like. In some variations, it may be advantageous to avoid use of sharp corner that could inadvertently damage tissue and accordingly, in some embodiments, one or more ends or edges of the first and/or second segments or arms **462**, **464**

may be rounded. For example, in some variations, the first and/or second segments or arms **462**, **464** may have a square or rectangular shape with rounded corners. The first and second segments or arms **462**, **464** may also have any suitable cross-sectional shape, including, for example be a square, a circle, a trapezoid, a triangle, or combinations thereof.

b. Friction Elements

In some variations, the first and/or second segments or arms **462**, **464** may be configured to apply a friction force. The friction force may be configured to maintain the implant **460** at the intended treatment site without slipping or otherwise moving. For example, each of the first and second segments or arms **462**, **464** may comprise one or more (e.g. a plurality of) friction elements **470** configured to apply a friction force to non-destructively engage heart valve tissue. The friction elements **470** may include, but are not limited to, tines, teeth, prongs, hooks, bumps (e.g., rounded bumps), surface roughening, or the like. The friction elements **470** may be positioned on or otherwise carried by any portion of a tissue-facing surface of the first and/or second segments or arms **462**, **464**. More specifically, the friction elements **470** may be positioned along one or more edges of one or more of the first and second segments or arms **462**, **464**, along a central axis of a tissue-facing surface **462**, **464**, and/or or between one or more edges and the central axis.

In some embodiments, the friction elements **470** may be arranged along a longitudinal dimension of one or more of the first and/or second segments or arms **462**, **464**. The longitudinal arrangement may be collinear or, in some variations, may include one or more offsets. The friction elements **470** may be uniformly spaced along the tissue-facing surface (e.g., along an edge of the tissue-facing surface of the first and/or second segment or arm **462**, **464**) or may be non-uniformly spaced. In some embodiments, the friction elements **470** may be formed integrally with the first and/or second segment or arm **462**, **464**. Alternatively and/or additionally, the friction elements **470** may be formed separately from the first and/or second segment or arm **462**, **464** and may be attached thereto (e.g., via a plate or other structural element on which the friction elements **470** are formed). The friction elements **470** may have a cross-sectional shape, such as a triangle, a circle, or a rectangle, etc. In some embodiments, the friction elements **470** may all have the same size and shape. In some embodiments, one or more friction elements **470** may have a first size and/or shape and one or more other friction elements **470** may have a different size and/or shape. Similarly, the first and/or second segments or arms **462**, **464** may comprise the same or different types of friction elements **470**. For example, one or more friction elements **470** on the first and/or second segment or arm **462**, **464** may be of a first type (e.g., tines) and one or more friction elements on the first and/or second segment or arm **462**, **464** may be of a second type (e.g., hooks). In some variations, all of the friction elements **470** on the first segment or arm **462** may be of a first size, shape, and/or type, and all of the friction elements **470** on the second segment or arm **464** may be of a second, different, size, shape and/or type. In some embodiments, an angle at which the friction elements **470** extend from the first and/or segment or arms **462**, **464** may vary based on a location of the implant **460**. For example, the friction elements **470** have equivalent shapes and sizes but may be bent to different angles based on their respective location on the implant **460**, described in further detail in FIGS. 16A-16C.

In some embodiments, the friction elements **470** may include one or more surface roughening features. For

example, each of the first and second segments or arms **462**, **464** may comprise a plurality of protrusions configured to increase a friction coefficient between the tissue-facing surface of the respective segment or arm **462**, **464** and heart valve tissue. In some embodiments, each protrusion may be rounded. In some embodiments, each protrusion may be pointed (e.g., tines). The plurality of protrusions may be arranged in one or more lines, which may form a grid. In some variations, the plurality of protrusions may be arranged randomly or may be arranged in one or more lines, which may extend along the longitudinal and/or lateral dimension of the respective segment or arm **462**, **464**. The plurality of protrusions may be combined with any other configuration of friction elements **470** described herein.

The friction elements **470** may extend from a tissue-facing surface of the first and/or second segment or arm **462**, **464** and may form an angle therewith. For example, in some embodiments, the angle between one or more friction element **470** and the tissue-facing surface of the first and/or second segment or arm **462**, **464** at a base of or otherwise adjacent to the friction element **470** may be about 5 degrees to about 90 degrees, about 10 degrees to about 75 degrees, about 15 degrees to about 60 degrees, or about 40 degrees to about 50 degrees, inclusive of all ranges and subranges therebetween. For example, in some variations, the angle may be about 5 degrees, about 15 degrees, about 30 degrees, about 45 degrees, or about 60 degrees, inclusive of all ranges and subranges therebetween. In some embodiments, all of the friction elements **470** may extend from the tissue-facing surface of the respective arm **462**, **464** on which they are positioned at the same angle. In other embodiments, some friction elements **470** on the first and/or second segment or arm **462**, **464** may extend at different angles than other friction elements **470** on that same segment or arm or on the opposing segment or arm.

The implant **460** may include any suitable number of friction elements **470** to maintain a position of the implant **460** relative to patient anatomy while the implant **460** is implanted. For example, each of the first and second segment or arm **462**, **464** may include 1 to 20, 2 to 15, or 3 to 10 friction elements **470**, inclusive of all ranges and subranges therebetween. In some embodiments, the first and/or second segment or arm **462**, **464** may include between about 6 and 12 friction elements.

In some embodiments, a first set (e.g., 2 or more such as 3, 4, 5, 6, 7, 8, 9, 10, or more) of friction elements **470** may be positioned along one or more edges of a tissue-facing surface of a first portion of the first segment or arm **462** and a second set of friction elements **470** may be positioned along one or more edges of the tissue-facing surface of a second portion of the first segment or arm **462**. The first set of friction elements **470** may define a first angle relative to the tissue-facing surface of the first portion of the first segment or arm **462** and the second set of friction elements **470** may define a second angle relative to the tissue-facing surface of the second portion of the first segment or arm **462**. Similarly, a third set of friction elements **470** may extend along one or more edges of a tissue-facing surface of the second segment or arm **464**, and may define a third angle relative to the tissue-facing surface of the second segment or arm **464**. When the implant **460** is in a closed configuration (e.g., when the first and second segments or arms **462**, **464** may be coplanar), the first and/or second sets of friction elements **470** may extend beyond the second segment or arm **464** and/or the third set of friction elements **470** may extend beyond the first segment or arm **462**. That is, the first and/or second sets of friction elements **470** on the first segment or

arm **462** may not be coplanar with the second segment or arm **464**, and the third set of friction elements **470** on the second segment or arm **464** may not be coplanar with the first segment or arm **462**.

The first, second, and/or third angles defined by each respective set of friction elements **470** may determine the friction force. For example, increasing one or more of the first, second, and third angles may increase the friction force, and decreasing one or more of the first, second, and third angles may decrease the friction force. In this way, the friction force may be adjusted via manipulation of the first, second, and/or third angles, where the manipulation may occur during the manufacturing phase and/or by a physician prior to performing the surgical procedure described herein. The first angle may be about 0 degrees to about 90 degrees, about 10 degrees to about 60 degrees, or about 20 degrees to about 45 degrees, including about 30 degrees, inclusive of all ranges and subranges therebetween. The second angle may be about 0 degrees to about 90 degrees, about 10 degrees to about 60 degrees, or about 20 degrees to about 45 degrees, including about 15 degrees, inclusive of all ranges and subranges therebetween. The third angle may be about 0 degrees to about 90 degrees, about 10 degrees to about 60 degrees, or about 20 degrees to about 45 degrees, including about 40 degrees, inclusive of all ranges and subranges therebetween.

The one or more sets of friction elements **470** may comprise a height sufficient to couple heart valve tissue. In some variations, the height of the one or more sets of friction elements **470** may be about 10 microns to about 100 microns, 25 microns to about 75 microns, or about 30 microns to about 60 microns, inclusive of all ranges and subranges therebetween. For example, in some variations, the height of the one or more sets of friction elements **470** may be about 10 microns, about 25 microns, about 35 microns, about 45 microns, or about 75 microns. The height of one set of friction elements **470** may be the same or different as any other set of friction elements **470**, which may facilitate optimizing the friction force applied by the implant **460**.

In some embodiments, the friction elements **470** may be formed by creating slots in the first and/or second segment or arm **462**, **464** or by creating slots in a structure from which the friction elements are formed (e.g., a plate). In some embodiments, the first and/or second segments or arms **462**, **464**, or the plate from which the friction elements **470** are formed, may include a plurality of slots. Each slot may be adjacent to a respective friction element **470** and may define a length corresponding to a height of the friction element **470**. The slots may facilitate forming (e.g., bending) the friction element **470** to the desired angle that corresponds to a desired friction force. That is, each slot may separate an edge of a friction element **470** from a tissue-facing surface of the segment or arm **462**, **464**. In some embodiments, one or more, including all, slots may terminate in a circular portion. It may be advantageous to utilize a slot with a circular portion as the circular portion may reduce mechanical stresses associated with forming (e.g., bending) the friction elements. Additionally, in some variations, the circular portion may prevent cracks from propagating from the slot.

c. Central Member

As previously described, the implant **460** may include the central member **466** positioned between the first segment or arm **462** and the second segment or arm **464**. The central member **466** may be configured to provide a biasing force to one or more of the first and second segments or arms **462**,

464, such that the first and second segments or arms 462, 464 may be biased towards one another. The biasing force may be sufficient to maintain coupling of the first and second segments or arms to the heart valve tissue. In some embodiments, the central member 466 may be curved. For example, the central member 466 may form an arc between the first segment or arm 462 and the second segment or arm 464. Accordingly, the biasing force may be at least partially based on the configuration (e.g., length, a radius of curvature, and/or thickness) of the central member. Therefore, the configuration of the central member 466 may facilitate movement of the second segment or arm 464 towards or away from the first segment or arm 462 and may define the extent to which the first and second segments or arms 462, 464 may separate.

A length of the central member 466 may correspond to the compressive force applied by the first and second segments or arms 462, 464 according to a moment of force (e.g., force multiplied by distance). In some variations, the length of the central member 466 may be about 1 mm to about 10 mm, about 2 mm to about 9 mm, about 3 mm to about 8 mm, or about 4 mm to about 7 mm, including about 1 mm, about 2 mm, about 3 mm, about 4 mm, about 5 mm, about 6 mm, about 7 mm, about 8 mm, about 9 mm, or about 10 mm, inclusive of all ranges and subranges therebetween.

A thickness of the central member 466 may be configured to allow the movement of the first and second segments or arms 462, 464 relative to one another without inelastically deforming or fracturing. In some embodiments, the thickness of the central member 466 may vary along its length. For example, a thickness of a section adjacent to the second segment or arm 464 may be greater than other sections of the central member 466. The thicker section of the central member 466 may correspond to a thicker section of the second segment or arm 464. The thicker sections of the central member 466 and/or second segment or arm 464 may facilitate deflection of the second segment or arm 464 relative to the first segment or arm 462 while reducing excessive force or fracturing of the central member 466. In some embodiments, the thickness of the central member 466 may be between about 50 microns to about 500 microns, about 100 microns to about 300 microns, or about 200 microns to about 300 microns. In some embodiments, the thickness may be about 50 microns, about 100 microns, about 200 microns, about 250 microns, about 300 microns, or about 400 microns.

A width of the central member 466 may be configured such that the mechanical stresses associated with adjusting the central member 466 (e.g., during transitions between configurations) may be minimized. In some embodiments, the width of the central member 466 may be less than the width of one or more of the first and second segments or arms 462, 464. The width of the central member 466 may be about 1 mm to about 10 mm, about 2 mm to about 7 mm, or about 2 mm to about 5 mm. For example, in some variations, the width of the central member 466 may be about 1 mm, about 2 mm, about 3 mm, about 4 mm, or about 5 mm. In some embodiments, a radius of curvature of the central member 466 may be in a range of about 2 mm to about 10 mm or beyond. In some embodiments, the radius of curvature of the central member 466 may be about 4 mm.

The biasing force generated by the central member 466 may correspond to a compressive force applied by the first and/or second segments or arms 462, 464 to heart valve tissue. The compressive force may not be so great as to damage the heart valve tissue. In some embodiments, the compressive force may be characterized by the amount of

force required to separate the first and second segments or arms 462, 464. In some embodiments, the biasing force may be about 100 mN to about 1000 mN, about 150 mN to about 600 mN, or about 200 mN to about 450 mN, including about 150 mN, about 200 mN, about 250 mN, about 300 mN, about 350 mN, about 400 mN, about 450 mN, about 500 mN, about 550 mN, or about 600 mN. The force applied by the first and second segments or arms 462, 464 may be configured to counter a force generated by the transmitral systolic pressure gradient on the mitral valve.

d. Support Members

The implant 460 may include one or more support members 480 configured to absorb impingement (e.g., via coaptation) from one or more portions of a heart valve and, in some cases, provide support for the cover 468 that may cover the support members 480 or otherwise be coupled thereto. Accordingly, the support members 480 may each have a radius of curvature to facilitate supporting the cover 468 and/or forming an appropriate coaptation surface for a native leaflet. The radii of curvature may allow the support members 480 and/or cover 468 to deflect such as, for example, upon impingement (e.g., coaptation) by another portion of the heart valve. In some embodiments, the support members 480 may deflect (e.g., bend and/or extend) in one or more directions such that the support members 480 are in an elongated configuration (e.g., in a delivery configuration). Each support member 480 may have a plurality of sections, where each section may have a different shape and/or radius of curvature. In some variations, each section may be linear. In some embodiments, one or more sections of each support member 480 may be linear and one or more sections may be curved. In some embodiments, each section of the support members 480 may be curved. For example, one or more support members 480 may have one or more of a convex section and a concave section. In some embodiments, one or more of the support members 480 may have a section with a radius of curvature greater than the radius of curvature of the central member 466. The radius of curvature of one support member 480 may be different than any other radius of curvature of another support member 480. Additionally, or alternatively, one or more of the support members 480 may include a free end, as described previously. The free ends of the support members 480 may allow the support members 480 to extend or contract even further. In some embodiments, the free ends of the support members 480 may each include an atraumatic distal tip such that the support members 480 may avoid damaging or otherwise traumatizing heart valve tissue. In some embodiments, the free end of each of the support members 480 may be coplanar with a proximal surface of the central member 466.

In some embodiments, at least one of the support members 480, such as a first support member, may be configured to contact a first native leaflet (i.e., the leaflet to which the attachment section is coupled) while at least one other support member 480, such as a second support member, may be configured to avoid contact with the first native leaflet. The second support member may be configured to contact a second native leaflet (e.g., via coaptation). At least one cover (e.g., cover 468) may be coupled to the coaptation section. The cover 468 may be a flexible (e.g., stretchable, malleable) material configured to move with the support members 480, such that the cover 468 may deflect when a leaflet applies a force to the coaptation section. In this way, a heart valve may be altered such that complete closure may be achieved, which may facilitate durably correcting heart valve regurgitation.

The support members **480** may include one or more dimensions configured to provide the flexibility described herein. For example, each support member **480** may comprise a length, which may, in some embodiments, may determine the volume of the coaptation section. Each support member **480** may have the same length, or a different length, as any of the other support members **480**. In some embodiments, a length of the support members **480** may be about 1 mm to about 20 mm, about 3 mm to about 15 mm, or about 5 mm to about 12 mm, including about 3 mm, about 5 mm, about 7 mm, about 9 mm, about 11 mm, or about 13 mm. Each support member may have a width, which may be determined to minimize a stress concentration during deflection. In some embodiments, the width may be about 10 microns to about 500 microns, about 50 microns to about 250 microns, or about 100 microns to about 200 microns, including about 50 microns, about 100 microns, or about 150 microns.

The number of support members **480** may be determined by the treatment site and/or the size of the heart valve. In some embodiments, the number of support members **480** may alternatively or additionally be determined by a size (e.g., an area) of the cover **468**. For example, the number of support members **480** may be increased to accommodate a larger cover **468**, which may facilitate treating relatively severe heart valve regurgitation associated with a relatively large gap between valve leaflets. In some instances, it may be advantageous to have an even number of support members **480** as this may assist in evenly distributing the mass of the implant **460** when coupled to heart valve tissue. In some embodiments, the implant **460** may include between 2 and 10, 2 and 9, 2 and 8, 2 and 7, 2 and 6, 2 and 5, 2 and 4, and 2 and 3 support members **480**. In some embodiments, the implant **460** may include 4 support members **480**.

In embodiment with an even number of support members **480**, the support members **480** may be symmetric (e.g., mirrored) about the central member **466**. That is, there may be the same number of support members **480** on either side of the central member **466** and symmetric pairs of support members **480** may have the same dimensions (e.g., length, width) and/or shape (e.g., radius or radii of curvature). For example, there may be a first and second support member on a first side of the central member **466** and third and fourth support members on a second side of the central member **466**, where the third support member may be symmetric to the second support member about the central member **466** and the fourth support member may be symmetric to the first support member about the central member. In some embodiments, the support members **480** need not be symmetric, which may facilitate customizing the coaptation section to optimize treatment for a given patient.

In some embodiments, the support members **480** may attach to the central member **466** and/or the first or second arm **462**, **464** and are free at their distal end. Therefore, when the implant **460** is compressed into the implant catheter of the implant delivery system (e.g., in the compressed configuration), the support members **480** compress inwards, but also elongate longitudinally to accommodate that compression. When the implant **460** is advanced out of the implant catheter, the support members **480** may transition to their baseline configuration or state (e.g., an expanded configuration), longitudinal elongation may be decreased. In other words, when the support members **480** are in the compressed configuration (e.g., when the implant **460** is in a delivery configuration), a transverse protrusion length of the support members **480** is reduced, and when the support members **480** are in the expanded configuration (e.g., when the

implant **460** is in the open configuration and/or the implanted configuration), the transverse protrusion length is increased (e.g., restored to a base condition).

e. Openings

The implant **460** may define one or more openings configured to receive one or more segments or arms **462**, **464**, as described herein. For example, the first segment or arm **462** may define an intermediate opening configured to receive the second segment or arm **464**. The intermediate opening may be configured to receive at least a portion of the second segment or arm **464** when the portion of the second segment or arm **464** is at least partially coplanar with at least a portion of the first segment or arm **462**. Accordingly, in some embodiments, the intermediate opening may comprise a size and/or shape corresponding to a portion of the second segment or arm **464**. The intermediate opening may comprise a length and a width. The length of the intermediate opening may be about 1 mm to about 10 mm, about 2 mm to about 8 mm, or about 3 mm to about 6 mm. For example, in some variations, the length of the intermediate opening may be about 1 mm, about 3 mm, about 4 mm, about 5 mm, or about 6 mm. The width of the intermediate opening may be about 1 mm to about 10 mm, about 2 mm to about 8 mm, about 2 mm to about 6 mm, or about 2.5 mm to about 5.5 mm. For example, in some embodiments, the width of the intermediate opening may be about 1 mm, about 2 mm, about 3 mm, about 4 mm, about 5 mm, or about 6 mm. In some variations, the length and/or width of the intermediate opening may maintain a gap between the first segment or arm **462** and second segment or arm **464**, such that a cover **468** may be routed through the gap while allowing the segments or arms **462**, **464** to freely move relative to each other. The intermediate opening may be positioned at a midline of the lateral (e.g., width) dimension of the first segment or arm. In some embodiments, the intermediate opening may be positioned at any point along the lateral dimension of the first segment or arm.

The implant **460** may further define one or more openings (e.g., positioning openings) **475** configured to facilitate adjusting the position of each of the first and second segments or arms **462**, **464** (e.g., opening and/or closing the implant **460**). The one or more openings **475** may be configured to receive one or more elongate members (e.g., the atrial tether **140** and/or the ventricular tether **150**) to assist in adjusting the position of one or more of the first and second segments or arms **462**, **464**. As described previously, an implant delivery system (e.g., implant delivery system **100**) may be used to assist in opening and/or closing the implant **460**. The one or more elongate members (e.g., wires, sutures, etc.) of the implant delivery systems may be disposed through one or more of the positioning openings **475** of each of the first and second segments or arms **462**, **464**. For example, a first suture or tether (e.g., the atrial tether **140**) may be routed through one or more positioning openings **475** (e.g., suture holes) of the first segment or arm **462** and back through an implant catheter of the implant delivery system, such that both ends of the first suture or tether are controllable by the implant delivery system. Additionally, the second suture or tether may be disposed through one or more positioning openings **475** of the second segment or arm **464** and back into the implant catheter of the implant delivery system, such that the ends of both the first and second sutures or tethers are controllable by the implant delivery system. In some embodiments, the first suture or tether may be disposed through one or more positioning openings **475** of the first and/or second segment or arm **462**, **464** and subsequently tied (e.g., to itself or a portion of the

35

catheter) to form a first loop, and the second suture or tether may be disposed through the first loop and back into the implant catheter of the implant delivery system to form a second loop, wherein the first loop may be smaller than the second loop.

Accordingly, the one or more positioning openings may comprise a size and/or shape corresponding to one or more elongate members (e.g., the atrial tether **140** and/or the ventricular tether **150**). For example, the shape of the positioning opening **475** may be a circle, oval, rectangle, or combination thereof. The shape may be determined to reduce friction between the segment or arm **462, 464** and the elongate member. The size of the positioning opening **475** may correspond to a diameter of the elongate member. In some variations, the size may correspond to multiple elongate members routed through the same positioning opening **475**. The width and length dimensions of each positioning opening **475** may be the same or may be different, and may depend on the number of elongate members routed there-through.

The plurality of positioning openings **475** of the first and/or second segments or arms **462, 464** may be arranged to facilitate application of a pre-determined force profile to the first and/or second segments or arms **462, 464** via the elongate member(s). For example, a plurality of positioning openings **475** may be arranged collinearly across a width of the first and/or second segment or arm **462, 464**, such that an elongate member may apply a force evenly across the first and/or second segment or arm **462, 464**. In some embodiments, the plurality of positioning openings **475** may not be arranged collinearly such that one or more positioning openings **475** are offset from at least one other positioning opening **475**. The offset may reduce a friction force between an elongate member and the positioning opening(s) **475**, such as when the elongate member may be routed through more than one positioning opening **475**. In some embodiments, the plurality of positioning openings **475** may be arranged along a longitudinal dimension of one or more of the first and/or second segments or arms **462, 464**. The longitudinal arrangement may be collinear or, in some embodiments, may include one or more offsets. The arrangement may be determined to evenly distribute the force applied by one or more elongate members routed through the plurality of positioning openings **475** and/or may reduce a friction force between the elongate member and the positioning openings **475**. In some embodiments, the longitudinal arrangement may be centered with respect to the width of one or more of the first and/or second segment or arms **462, 464**, which may further facilitate even distribution of the force applied by the one or more elongate members.

In some embodiments, each of the first and second segments or arms **462, 464** may include a number of positioning openings **475** determined by the size and/or shape of the respective segment or arm **462, 464**. For example, each of the first and second segments or arms **462, 464** may include 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 positioning openings **475** configured to receive an elongate member. In some embodiments, the first and/or second segments or arms **462, 464** may include openings used during manufacturing to hold a portion of the implant **460** with a shape setting fixture. These openings may not be used in the function of the implant **460**. In some embodiments, the first and second segments or arms **462, 464** may include the same number and/or configuration of positioning openings **475**. In some embodiments, the first and second segments or arms **462, 464** may include different numbers and/or configurations of positioning openings **475**. For example, the number of

36

positioning openings **475** of the first segment or arm **462** may not be equivalent to the number of positioning openings **475** of the second segment or arm **464**, which may be determined by the size of each segment or arm **462, 464** and/or the amount of force required to separate the segments or arms **462, 464** from one another. That is, the number of positioning openings **475** may be increased if a greater force is required to separate the segments or arms **462, 464** and/or more elongate members are required to apply the necessary separation force to avoid breaking the one or more elongate members. In some embodiments, the first segment or arm **462** may include four positioning openings **475** and the second segment or arm **464** may include two positioning openings **475**.

Utilizing the one or more elongate members, the physician may manipulate the configuration of the segments or arms **462, 464** such that the segments or arms **462, 464** may be separated prior to receiving and/or coupling to heart valve tissue, released to couple to heart valve tissue, and optionally re-separated to reposition the implant **460**. In some embodiments, the first and second segments or arms **462, 464** of the implant **460** may be coplanar in a closed configuration, a delivery configuration, and/or an implanted configuration. The physician may apply tension to the one or more elongate members via the implant delivery system, which may apply a force to the first and second segments or arms **462, 464** sufficient to overcome the biasing force provided by the central member **466** and thus separate the first and second segments or arms **462, 464** to the open configuration. The separation of the first and second segments or arms **462, 464** in the open configuration may define a separation angle relative to the closed configuration (e.g., coplanar). The separation angle may be about 1 degree to about 180 degrees, about 10 degrees to about 150 degrees, or about 15 degrees to about 120 degrees. The separation angle may correspond, at a minimum, to an uncompressed thickness of the targeted heart valve tissue (e.g., at the treatment site). The separation angle may determine a separation distance between portions of the first and second segments or arms **462, 464**. For example, the separation distance may be defined by the maximum distance between the distal end of the first segment or arm **462** relative to the distal end of the second segment or arm **464**. In some embodiments, the separation distance may be about 1 mm to about 20 mm, about 5 mm to about 18 mm, or about 10 mm to about 15 mm. For example, in some variations, the separation distance may be about 1 mm, about 5 mm, about 15 mm, about 17 mm, or about 20 mm.

The physician may also reduce tension in the one or more elongate members via the implant delivery system **100**, which may allow the biasing force to bring the first and second segments or arms **462, 464** towards one another to an implanted configuration. The implanted configuration may correspond to the first and/or second segments or arms **462, 464** being coupled to heart valve tissue. Accordingly, the implanted configuration may correspond to a compressed thickness of the targeted heart valve tissue. Depending on the thickness of the compressed heart valve tissue the first and second segments or arms **462, 464** may not be coplanar relative to each other in the implanted configuration. In some embodiments, the first and second segments or arms **462, 464** may be partially or fully coplanar relative to each other in the implanted configuration.

The implant **460** described herein may further include one or more visualization openings configured to receive visualization markers, which may assist with visualization of the implant **460** (e.g., the position and/or configuration) before,

during, and/or after implantation. The implant **460** may define one or more visualization openings in any location, such as in one or more of the first segment or arm **462**, the second segment or arm **464**, and the central member **466**. For example, the first and/or second segment or arm **462**, **464** may each include one or more visualization openings configured to receive a visualization marker (e.g., a radiopaque marker). In some embodiments, the visualization openings of the first and/or second segment or arm **462**, **464** may be located in a distal portion thereof, such that the separation distance between the first arm **462** and the second segment or arm **464** may be determined. When coupled to the first and/or second segments or arms **462**, **464**, the visualization markers may be indirectly visible to the physician while the implant **460** is within the patient. In some embodiments, the visualization markers may be directly visible, e.g., via fluoroscopy, echocardiography, and/or other imaging modality. In this way, the position of the first and second segments or arms **462**, **464** relative to each other may be determined via the radiopaque markers attached thereto, which may be used to determine the configuration of the implant **460** (e.g., delivery, closed, open, or implanted). Additionally, if the segments or arms **462**, **464** are separated (e.g., in the open configuration), the visualization markers may be used to determine the separation distance between the segments or arms **462**, **464**.

The visualization openings described herein may include dimensions corresponding to the shape and size of the visualization markers. A visualization opening may have the same shape and size, or a or different shape and size, as any of the other visualization openings. In some embodiments, the openings and associated visualization markers may have any shape, such as a circle, an oval, a triangle, a rectangle, a square, or a trapezoid. For example, a substantially circular opening may correspond to a substantially circular radiopaque marker. The visualization openings may comprise a diameter or length. The diameter or length may correspond to a diameter or width of the visualization marker. The diameter or length of the visualization opening may be about 250 microns to about 2 microns, about 500 microns to about 1.5 mm, or about 500 microns to about 1 mm. For example, in some variations, the diameter or length may be about 250 microns, about 500 microns, about 750 microns, or about 1 mm. In some embodiments, the visualization opening may extend through the thickness of segments or arms **462**, **464** described herein (i.e., are through holes).

In some embodiments, a first visualization opening or a first set of visualization openings defined by the first arm or segment **462** may have a different shape and/or orientation than a second visualization opening or second set of visualization openings defined by the second arm or segment **464**. For example, a first visualization opening on the first arm or segment **462** may be elongated vertically (e.g., an oval, slit, or rectangle, that is longer along the longitudinal axis of the first arm or segment **462**) and a second visualization opening on the second arm or segment **464** may be elongated horizontally (e.g., an oval, slit, or rectangle that is longer along the transverse axis of the second arm or segment **464**). When the first arm or segment **462** and the second arm or segment **464** are coplanar, the first visualization opening and second visualization opening form a “+” shape. When the first arm or segment **462** and the second arm or segment **464** are not coplanar, the first and second visualization markers form a series of horizontal and vertical shapes.

In some embodiments, each of the first and second segments or arms **462**, **464** may comprise 0 to 20 visualization openings, such as 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, or 20 visualization openings configured to receive a visualization marker. The number of visualization openings of the first segment or arm **462** may not be equivalent to the number of visualization openings of the second segment or arm **464**. For example, the number of visualization openings of each segment or arm **462**, **464** may be determined by the size of each segment or arm **462**, **464**. For example, the number of visualization openings of the first segment or arm **462** may be greater than may be the number of visualization openings of the second segment or arm **464** because of the larger surface area of the first segment or arm **462** relative to the second segment or arm **464**. In some embodiments, the number of visualization openings of a segment or arm **462**, **464** may be increased if a surface area thereof, is increased to enable a physician to identify the position of the larger segment or arm **462**, **464**. In some embodiments, one or more visualization markers may be coupled to the first and/or second segments or arms **462**, **464** without being received in a corresponding visualization opening, which may facilitate coupling additional visualization markers to the device after the initial assembly of the implant **460**.

f. Plates

The implants **460** may further include one or more plates configured to couple to the implant **460** (e.g., the first segment or arm **462** and/or the second segment or arm **464**). The one or more plates may be coupled to the implant **460** such that the plates remain fixedly attached to the implant **460** during the initial implant process and any subsequent repositioning. For example, a first plate may be coupled to the first segment or arm **462** and a second plate may be coupled to the second segment or arm **464**. The first plate may comprise a shape corresponding to at least a portion of the first segment or arm **462** and/or the second plate may comprise a shape corresponding to at least a portion of the second segment or arm **464**. Accordingly, in some embodiments, each of the first and second plates may comprise one or more of a planar section and a curved section. The curved sections of the first and second plates may be shaped to match the curved sections of the first and second segments or arms **462**, **464** described previously. The one or more plates may be further configured to releasably couple to heart valve tissue, which may facilitate repositioning of the implant **460**. The one or more plates may include any of the friction elements **470** described above in detail and in any of the configurations described herein.

The one or more plates may further include one or more openings (e.g., positional openings **475**) configured to receive one or more elongate members, which may facilitate opening and/or closing the implant **460**. For example, the first plate may comprise one or more positioning openings **475** with a similar size and position as the one or more positioning openings **475** of the first segment or arm **462**. Additionally, or alternatively, the second plate may comprise one or more positioning openings **475** with a similar size and position as the one or more positioning openings **475** of the second segment or arm **464**. In this way, one or more elongate members may be routed through the positioning openings of at least one of the first plate, first segment or arm **462**, second plate, and/or second segment or arm **464**. In some embodiments, the first and/or second plates may each include one or more visualization openings configured to receive a visualization marker. In some embodiments, the first and/or second plates may not include one or more visualization openings, such that the first and/or second

plates may cover one side of the visualization openings of the respective first and second segments or arms **462**, **464** and thus provide a mounting surface (e.g., backstop). In this way, a visualization marker received within one of the visualization opening of the first and/or second segments or arms **462**, **464** may be securely mounted to the first or second plates.

Alternatively or additionally, the implant may include one or more attachments, couplers, loops, and/or the like, coupled to the first segment or arm **462** and/or the second segment or arm **464**. For example, the implant may include one or more suture loops through one or more (e.g., two) positional openings in the atrial arm, and one or more suture loops through one or more (e.g., two) positional openings in the ventricular arm. In some embodiments, the atrial tether and the ventricular tether may be configured to couple to the suture loops in addition to or instead of the positional openings **475**. For example, the atrial tether and/or ventricular tether may be routed through the one or more suture loops and coupled to the proximal end of the implant delivery system. In some embodiments, the atrial tether may be routed through one or more suture loops on the atrial arm. In some embodiments, the ventricular tether may be routed through one or more suture loops on the ventricular arm. In some embodiments, the suture loop may only be on one of the segments or arms of the implant **460** to provide less friction or resistance when releasing and/or withdrawing the tethers. In some embodiments, coupling the tethers to suture loops on the implant **460** may provide a larger moment arm than coupling the tethers directly to the implant. In some embodiments, the tethers routed through the suture loops may be easier to decouple from the implant **460**. In some embodiments, coupling the tethers to the suture loops may also cause less deformation of the tethers than when the tethers are directly coupled to the implant **460**. In some embodiments, the suture loops may remain on the implant **460** permanently. In some embodiments the suture be removed from the implant after implantation.

The first and second plates described herein may be coupled to the first and second segments or arms **462**, **464**, respectively. For example, in some embodiments, the first and second plates may be fixedly attached (e.g., permanently) to the first and second segments or arms **462**, **464**, respectively, such that the implant **460** may maintain its structural integrity before, during, and after the implantation process. The first and second plates may be coupled to the first and second segments or arms **462**, **464** via any suitable fastening technique, such as, for example, via welding, adhesives, and/or mechanical fasteners (e.g., screws, bolts). The means of coupling may be at least partially dependent upon the material of the plates and/or segments or arms **462**, **464**.

In some embodiments, the plates and segments or arms **462**, **464** may each be manufactured using metal (e.g., stainless steel, nitinol) such that the first plate may be welded to the first segment or arm **462**, and the second plate may similarly be welded to the second segment or arm **464**. In some embodiments, the plates and arms may each be manufactured using a plastic such that the plates may be coupled to the respective segments or arms **462**, **464** using a polyurethane adhesive, which may be non-toxic to humans and thus suitable for a medical device. The first and second plates may each have a thickness between about 50 microns to about 500 microns, about 50 microns to about 300 microns, about 50 microns to 200 microns, or about 75 microns to about 125 microns. For example, in some variations, the first and second plates may each have a thickness

of about 50 microns, about 75 microns, about 100 microns, about 200 microns, or about 250 microns.

g. Visualization Markers

The implant **460** may be implanted within a patient without direct visualization of the implant **460**, which may reduce the number of components associated with the surgical procedure, facilitate a minimally invasive procedure, and/or reduce a size of the implant **460** and/or the delivery system. For example, the implant **460** may be implanted within the patient using visualization via X-ray, CT, or similar medical imaging means. In some embodiments, the implant **460** may include one or more visualization markers (e.g., one or more radiopaque markers) configured to allow a user to indirectly visualize the implant **460** and thereby determine a position of one or more elements of the implant **460**. As described above, the implant **460** may define one or more visualization openings configured to receive one or more visualization markers. In some embodiments, the first and second plates may be configured to provide a mounting surface for one or more visualization markers. In this way, the precise position of implant **460** may be determined relative to a heart valve. Furthermore, the position of the first and second segments or arms **462**, **464** relative to each other may also be determined via the visualization markers attached thereto. Therefore, it may be possible to determine whether the first and second segments or arms **462**, **464** are coplanar (e.g., in a closed configuration) or if they are separated (e.g., in an open configuration) and, if so, the distance by which the segments or arms **462**, **464** are separated.

The visualization markers described herein may comprise dimensions corresponding to the shape and size of the visualization openings receiving them. In some embodiments, a width of the visualization marker may be about 250 microns to about 2 mm, about 500 microns to about 1.5 mm, or about 500 microns to about 1 mm. For example, in some variations, the width may be about 250 microns, about 500 microns, about 750 microns, or about 1 mm. A height of the visualization marker may be about 50 microns to about 300 microns, about 100 microns to about 300 microns, or about 150 microns to about 300 microns. In embodiments in which a visualization opening extends through the segment or arm **462**, **464**, the height of the visualization marker may be about 200 microns to about 300 microns, including about 250 microns.

The visualization markers may be coupled to the implant **460** such that the visualization markers remain fixedly attached to the implant **460** for the entire duration that the implant remains within the patient. For example, the visualization markers may be coupled to the arm(s) **462**, **464** and/or plate(s) using welding, adhesives, and/or mechanical fasteners (e.g., screws, bolts). The means of coupling may be at least partially dependent upon the material of the visualization markers, plates, and/or arms. In some embodiments, the visualization markers may be manufactured from gold, platinum, iridium, tantalum, or other radiopaque material. In some embodiments, the visualization markers (e.g., radiopaque marker) may include platinum and may be welded to an implant **460** made from nitinol.

While described above as received within a visualization opening, in some embodiments, one or more visualization markers may be coupled to a surface of the first and/or second segment or arm **462**, **464**, or the central member **466**, without the use of an opening. Moreover, in some instances, a visualization marker may be in the form of a visual marker on the surface of the implant **460** (e.g., first and/or second

segment or arm **462**, **464**, central member **466**) that is intended to be viewed with an endoscope without the use of radiology.

h. Covers

The implant **460** may include one or more covers **468** configured to couple to and in some instances surround at least a portion of the implant **460**. In some embodiments, the cover **468** may extend along at least a portion of the attachment section and coaptation section of the implant **460**. For example, the cover **468** may cover at least a portion of each of the first and/or second segments or arms **462**, **464**. In some embodiments, the cover **468** may cover each of the first and second segments or arms **462**, **464** except for the plurality of positioning openings **475** and/or friction elements **470** thereof. The cover **468** may cover the central member **466** and/or may cover the support members **480**. When used to cover the support members **480**, the cover **468** may be configured to alter one or more physical characteristics of heart valve tissue, such as a thickness, length, width, curvature, and/or stiffness thereof. The cover **468** may also assist in providing a malleable coaptation surface for receiving heart valve tissue and may, in combination with the support members **480**, advantageously provide a coaptation surface for heart valve tissue. The cover **468** may also provide a surface upon which native tissue may grow, which may facilitate improved outcomes and/or further reduce heart valve regurgitation as native tissue naturally connects portions of the heart valve after the implanting process. Additionally or alternatively, the cover **468** may operate as a flexible extension of the implant **460** such that a greater surface area of heart valve tissue (e.g., a second native leaflet) may contact the implant **460** and/or a larger gap between the native leaflets may be filled. In these embodiments, the cover **468** may further reduce the precision required by physician during the implanting process because the cover may increase the overall size of the implant **460**.

The cover **468** may be made from a material with sufficient pliability to match the movements of the attachment and/or coaptation sections without tearing, breaking, or otherwise being damaged. As described above, the material of the cover **468** may facilitate native heart valve tissue growth. Accordingly, in some embodiments, the cover **468** may comprise one or more of a fabric (e.g., textile) and a polymer (e.g., polyester, polytetrafluoroethylene). Additionally, or alternatively, the cover **468** may be woven, braided, and/or layered using one or more materials. The material and texture of the cover **468** may be determined based on biocompatibility with heart valve tissue and/or durability. The cover **468** may be coupled to the implant **460** using any suitable technique, such as, for example, sutures, ties, knots, adhesives, and/or elastic bands.

In some embodiments, the cover **468** may include three portions, such as a first portion configured to couple to the support members **480**, a second portion configured to extend therefrom, and a third portion configured to couple to the attachment section (e.g., at least a portion of the first and second segment or arms **462**, **464**). The portions may be integrally formed or may be separate from one another. For example, the first and second portions may be integrally formed (e.g., a first cover) and the third portion may be formed separately (e.g., a second cover), such that the first and second covers may be coupled to the implant **460** during assembly.

In another example, the cover **468** may include a first portion configured to couple to the plurality of support members **480** of the coaptation section, with one or more (e.g., all) of the support members **480** supporting the cover

468. The cover **468** may further include a second portion that may extend from a proximal end of the first portion, such that the second portion may not be in direct contact with or otherwise covering the implant body (e.g., the second portion may be devoid of support members **480**). Accordingly, the second portion may increase one or more dimensions of the implant **460** (e.g., of the coaptation section) and may provide additional volume to the implant body by defining a 3-dimensional shape. In some embodiments, the first portion may be coupled to the second portion along a distal edge of the second portion and/or along a width of the second portion. The second portion may be rolled and/or folded (e.g., accordion fold), and may be curved along a lateral dimension (e.g., width) of the implant **460**. For example, in some embodiments, the second portion may be or may include a cylindrical cuff that may be formed from rolled material (e.g., cover material such as fabric) and secured (e.g., using a mechanical fastener such as a suture, adhesive, and/or the like) such that it retains a rolled configuration when in use. In these embodiments, the second portion may provide a flexible and/or unrollable extension that may be configured to increase the size of the implant **460**. The third portion may be configured to couple to the attachment section, such that the third portion covers at least a part of the first and second segments or arms **462**, **464**. The third portion may cover the friction elements **470** of the first and/or second segments or arms **462**, **464**. In some embodiments, some or all of the friction elements **470** of the first and/or second segments or arms **462**, **464** may extend through (e.g., pierce) the cover **468** such that the friction elements **470** may directly engage native leaflet tissue.

FIG. 5A is an illustration of a short catheter implant delivery system **500** including a handle assembly **505**, an implant catheter **530**, and an implant holder **535**, according to embodiments. FIG. 5B shows a close up of a distal end of the implant delivery system **500**, according to embodiments. As shown, the implant catheter **530** includes four hypotubes extending therethrough. Each hypotube may define a lumen **533** configured to receive an elongate member (e.g., the tethers not shown) that extends therethrough. Each tether extends along the length of the implant catheter **530** inside each lumen **533** of a respective hypotube with at least one end or portion of the tether anchored (e.g., removably anchored) to the handle assembly **505**. The implant holder **535** is coupled to a distal end of the implant catheter **530** and includes four channels, each channel configured to receive a respective portion of a tether. For example, each tether may extend from a distal end of a respective hypotube and through a respective channel in the implant holder **535** such that each tether can be coupled to a portion of an implant disposed in the implant holder **535**. Although not shown, in some embodiments, a first tether (e.g., the atrial tether) may be configured to couple to the first arm (e.g., atrial arm) of the implant and a second tether (e.g., the ventricular tether) may be configured to couple to the second arm (e.g., ventricular arm) of the implant.

FIG. 5C shows the handle assembly of the short catheter implant delivery system **500**. As shown, the handle assembly **505** includes an AT actuator **542** and a VT actuator **552**. The AT actuator **542** is configured to be coupled to the atrial tether (not shown) and the VT actuator **552** is configured to be coupled to the ventricular tether (not shown). The AT actuator **542** and the VT actuator **552** may be configured to move linearly along the length of the handle assembly **505** in response to being manipulated by a user to apply tension and/or to introduce slack to the tethers in order to control a configuration of the implant **560**. For example, to dispose

the implant **560** around a portion of a heart valve, the AT actuator **542** may be moved proximally to tension the atrial tether such that the first arm of the implant **560** is stabilized in the implant holder **535**. Then, the VT actuator **552** may be moved proximally to tension the ventricular tether, thereby transitioning the implant **560** to the open configuration in which the second arm (not shown) of the implant **560** is moved away from the first arm (not shown) of the implant **560**. Once the implant **560** is positioned to clamp a portion of the heart valve tissue, the VT actuator **552** can be moved distally to transition the implant **560** to the implanted configuration in which the first arm and the second arm exert a clamping force on the heart valve tissue. In order to confirm a position of the implant **560** relative to the heart valve, the AT actuator **542** and the VT actuator **552** can be moved distally to introduce slack to the atrial and ventricular tethers such that the distal end of the implant delivery system can be moved away from the implant **560** without unclamping the implant **560** from the heart valve tissue. This partial disengagement allows for the implant **560** position to be inspected after initial deployment. Moving the AT actuator **542** and the VT actuator **552** proximally can enable repositioning of the implant **560** if desired. For complete disengagement of the implant **560** from the implant holder **535**, the atrial and ventricular tethers can be removed (e.g., cut, unlooped, decoupled, etc.) and pulled out of the proximal end of the implant delivery system (e.g., by a user). The implant delivery system **500**, the implant holder **535**, and the implant **560** may be structurally and/or functionally similar to the implant delivery system **100**, the implant holder **135**, **335**, and the implant **160**, **360**, **460**; and therefore, certain aspects of the delivery system **500**, the implant holder **535**, and the implant **560** are not described in further detail herein with respect to FIGS. **5A-5C**.

FIGS. **6A-6J** show images of proximal control mechanisms for each catheter of an implant delivery system. The proximal control mechanisms may be separate or may be included in one handle assembly. As shown in FIGS. **6A-6J**, the implant delivery system may include a guide catheter **610** coupled to a guide catheter (GC) actuator **612**, a delivery catheter **620** coupled to a delivery catheter (DC) actuator **622**, and an implant catheter **630** coupled to an implant catheter (IC) actuator **632**. A distal end of the implant catheter **630** is coupled to an implant holder **635**, and the implant holder **635** is configured to receive an implant **660**. The delivery catheter **630** and the guide catheter **610** may be steerable via the DC actuator **622** and the GC actuator **612**, respectively. In some embodiments, the catheters **610**, **620**, **630** may have varying thickness along their length. For example, the catheters **610**, **620**, **630** may be more rigid at a proximal end and less rigid at a distal end such that a portion of the catheter **610**, **620**, **630** that is less rigid may be bent while the more rigid portion of the catheter **610**, **620**, **630** remains straight and stabilizes the delivery system. In some embodiments, the guide catheter **610** and the delivery catheter **620** may each have a proximal portion having a first hardness, a middle portion having a second hardness lower than the first hardness, and a distal portion having a third hardness smaller than the second hardness. The delivery catheter **620** and the guide catheter **630** may each include a weld ring at a distal end thereof and two articulating members (e.g., pull wires) including a first end coupled to the weld ring and a second end coupled to a proximal control mechanism. In some embodiments, the articulating members may be coupled to the weld ring 180 degrees apart. In some embodiments, the DC actuator **622** and/or the GC actuator **612** may each include a rack and

pinion assembly. The gears on each rack and pinion assembly may be calibrated to enable precise steering of each catheter independently. The user may turn a knob **613** on the GC actuator **612** to steer the guide catheter **610**. The user may turn a knob **623** on the DC actuator **622** to steer the delivery catheter **620**. Each of the catheters **610**, **620**, **630** may include a flushing port connected to an inner lumen thereof to help reduce frictional contact between the catheter shafts and to prevent air from moving into the patient's circulation.

In some embodiments, the guide catheter **610** may be a 24Fr catheter, the delivery catheter **620** may be a 16Fr catheter, and the implant catheter **620** may be a 12Fr catheter. In some embodiments, a maximum angle the delivery catheter **620** may be steered is larger than a maximum angle the guide catheter **610** may be steered. In some embodiments, a length of the steerable portion of the delivery catheter **620** may be smaller than a length of the steerable portion of the guide catheter **610**. In some embodiments, the guide catheter **610** may be steered up to an angle between about 0 degrees to about 150 degrees from the longitudinal axis of the catheter **610**, inclusive of all ranges and subranges therebetween. In some embodiments a length of the steerable portion of the guide catheter **610** may be in a range of about 30 mm to about 80 mm, inclusive of all ranges and subranges therebetween. In some embodiments, the delivery catheter **620** may be steered up to an angle between about 0 degrees and about 150 degrees, inclusive of all ranges and subranges therebetween. In some embodiments, a length of the steerable portion of the delivery catheter **620** may be in a range of about 20 mm to about 70 mm, inclusive of all ranges and subranges therebetween.

The implant catheter **630** can be moved linearly (e.g., in a proximal and distal direction) within the delivery catheter **620** via a linear slider **633** on the IC actuator **632**. The implant catheter **630** may have a stiffness or hardness of 35D. The implant catheter **630** may be configured for pushability and torquability such that the implant catheter **640** may extend linear out of the delivery catheter **620** even when the delivery catheter **620** is in a curved configuration. Therefore, the trajectory of the implant **660** may be independent of the curvature of the implant catheter **630**, which may allow precise positioning and leaflet capture. The implant catheter **630** may include a plurality of hypotubes (e.g., 4 hypotubes) extending therethrough. Each hypotube may include an elongate member (e.g., the atrial and/or ventricular tethers) extending therethrough. As shown, the IC actuator **632** may further include one or more actuators each coupled to an elongate member to control tension of the elongate members. The one or more actuators may be any suitable actuator such as, for example, a knob, rotating screw, button, slider, etc. In some embodiments, the IC actuator **632** may include a first rotational screw configured to control tension on the atrial tether and a second rotational screw configured to control tension on the ventricular tether. Rotation of the rotating screws in a first direction may apply tension to the tethers and rotation of the rotating screws in a second direction may release tension from the tethers. The implant catheters **610**, **620**, **630**, the implant holder **635**, and the implant **660** may be structurally and/or functionally similar to the implant catheters **110**, **120**, **130**, the implant holder **135**, **335**, and the implant **160**, **360**, **460**; and therefore, certain aspects of the implant catheters **610**, **620**, **630**, the implant holder **635**, and the implant **660** are not described in further detail herein with respect to FIGS. **6A-6J**.

FIGS. 7A-7C shows steerability of a three-catheter system of an implant delivery system, according to embodiments. As shown, the implant delivery system includes a guide catheter 710, a delivery catheter 720, and an implant catheter 730. The implant catheter 730 is coupled to an implant holder 735 holding an implant 760. The guide catheter 710 is steerable, the delivery catheter 720 is steerable, and the implant catheter 730 may be non-steerable. However, the implant catheter 730 may be translated linearly relative to the delivery catheter 720 and guide catheter 710 and rotated about its own longitudinal axis. In some embodiments, the implant catheter 730 may be torqued (e.g., rotated about its longitudinal axis). In some embodiments, the guide catheter 710 is steerable in one plane and configured to achieve a curvature corresponding to a path of access to the left atrium by crossing the septum. Although not shown, the delivery catheter 720 may have biplane steerability (e.g., steerability in two planes). The three-catheter system, enables the user to achieve a compound curve at a distal end of the delivery system that may be in more than one plane(s). In some embodiments, the three-catheter system may achieve an angle at the distal end up to about 180 degrees, about 175 degrees, about 170 degrees, about 165 degrees, about 160 degrees, about 155 degrees, about 150 degrees, inclusive of all ranges and subranges therebetween. The implant catheters 710, 720, 730, the implant holder 735, and the implant 760 may be structurally and/or functionally similar to the implant catheters 110, 120, 130, 530, 630 the implant holder 135, 335, 535, 635 and the implant 160, 360, 460, 560, 660; and therefore, certain aspects of the implant catheters 710, 720, 730, the implant holder 735, and the implant 760 are not described in further detail herein with respect to FIGS. 7A-7C.

FIG. 8 is a diagram of an implant holder 835 of an implant delivery system, according to embodiments. The implant holder 835 includes a proximal end 835A configured to couple to a distal end of an implant catheter (not shown) of the delivery system. The implant holder 835 has a curved distal end 835B. The curved distal end 835B may be atraumatic to prevent damage to tissue during delivery of the implant. The implant holder 835 may further define a cavity configured to receive a first portion the implant (not shown in FIG. 8) and a proximal engagement surface (e.g., a flat portion 837) configured to receive a second portion of the implant. The implant holder 835 includes a pair of atrial channels 836A, 836B and a pair of ventricular channels 838 (only one ventricular channel 838 is shown in FIG. 8 because the other channel sits directly behind the shown channel in side view). Each channel may be configured to receive an elongate member (e.g., a tether) to couple the implant to the implant holder 835 and transition the implant between configurations. A first atrial channel 836A may define a first opening at a proximal end of the flat portion 837 of the implant holder 835. A second atrial channel 836B may define a second opening at a proximal end of the cavity of the implant holder 835. The first opening and the second opening may correspond to positional openings on the first arm of the implant when the implant is coupled to the implant holder 835. The ventricular channels 838 may define a third and fourth opening, respectively, at a distal end of the implant holder 835. The implant holder 835 may be structurally and/or functionally similar to the implant holder 135, 335, 535, 635, 735; and therefore, certain details of the implant holder 835 are not described herein with respect to FIG. 8.

FIGS. 9A-9C show images of an implant holder 935 of an implant delivery system including an implant 960 coupled

thereto, according to embodiments. As shown, the implant holder 935 includes a proximal end coupled to an implant catheter and a curved distal end. The implant holder 935 defines a cavity in which a first portion of the implant 960 (e.g., the central portion and a first portion of the second arm) is disposed and an engagement surface (e.g., a flat portion, a surface feature etc.) in which a second portion of the implant 960 (e.g., the first arm 962 and a second portion of the second arm) may be disposed. In some embodiments, the engagement surface may be a flat portion of the implant holder 935. In some embodiments, the engagement surface may be curved or shaped corresponding to a portion of the implant 960. In some embodiments, the engagement surface may include one or more surface features to help engage the portion of the implant 960. The atrial tether 940 is configured to extend through a first atrial channel defining a first opening at a proximal end of the flat portion, through a positional opening on the first arm 962 of the implant 960, and into a second atrial channel defining a second opening at a proximal end of the cavity. The ventricular tether 950 may be configured to extend through a first ventricular channel defining a third opening at the distal end of the implant holder 935, through a positional opening on the second arm 964 of the implant 960, and into a second ventricular channel defining a fourth opening at the distal end of the implant holder 935 adjacent to the third opening.

As shown in FIG. 9B, the atrial tether 940 can be tensioned such that the first arm 962 of the implant abuts the engagement surface of the implant holder 935. Tensioning the atrial tether 940 stabilizes the first arm of the implant 960 such that the first arm 962 is held in place when a force is applied to the second arm 964 of the implant 960. When the ventricular tether 950 is tensioned, the second arm 964 of the implant is moved away from the first arm 962 of the implant 960, thereby transitioning the implant 960 to the open configuration, as shown in FIG. 9C. The implant holder 935 and implant 960 may be structurally and/or functionally similar to the implant holder 135, 335, 535, 635, 735, 835 and implant 160, 360, 460, 560, 660; and therefore, certain details of the implant holder 935 are not described herein with respect to FIGS. 9A-9C.

FIGS. 10-15 depict implants 1060, 1160, 1260, 1360, 1460, 1560 for treating heart valve regurgitation, according to various embodiments. As shown in FIG. 10, an implant 1060 includes a first arm 1062, a second arm 1064, and a central portion 1066. The first arm 1062 may include three positional openings 1075 configured to receive one or more atrial tethers to couple the first arm 1062 to an implant holder. Although not shown, the first arm may further include one or more friction elements configured to secure the first arm 1062 relative to tissue of the heart valve when the implant 1060 is implanted. The second arm 1064 may also include two positional openings 1077 configured to receive one or more ventricular tethers to couple the second arm 1064 to the implant holder. The first arm may further include positional openings 1079A, 1079B on a proximal end thereof. Positional opening 1079A may be centrally located on the proximal end of the first arm 1062, and positional openings 1079B may be on opposing sides of the proximal end of the first arm 1062. The second arm 1064 may further include one or more frictional elements 1070 to secure the implant 1060 to the heart tissue. The central portion 1066 may include one or more horizontal struts 1061 disposed along the length of the central portion 1066. In some embodiments, a cover may be disposed over the central portion 1066 and a portion of the second arm 1064. The implant 1060 may be formed from a material including

Nitinol such that the implant **1060** can deform between configurations (e.g., delivery configuration, open configuration, closed configuration, implanted configuration, etc.). The implant **1060** may be structurally and/or functionally similar to any of the implants described herein, and therefore, certain aspects of the implant **1060** are not described in detail with respect to FIG. 10.

Similarly to the implant **1060** of FIG. 10, FIG. 11 shows an implant **1160** including a first arm **1162** including three positional openings **1175**, a second arm **1164** including two positional openings **1177**, and a central portion **1166**. The first arm may include additional positional openings **1179A**, **1179B** on a proximal end thereof. The first and second arm may include one or more frictional elements **1170** configured to secure the implant **1160** to the heart tissue. The central portion **1166** includes horizontal struts **1161** disposed along its length. In contrast to the implant **1060** of FIG. 10, implant **1160** includes four support members **1180** welded to the implant frame. For example, a first support member and a second support member may be disposed on a first side of the central portion **1166** and a third support member and a fourth support member may be disposed on a second side of the central portion **1166** opposite the first side. Each support member may include a first end that may be welded to the proximal end of the first arm and a second end that may be welded to a proximal end of the central portion **1166**. In some embodiments, a cover can be mounted over the support members **1180**.

The implant **1260** of FIG. 12 may also include a first arm **1262** and a second arm **1264** including positional openings **1275**, **1277** and friction elements **1270**; a central member **1266** including horizontal struts **1261**; and four support members **1280**. The first arm **1262** may include additional positional openings **1279A**, **1279B** on a proximal end thereof. The implant **1260** may differ from the implant **1160** of FIG. 11, in that the support members **1280** of implant **1260** may be integrated into the implant frame (rather than welded) and may extend further away from the central member **1266** such that the implant **1260** has a larger maximum width. FIG. 13 shows an implant **1360** including a first arm **1362** and a second arm **1364** including positional openings **1375**, **1377**; a central member **1366**; and four support members **1380**. The implant **1360** may include additional positional openings **1379A**, **1379B** on a proximal end of the first arm **1362**. The implant **1360** may differ from the implant **1260** of FIG. 12 in that the central member **1366** does not include horizontal struts, a thickness of the support members **1380** is reduced, and/or an overall radius of curvature of the support members **1380** is larger than that of implant **1260**. Furthermore, each support member **1380** may include a free end (e.g., a free proximal end) that is not coupled to the implant frame. Although not shown, in some embodiments, the implant **1360** may optionally include friction elements on the first arm **1362** and/or the second arm **1364**. In some embodiments, the implant **1360** may not include friction elements.

FIG. 14 shows an implant **1460** including a first arm **1462** and a second arm **1464** including positional openings **1475**, **1477** and friction elements **1470**; a central member **1466**; and four support members **1480**. The implant **1460** may differ from the implant **1360** of FIG. 13 in that a width of the second arm **1464** (e.g., the ventricular arm) is increased and a width of an opening defined by the first arm (e.g., the atrial arm) is increased to accommodate the second arm **1464**. Therefore, a width of the sides of the first arm **1462** that frame the second arm **1464** is decreased accordingly. Additionally, the positional openings located on opposing sides of

the proximal end of the first arm (e.g., **1379B**) are removed. Therefore, implant **1460** may only include one positional opening **1479** on the proximal end of the first arm **1462**.

FIG. 15 shows an implant **1560** including a first arm **1562** and a second arm **1564** including positional openings **1575**, **1577** and friction elements **1570**; a central member **1566**; and four support members **1580**. The implant **1560** may differ from the implant **1460** of FIG. 14 in that a width of the first arm **1562** (e.g., the atrial arm) is larger than a width of the first arm **1462**. Additionally, the second arm **1564** has been modified to have two portions having two different widths. A first portion of the second arm **1564** that biases into the opening of the first arm **1562** has a first width, and a second portion of the second arm **1564** that is not disposed in the opening of the first arm **1562** has a second width larger than the first width. Implant **1560** further includes openings **1576** disposed at a distal end of the first arm **1562** and configured to receive radio-opaque markers. The implant **1560** may further include openings on the second arm **1564** configured to receive radio-opaque markers. These radio-opaque openings may enable the user to locate the implant **1560** by imaging during implantation of the implant **1560**. The implant design will be described in further detail with respect to FIGS. 16A-16C.

FIGS. 16A-16C illustrate a front perspective view (FIG. 16A), a back perspective view (FIG. 16B), and a bottom view (FIG. 16C) of an implant **1660**, according to an embodiment. The implant **1660** may comprise an attachment section **1602** and a coaptation section **1604**. The attachment section **1602** may include a first segment or arm **1662** and a first portion of a second segment or arm **1664**. The coaptation section **1604** may include a second portion of the second segment or arm **1664** and a central member **1666**. The first arm **1662** may extend from a first end of the central member **1666** and the second arm **1664** may extend from a second end of the central member **1666**. As shown, at least a portion of each of the first arm **1662** and second arm **1664** may be coplanar when in the closed configuration and/or implanted configuration. In some embodiments, this coplanarity may facilitate securely coupling the implant **1660** to the heart valve tissue. Additionally, as depicted, the first arm **1662** and the second arm **1664** may each have a plurality of friction elements **1672**, **1673**, **1674a**, **1674b** extending therefrom. The friction elements **1672**, **1673**, **1674a**, **1674b** may be configured to apply a friction force to the heart valve tissue. The friction elements **1672** and **1673** of the first arm **1662** may be coupled to or integrally formed with a tissue-facing surface of the first arm **1662**. Similarly, the friction elements **1674a**, **1674b** of the second arm **1664** may be coupled to or integrally formed with a tissue-facing surface of the second arm **1664**. Accordingly, the friction elements **1672**, **1673**, **1674a**, **1674b** may be configured to non-destructively engage heart valve tissue when heart valve tissue is received between the first and second arms **1662**, **1664**. For example, a flat portion of the first arm **1662** and the second arm **1664** may be coplanar in a plane while the friction elements **1672** may extend or protrude beyond the plane. Therefore, when the first arm **1662** and the second arm **1664** are coplanar and clamped around the leaflet, the friction elements **1672** may at least partially extend into the leaflet tissue. As shown, the friction elements **1672**, **1673** may extend beyond the second arm **1664** and/or the friction elements **1674** may extend beyond the first arm **1662** when the implant may be in the closed configuration. That is, the friction elements **1672**, **1673** of the first arm **1662** may not be coplanar with second arm **1664** and the friction elements **1674** may not be coplanar with the first arm **1662**.

In some embodiments, the friction elements **1674a**, **1672** in the attachment section **1602** may have an angle with a surface of the respective arm or segment **1662**, **1664** than the friction elements **1674b**, **1673** in the coaptation section **1604**. For example, the friction elements **1672** may form a first angle with a surface of the first segment or arm **1662** and the friction elements **1674a** may form a second angle from a surface of the second segment or arm **1662**. In some embodiments, the first angle and the second angle may be equivalent. In some embodiments, the friction elements **1674b** may form a third angle with the surface of the first segment or arm **1662** and the friction elements **1673** may form a fourth angle with the surface of the second segment or arm **1664**. In some embodiments, the third angle and the fourth angle may be equivalent. The first angle and the second angle may be greater than the third angle and the fourth angle such that the friction elements **1674b** and **1673** do interfere with one another when the implant **1660** is in the closed configuration.

Extending from the first arm **1662** may be a plurality of support members, such as a first support member **1681**, a second support member **1682**, a third support member **1683**, and a fourth support member **1684**. The plurality of support members **1681**, **1682**, **1683**, **1684** may be configured to elastically deflect, including during the delivery process and/or coaptation by a native leaflet. One or more of the support members **1681**, **1682**, **1683**, **1684**, including all of the support members, may extend from the first arm **1662** to a free end, which may further facilitate the deformability of each of the support members **1681**, **1682**, **1683**, **1684**. In some variations, the free end of each of the supporting members **1681**, **1682**, **1683**, **1684** may be positioned relative to the central member **1666** to facilitate delivery, coaptation, and/or contacting heart valve tissue. For example, when the implant **1660** is in a closed configuration, for example, the free end of each of the support members **1681**, **1682**, **1683**, **1684** may be coplanar with a proximal surface of the central member **1666**. The coplanarity of the free ends of the support members **1681**, **1682**, **1683**, **1684** with a proximal surface of the central member **1666** may maximize the volume defined by coaptation section **1604**. In some instances, a first support member may mirror or otherwise have corresponding and opposite shapes as a second support member. For example, in some embodiments, a first support member on a first side of the central member may mirror a second support member on a second side of the central member. As shown, the first support member **1681** and the second support member **1682** may be mirrored counterparts to the third support member **1683** and the fourth support member **1684**. When implanted, the support members **1681**, **1682**, **1683**, **1684** may deform or otherwise flex independently, such that the support members **1681**, **1682**, **1683**, **1684** need not maintain a mirrored configuration. In some embodiments, the implant **1660** may further include one or more positional openings **1675A**, **1675B**, **1675C** defined by the first arm or segment **1662**, as described in further detail with respect to FIG. **25-26**.

FIGS. **17A-17G** show different views of an implant **1760** for treating heart valve regurgitation in a closed configuration, according to an embodiment. The closed configuration may correspond to a pre-set configuration formed by the manufacturing process, and the implant **1760** may be manufactured from a material (e.g., nitinol) capable of returning to the pre-set configuration after deforming during, for example, a delivery process. The pre-set configuration of the implant **1760** may facilitate the biasing forces described herein. The implant **1760** may comprise an attachment

section **1702** and a coaptation section **1704**, as previously described. The attachment section **1702** may include a first segment or arm **1762** and a first portion of a second segment or arm **1764**, such as a planar section **1724**. The coaptation section **1704** may include a second portion of the second segment or arm **1764**, such as a curved section **1726**, and a central member **1766**.

The attachment section **1702** may comprise a length between about 1 mm to about 15 mm, about 2 mm to about 12, about 4 mm to about 10 mm, or about 5 mm to about 7 mm. For example, in some variations, the length of the attachment section **202** may be about 4 mm, about 5 mm, about 6 mm, about 6.5 mm, about 7 mm, or about 10 mm. The coaptation section **1704** may comprise a length between about 1 mm to about 15 mm, about 2 mm to about 12, about 4 mm to about 10 mm, or about 5 mm to about 7 mm. For example, in some variations, the length of the coaptation section **1704** may be about 4 mm, about 5 mm, about 6 mm, about 6.5 mm, about 7 mm, or about 10 mm. A ratio of the length of the attachment section **1702** to the length of coaptation section **1704** may be between about 1:1 to about 1.5:1. In some variations, each of the first and second segments or arms **1762**, **1764** may comprise one or more of a planar section and a curved section. As illustrated in FIG. **17E**, the first arm **1762** may comprise a planar section **1714** and a curved section **1716**. The curved section **1716** may extend from the central member **1766**. In some variations, the attachment section **1702** may comprise the planar section **1714** and the curved section **1716**. Similarly, the second arm **1764** may comprise a planar section **1724** and the curved section **1726**, as shown in FIG. **17D**. The curved section **1726** may extend from the central member **1766**.

As illustrated, the first arm **1762** may comprise a size greater than the second arm **1764**, which may allow the second arm **1764** to interpose the first arm **1762**. For example, the second arm **1764** may be received within an intermediate opening **1763** defined by the first arm **1762**. The intermediate opening **1763** may be sized such that a gap may form between the first arm **1762** and second arm **1764**. The gap may facilitate extending a cover (not shown) through the opening **1763** and between the first arm **1762** and second arm **1764**. Accordingly, the first arm **1762** may comprise a width **1711** and a length **1712**. In some variations, the width **1711** may be between about 2 mm to about 10 mm, about 2 mm to about 9 mm, about 4 mm to about 8 mm, or about 4 mm to about 7 mm. For example, in some variations, the width **1711** may be about 2 mm, about 4 mm, about 5 mm, about 5.5 mm, about 6 mm, or about 7 mm. The length **1712** may be between about 2 mm to about 12 mm, about 3 mm about 11 mm, about 5 mm to about 10 mm, or about 6 mm to about 8 mm. For example, in some variations, the length **1712** may be about 4 mm, about 5 mm, about 6 mm, about 7 mm, or about 8 mm. In comparison, the second arm **1764** may comprise a first width **1720**, a second width **1721**, and a length **1722**. As described previously, the second arm **1764** may comprise a varying width to reduce mechanical stresses associated with coupling heart valve tissue. The first width **1720** may be greater than the second width **1721**. Each of the first width **1720** and second width **1721** may be less than the width **1711** of the first arm **1762**. The transition from the first width **1720** to the second width **1721** may occur at any point along the longitudinal dimension (e.g., length) of the second segment or arm **1764**, such as about $\frac{1}{2}$ from the distal edge of the second segment or arm **1764**.

In some variations, the transition between widths **1720**, **1721** may be gradual such that stress concentrations may be reduced. In some variations, the first width **1720** may be

between about 0.25 mm to about 8 mm, about 1 mm to about 5 mm, about 2 mm to about 4 mm, or about 2.5 mm to about 3.5 mm. For example, in some variations, the first width **1720** may be about 2 mm, about 2.5 mm, about 3 mm, or about 3.5 mm. In some variations, the second width **1721** may be between about 1 mm to about 5 mm, about 2 mm to about 4 mm, about 2.5 mm to about 3.5 mm, or about 2.5 mm to about 3 mm. For example, in some variations, the second width **1721** may be about 2 mm, about 2.5 mm, about 2.75 mm, or about 3 mm. The length **1722** may be less than the length **1712**. In some variations, the length **1722** may be between about 2 mm to about 12 mm, about 3 mm to about 12 mm, about 5 mm to about 10 mm, or about 6 mm to about 8 mm. For example, in some variations, the length **1722** may be about 5 mm, about 6 mm, about 7 mm, or about 8 mm.

The first arm **1762** may define a plurality of openings in a distal portion of the first arm **1762**, which may be configured to receive one or more elongate members (e.g., sutures) and/or visualization markers (e.g., radiopaque markers). As shown, the first arm **1762** may comprise a plurality of positioning openings (e.g., suture holes) **1775A**, **1775B**, **1775C** that may each be configured to receive an elongate member. In the variation shown, the positioning openings **1775A** and **1775C** may be arranged collinearly and the positioning opening **1775B** may be slightly offset therefrom. The slight offset of positioning opening **1775B** may allow an elongate member to be routed through each of the positioning openings **1775A-1775C** in a looped configuration such that a number of contact points between the suture is limited, which may reduce a friction force associated with the suture contacting itself. Defined in a proximal portion (e.g., the curved section **1716**) of the first arm **1762** may be a positioning opening **1779**, which may be configured to receive an elongate member in a similar fashion to the positioning openings **1775A-1775C**. The positioning openings **1775B** and **1779** may be colinear with the longitudinal axis of the first arm **1762** and may be centered with respect to the width of the first arm **1762**. The placement of the positioning openings **1775A-1775C**, **1779** may allow for applying an even force to the first arm **1762** via the suture(s) routed therethrough. In some variations, a single elongate member may be routed through each of the positioning openings **1775A-1775C**, **1779**, which may allow a user to manipulate the entire first arm **1762**. In further variations, a first elongate member may be routed through each of the positioning openings **1775A-1775C** and a second elongate member may be routed through the positioning opening **1779**. In such a variation, a user (e.g., physician) may control the proximal and distal portions of the first arm **1762** separately, which may allow the user to adjust the position of the first arm **1762** with increased precision relative to the variation using a single elongate member. The plurality of openings defined in the distal portion of the first arm **1762** may include a plurality of visualization openings, such as the visualization openings **1776A**, **1776B**. The visualization openings **1776A**, **1776B** may each be configured to receive a visualization marker (e.g., radiopaque marker). As described previously, the visualization markers received within the visualization openings **1776A**, **1776B** may be indirectly visible to the physician while the implant is within the patient.

Similar to the first arm **1762**, the second arm **1764** may define a plurality of openings in a distal portion of the second arm **1764**, which may be configured to receive one or more elongate members and/or visualization markers. In turn, the openings defined by the second arm **1764** may facilitate opening, closing, and/or locating the second arm

1764. As shown, a plurality of positioning openings (e.g., suture holes) may be defined in a distal portion of the second arm **1764**. The plurality of positioning openings may include the positioning openings **1777A**, **1777B**. The positioning openings **1777A**, **1777B** may be collinear relative to each other and may also be collinear with the longitudinal axis of the second arm **1764** and/or central member **1766**. The collinear configuration of the positioning openings **1777A**, **1777B** may facilitate routing an elongate member through each of the positioning openings **1777A**, **1777B**. In this way, the elongate member may form a loop such that the elongate member may, for example, be routed back through the delivery device. Further defined in the distal portion of the second arm **1764** may be a plurality of visualization openings, such as visualization openings **1778A**, **1778B**. The visualization openings **1778A** may each be configured to receive a visualization marker (e.g., radiopaque marker) and may be colinear with a transverse axis of the second arm **1764**.

The first arm **1762** may comprise a plurality of support members **1781**, **1782**, **1783**, **1784** extending from a curved portion of the first arm **1762**. The central member **1766** may separate a first support member **1781** and a second support member **1782** from a third support member **1783** and a fourth support member **1784**. As shown, each of the plurality of support members **1781-1784** may comprise a plurality of curves. For example, the first support member **1781** may comprise a first section **1781A** that is connected to a second section **1781B**, which may be connected to a third section **1781C**. The first section **1781A** may comprise a first radius of curvature forming a convex shape. The second section **1781B** may comprise a second radius of curvature that may form a concave shape. The first radius of curvature may be greater than the second radius of curvature. The third section **1781C** may extend along a plane parallel to a plane defined by the first arm **1762**. Extending from the third section **1781C** may be a fourth section **1781D**, which may comprise a curve with a third radius of curvature. A fifth section **1781E** may extend from the fourth section **1781D**. A sixth section **1781F** may extend from the fifth section **1781E**, and the sixth section **1781F** may comprise a curve with a fourth radius of curvature. The sixth section **1781F** may define a free distal end of the support member **1781**. The free distal end of the sixth section **1781F** may be curved towards the central member **1766**. The free distal end of the support members **1781** may be dulled such that it may not pierce heart valve tissue. Accordingly, as described and illustrated, the support members **1781** may comprise a concave and convex configuration. In some variations, the fourth support member **1784** may be symmetric to the first support member **1781** about the central member **1766**. The fourth support member **1784** may include a first section **1784A**, a second section **1784B**, a third section **1784C**, a fourth section **1784D**, a fifth section **1784E**, and a sixth section **1784F** that are similar to those of the first support member **1781**. The first support member **1781** and fourth support member **1784** may together define a total width **1732** of the implant **1760**. In some variations, the total width **1732** may be between about 10 mm to about 20 mm, about 12 mm to about 20 mm, about 13 mm to about 18 mm, about 14 mm to about 17 mm, or about 15 mm to about 16 mm. For example, in some variations, the total width **1732** may be about 12 mm, about 14 mm, about 15 mm, about 15.5 mm, about 15.8 mm, about 16 mm, or about 17 mm. The second support member **1782** and/or third support member **1783** may define a total height **1733** of the implant **1760**. In some variations, the total height **1733** may be between about 3 mm to about 12 mm, about 4

mm to about 9 mm, about 5 mm to about 8, or about 6 mm to about 8 mm. For example, in some variations, the total height 1733 may be about 4 mm, about 5 mm, about 6 mm, about 7 mm, about 7.2 mm, about 7.5 mm, or about 8 mm. The total width 1732, total height 1733, and length of the coaptation section 1704 may be used (e.g., multiplied together) to determine a total volume of the coaptation section.

As further illustrated, the second support member 1782 may comprise a plurality of sections 1782A, 1782B, 1782C, 1782D. A first section 1782A may comprise a first radius of curvature that may be less than the first radius of curvature of the first section 1782A of the first support member 1782. Extending from the first section 1782A may be a second section 1782B that may comprise a curve with a second radius of curvature. A third section 1782C may extend from the second section 1782B and may connect to a fourth section 1782D. The third section 1782C may be configured to avoid contacting heart valve tissue. The fourth section 1782D may comprise a curve with a third radius of curvature. The third radius of curvature may be similar (e.g., substantially equivalent) to the second radius of curvature of the second section 1782B. A fifth section 1782E may extend from the fourth section 1782D. A sixth section 1782F may extend from the fifth section 1782E, and, similar to the description of 1782F above, may comprise a curve with a fourth radius of curvature and may define a free distal end of the support member 1782. The free distal end of the support member 1782 may be curved towards the central member 1766 and/or curved section 1726. In some variations, the third support member 1783 may be symmetric to the second support member 1782 about the central member 1766. In some embodiments, the third support member 1783 may include a first section 1783A, a second section 1783B, a third section 1783C, a fourth section 1783D, a fifth section 1783E, and a sixth section 1783F similar to those of the second support member 1782. The second support member 1782 may define an apex distance 1730 corresponding to the distance between the apex of the support member 1782 and the apex of the central member 1766. The apex of the central member 1766 may determine a central member height 1734, relative to a plane defined by the planar section 1714 of the first segment or arm 1762. The central member height 1734 may be between about 1 mm to about 7 mm, about 2 mm to about 6 mm, or about 3 mm to about 5 mm. For example, in some variations, the central member height 1734 may be about 3 mm, about 4 mm, or about 5 mm. Accordingly, the apex distance 1730 may be between about 1 mm to about 6 mm, about 2 mm to about 5 mm, or about 2.5 mm to about 3.5 mm. For example, in some variations, the apex distance 1730 may be about 1 mm, about 2 mm, about 3 mm, about 4 mm, or about 5 mm. Similarly, the third support member 1783 may define an apex distance 1731 corresponding to the distance between the apex of the support member 1783 and the apex of the central member 1766. The apex distance 1731 may be the same as the apex distance 1730. In some variations, the apex distances 1730, 1731 may be different, such as during deflection of one or more support members during coaptation by a native leaflet. Accordingly, the apex distance 1731 may be between about 1 mm to about 6 mm, about 2 mm to about 5 mm, or about 2.5 mm to about 3.5 mm. For example, in some variations, the apex distance 1730 may be about 1 mm, about 2 mm, about 3 mm, about 4 mm, or about 5 mm.

FIGS. 18A-18C illustrate a front perspective view, a back perspective view, and a bottom view, respectively, of a first plate configured to be coupled to a bottom surface of a first

arm of an implant, according to an embodiment. The first plate 1862 may comprise a planar section 1802, a curved section 1804, a plate extension 1806, an intermediate opening 1863, a plurality of positioning openings 1875A-1875C, 1879, and friction elements 1872, 1873. The planar section 1802 and the curved section 1804 may each comprise shapes corresponding to the respective sections of the first arm described herein. The planar section 1802 may define one or more openings, such as positioning openings 1875A-1875C, and the curved section 1804 may also define one or more openings, such as positioning opening 1879. As described above, the positioning openings 1875A-1875C, 1879 of the first plate 1862 may correspond to positioning openings of the first arm such that an elongate member may be routed through one or more the positioning openings 1875A-1875C, 1879 to facilitate opening and/or closing the implant. Interposing each of the planar section 1802 and curved section 1804 may be the intermediate opening 1863. The intermediate opening 1863 may comprise a size and shape corresponding to the size and shape of the second arm (not shown) such that the second arm may be received therein. For example, the opening 1863 may be sized such that the second arm may pass through the opening 1863 without contacting the first plate 1862.

Extending from the curved section 1804 may be the plate extension 1806, which may be configured to contact heart valve tissue and/or apply a compressive force thereto. While the planar section 1802 and curved section 1804 may contact the first arm, the plate extension 1806 may not contact the first arm (e.g., the plate extension 1806 may extend away from the first arm). The plate extension 1806 may advantageously apply a compressive force due to the protruding configuration of the plate extension 1806. The plate extension 1806 may comprise a smaller width than the planar section 1802 and/or curved section 1804, which may contribute to the elastic deformability of the plate extension 1806. In some variations, the width of the plate extension 1806 may be such that the plate extension 1806 may be positioned between (e.g., not in contact with) friction elements of the second segment or arm or second plate. Accordingly, the width of the plate extension 1806 may be between about 0.5 mm to about 3 mm, about 0.5 mm about 2 mm, about 1 mm to about 2 mm, or about 1 mm to about 1.5 mm. For example, in some variations, the width of the plate extension 306 may be about 0.5 mm, about 1 mm, about 1.2 mm, or about 1.5 mm.

The plates may comprise friction elements which, as described previously, may be configured to apply a friction force to heart valve tissue. As shown, the first plate 1862 may comprise a first set of friction elements 1872 along two edges of the planar section 1802 and a second set of friction elements 1873 along two edges of the plate extension 1806. The first set of friction elements 1872 may define a first angle relative to a tissue-facing surface of the planar section 1802 and the second set of friction elements 1873 may define a second angle relative to a tissue-facing surface of the plate extension 1806. The first and second angles may not be the same, which may facilitate easier disengagement of the friction elements 1872, 1873 from tissue such that the implant as described herein may be non-destructively removed from a first position of a heart valve (e.g., a first native valve leaflet) and moved to a second position of the heart valve. In further variations, the first and second angles may be the same, which may facilitate faster and/or easier manufacturing.

FIGS. 19A-19C illustrate a front view, a side view, and a bottom view of a second plate 1964 configured to be coupled

to a top surface of a second arm of an implant, according to an embodiment. The second plate **1964** may comprise a planar section **1902**, a curved section **1904**, a plurality of positioning openings **1977A**, **1977B**, and a plurality of friction elements **1974**. The planar section **1902** and the curved section **1904** may each comprise shapes corresponding to the second arm described herein. Similar to the first plate **1862**, the second plate **1964** may define one or more openings, such as positioning openings **1977A**, **1977B**. The positioning openings **1977A**, **1977B** may correspond to positioning openings of the second arm such that an elongate member may be routed through one or more the positioning openings **1977A**, **1977B** to facilitate opening and/or closing the implant. The plurality of friction elements **1974** may define a third angle relative to a tissue-facing surface of the second plate **1964**. The third angle may be the same or different as either the first angle or second angle associated with the plurality of friction elements **1872**, **1873**. Any of the first, second, or third angles may be adjusted during the manufacturing phase or by a physician prior to inserting the implant into the patient.

FIG. **20A** illustrate an exemplary embodiment of coupling the first plate to the first arm and the second plate to the second arm. The illustrated implant **2060** may be made of a metal (e.g., nitinol) so the components may be well-suited for welding. As illustrated, the first plate may be coupled to the first arm via a plurality of welds, such as along a first seam **2040** and a second seam **2042**. The first seam **2040** may extend along a portion of an outer perimeter of each of the first arm and first plate, and the second seam **2042** may extend along a portion of an inner perimeter of each of the first arm and first plate. The length of the weld seams **2040**, **2042** may be sufficient to withstand any forces associated with the surgical procedure, implanting process, and/or cardiac events described herein. Accordingly, the length of the weld seam **2040** may be between about 2 mm to about 15 mm, about 4 mm to about 12 mm, or about 5 mm to about 8 mm. For example, in some variations, the length of the weld seam **2040** may be about 2 mm, about 4 mm, about 5 mm, about 6 mm, about 7 mm, or about 8 mm. The length of the weld seam **2042** may be between about 3 mm to about 15 mm, about 4 mm to about 13 mm, about 5 mm to about 12 mm, or about 6 mm to about 10 mm. For example, in some variations, the length of the weld seam **2042** may be about 4 mm, about 6 mm, about 8 mm, about 10 mm, or about 12 mm. Additionally or alternatively, the second plate may be coupled to the second arm via a plurality of welds, such as along a third seam **2044** and a fourth seam **2046**. The third seam **2044** may extend along a portion of an outer perimeter of each of the second plate and the second arm, and the fourth seam **2046** may extend along the entirety of a perimeter of an opening (e.g., a suture hole) defined by each of the second arm and the second plate. Accordingly, the length of the weld seam **2044** may be between about 3 mm to about 15 mm, about 4 mm to about 12 mm, or about 5 mm to about 10 mm. For example, in some variations, the length of the weld seam **2044** may be about 4 mm, about 6 mm, about 8 mm, or about 10 mm. The length of the weld seam **2046** may be between about 0.5 mm to about 5 mm, about 1 mm to about 5 mm, or about 1 mm to about 3 mm. For example, in some variations, the length of the weld seam **2046** may be about 0.5 mm, about 1 mm, about 2 mm, about 3 mm, or about 5 mm.

FIGS. **21A-21B** illustrate an exemplary variation of visualization markers (e.g., radiopaque markers) received within visualization openings (e.g., radiopaque openings) of a first arm **2162**. The first arm **2162** may comprise a first visual-

ization opening **2179A** configured to receive a first visualization marker **2185A** and a second visualization opening **2179B** configured to receive a second visualization marker **2185B**. The visualization openings **2179A**, **2179B** may comprise substantially circular shapes and the visualization markers **2185A**, **2185B** may be similarly shaped. In some variations, the visualization markers **2185A**, **2185B** may comprise a diameter slightly less than the diameter of the respective visualization openings **2179A**, **2179B** such that the visualization markers **2185A**, **2185B** may utilize friction (e.g., via a friction fit) to fixedly couple to the respective visualization openings **2179A**, **2179B**. In other variations, the visualization markers **2185A**, **2185B** may comprise a diameter smaller than the respective visualization opening **2179A**, **2179B** such that a friction fit technique may not be possible, so an adhesive may be placed under, around, and/or on top of the visualization marker **2185A**, **2185B** to fixedly couple the radiopaque marker to the radiopaque opening. In either instance, the visualization markers **2185A**, **2185B** may be securely fastened in place using, for example, an adhesive and/or welding. The visualization markers **2185A**, **2185B** shown may be manufactured from a metal (e.g., platinum) and thus may be welded into place along a first marker seam **2148A** and a second marker seam **2148B**. The length of the marker seams **2148A**, **2148B** may correspond to the entire circumference of the respective visualization openings **2179A**, **2179B**. In some variations, the length of marker seams may correspond to only a portion of the circumference of the respective visualization openings (e.g., via spot welding). visualization markers may be coupled to visualization openings of the second arm in a similar fashion.

FIG. **22A** illustrate a cover for an implant for treating heart valve regurgitation, according to embodiments. The cover **2268** may be configured to cover portions of an attachment section, such as portions of the first arm, central member, and second arm described herein. For example, the cover **2268** may comprise a first section **2292** configured to cover portions of the first arm, and a second section **2296** configured to cover portions of the central member and the second arm. As shown, the first section **2292** may comprise a greater width than the second section **2296**, which may correspond to the greater width of the first arm relative to the second arm and/or central member. As also shown, the second section **2296** may comprise a greater length than the first section **2292**, which may correspond to the combined lengths of portions of the central member and the second arm relative to the length of the first arm. FIGS. **22B-22C** illustrate alternative views of the cover **2268** coupled to an implant. As shown, the section **2296** may cover a majority of the second arm and the central member and may extend through an opening defined by the first arm **2262**. The section **2292** may cover a majority of the first arm, while leaving the support members **2281**, **2282**, **2283**, **2284** extending from the first arm uncovered. The sections **2292**, **2296** may similarly leave uncovered friction elements extending from each of the first and second arms. The cover **2268** may be securely coupled to the implant via a plurality of sutures along a perimeter of the cover **2268**.

FIG. **23A** illustrates a top view of a cover in a flat configuration, according to embodiments. FIGS. **23B-23C** illustrate a front view of the cover coupled to the implant, according to embodiments. As illustrated, the cover **2368** may be used in combination with the cover **2268** shown in FIGS. **22A-22C**. The cover **2368** may be configured to cover portions of the support members and extend beyond the implant. For example, the cover **2368** may comprise a first

section **2392A** and a second section **2392B** that together define a channel **2395**. The first section **2392A** may be configured to cover a portion of support members on a first side of the central member, and the second section **2392B** may be configured to cover a portion of support members on a second side of the central member. The channel **2395** may be configured to receive the central member, which may already be covered by the cover **2268**. The cover **2368** may further comprise a third section **2394**, which may be configured to cover another portion of the support members on both sides of the central member. In this way, the third section **2394** may be coupled to each of the first section **2392A** and second section **2392B** such that the support members are covered therebetween. The cover **2368** may further comprise a fourth section **2396**, which may be initially formed as a rectangle before being rolled up into a cylinder and partially attached to other portions of the cover **2368**. For example, the fourth section **2396** may be coupled to the third section **2394** via one or more sutures and/or adhesives. In this way, the fourth section **2396** may extend beyond the implant body, which may increase the surface area of the coaptation section to provide a larger surface area for coaptation. The increased surface for coaptation may facilitate use of the implant described in a patient with a clinical condition (e.g., heart valve regurgitation) associated with a relatively large gap between leaflets

FIG. **24** illustrates a front view of the implant shown in FIGS. **17A-17G** in a substantially flat configuration, according to embodiments. The flat configuration of the implant body may be achieved via cutting the outline of the implant from a flat sheet of material. As shown, the implant body **2460** shown in a flat configuration may include a first arm **2462**, a central member **2466**, and a second arm **2464**. The first arm **2462** may extend from, or may be otherwise coupled to, a first end of the central member **2466** and the second arm **2464** may extend from, or may be otherwise coupled to, a second end of the central member **2466** that is opposite the first end. The first arm **2462** may define a plurality of openings in a distal portion of the first arm **2462**, such as positioning openings **2475A**, **2475B**, **2475C** that are each configured to receive an elongate members (e.g., suture), similar to the description for FIG. **17A-17G**. The plurality of openings in the distal portion of the first arm **2462** may also include a plurality of visualization openings, such as visualization openings **2476A**, **2476B**. As described previously in reference to FIG. **17A-17G**, the visualization openings **2476A**, **2476B** may each be configured to receive a visualization marker. Further defined by the first arm **2462** may be an intermediate opening **2463**, which may be configured to receive a portion of the second arm **2464**. The central opening **2463** may be further configured to receive a portion of a cover, such as the cover **2268**, **2368** described previously. Defined in a proximal portion of the first arm **2462** may be a positioning opening **2479**, which may be configured to receive an elongate member in a similar fashion to the positioning openings **2475A-2475C**.

Extending from the first arm **2462** may be a plurality of support members, such as the support members **2481**, **2482**, **2483**, **2484**. The support members **2481**, **2482** may be separated from the support members **2483**, **2484** by the central member **2466**. In some variations, the support members **2481**, **2482** may be symmetric to the support members **2483**, **2484** about a longitudinal axis of the first arm **2462** and/or the central member **2466**. Accordingly, the support members **2481**, **2484** may each comprise a first length and the support members **2482**, **2483** may each comprise a second length. The first length may be greater than the

second length. The support members **2481-2484** may comprise a thickness equivalent to the rest of the implant body and may each comprise a width of about 100 microns to about 200 microns, such as about 150 microns. The width may facilitate the flexibility of the support members described herein by reducing mechanical stresses generated at the connection points between the support members and the first arm **2462**.

Similar to the descriptions provided in reference to FIG. **17A-17G**, the second arm **2464** may comprise a plurality of openings to facilitate opening, closing, and/or locating the second arm **2464**. As shown, a plurality of positioning openings may be defined in a distal portion (relative to a physician during implanting when the implant body may be in a closed configuration) of the second arm **2464**. The plurality of positioning openings may include the positioning openings **2477A**, **2477B**. Further defined in the distal portion of the second arm **2464** may be a plurality of radiopaque openings, such as visualization openings **2478A**, **2478B**. The visualization openings **2478A**, **2478B** may each be configured to receive a visualization marker, similar to the descriptions provided previously.

FIGS. **25** and **26** illustrate exemplary variations of a first plate **2562** in a flat configuration and a second plate **2564** in a flat configuration, respectively, configured to couple to the body described herein. The first plate **2562** may be manipulated into the configuration shown in FIGS. **18A-18C**. Accordingly, the description of the positioning openings **2575A**, **2575B**, **2575C**, **2579** correspond to the positioning openings **1875A**, **1875B**, **1875C**, **1879**. The plate **1862** may further comprise a plurality of slots **2572** that may each be adjacent to a friction element **1872**, **1873** (not shown). Each slot **2572** may define a length corresponding to a height of the friction element **1872**, **1873** and may facilitate bending the friction element **1872**, **1873** (shown in FIGS. **18A-18C**) to the desired angle that corresponds to a desired friction force. Each slot **2572** may terminate in a circular portion **2572A**. The circular portion **2572A** may be configured to reduce mechanical stresses associated with bending the friction elements **1872** (shown in FIGS. **18A-18C**). The second plate **2664** may be manipulated into the configuration shown in FIGS. **19A-19C**. The second plate **2664** may comprise a plurality of positioning openings **2677A**, **2677B**, which may correspond to the positioning openings **1977A**, **1977B**. The second plate **2664** may further comprise a slot **2674** and a circular portion **2674A** that function similarly to the slot **2572** and circular portion **2572A**, respectively, described in reference to the first plate **2562** of FIG. **25**. The plates **2562**, **2664** may then be coupled to the implant body as described previously in reference to FIGS. **20A-20E**.

FIG. **27A** is a front view of an implant **2760**, according to an embodiment. The implant **2760** may include an attachment section and a coaptation section as previously described herein. The attachment section may include a first segment or arm **2762** and a first portion of a second segment or arm **2764**. The coaptation section may include a second portion of the second segment or arm **2764** and a central member **2766**. The first arm **2762** may include one or more positional openings **2775A**, **2775B**, **2775C** (e.g., three positional openings) disposed at a distal end thereof, and the second arm **2764** may include one or more positional openings **2777A**, **2777B** (e.g., two positional openings) disposed at a distal end thereof. The first arm **2762** may further include one or more visual marker openings **2776A**, **2776B**, as previously described herein, disposed distal to the positional openings **2775A**, **2775B**, **2775C**. Each of the first arm **2762** and the second arm **2764** may further include

friction elements 2770, as previously described herein. The first arm may further include a proximal positional opening 2779. The implant may further include a first support member 2780A, a second support member 2780B, a third support member 2780C, and a fourth support member 2780D. The implant 2760 may be structurally and/or functionally similar to any of the implants described herein (e.g., implant 460, implant 1660, etc.), and therefore certain details of the implant 2760 are not described in further detail with respect to FIGS. 27A-27B. The implant 2760 may be similar to the implants 1560 and 1660 of FIGS. 15-16; however, the implant 2760 may include a tear drop shape 2771 at the end of each tine or friction element 2770 to reduce stress concentration when they are bent into shape, as shown in FIG. 27B.

FIG. 28 are images of various implant sizes (e.g., 10 mm on left, 12 mm in center, and 16 mm on right) in a delivery configuration (bottom row) and a deployed configuration (top row). The implant in the delivery configuration is in a constrained state, in which the support members of the implant are elongated such that the implant fits in the inner lumen of the implant catheter. In some embodiments, in the constrained state, the support members of the implant are configured to fold across one another to decrease the overall width of (or cross-sectional area of) the implant such that the implant fits in the inner lumen of the implant catheter. The free ends of the support members may enable the support members to fold across one another to transition to the constrained state. In some embodiments, the support members of the implant may elongate in the constrained state to decrease the overall width of the implant. When the implant is deployed (e.g., advanced out of the implant catheter), the implant catheter no longer constrains the implant, and the implant returns to its relaxed state in which the support members expand away from the central member of the implant. While in FIG. 28, the implant is shown directly in the implant catheter, it should be appreciated that the implant can be coupled to an implant holder and transition between the constrained state and the relaxed state while coupled to the implant holder. Additionally, the implant may include a cover disposed over the support members and configured to conform with the support members as the implant transitions between the constrained and relaxed states.

FIG. 29 shows an implant delivery system 3000 including a handle assembly 3001 coupled to a proximal end of a catheter system (e.g., a guide catheter 3010, a delivery catheter 3020, and an implant catheter 3030) and an implant holder 3035 coupled to distal end of the catheter system, according to embodiments. The catheter system may include a guide catheter 3010 coupled to a GC actuator 3012, a delivery catheter 3020 coupled to a DC actuator 3022, and an implant catheter 3030 coupled to an IC actuator 3032. As shown, the handle assembly 3001 includes the GC actuator 3012 including a GC knob 3013 configured to control a degree of bending at a distal end of the guide catheter 3010, and a GC indicator 3014 configured to show a magnitude of the degree of bending at the distal end of the guide catheter 3010. The GC actuator 3012 may be coupled to a guide catheter (GC) flushing port 3018. The GC flushing port 3018 may be in fluid communication with a lumen of the guide catheter 3010 such that bodily fluid (e.g., blood) may be flushed from the lumen of the guide catheter 3010 during implantation. The handle assembly 3001 may further include the DC actuator 3022 including a DC knob 3023 configured to control a degree of bending at a distal end of the delivery catheter 3020, and a DC indicator 3024 con-

figured to show a magnitude of the degree of bending at the distal end of the delivery catheter 3020. The DC actuator 3022 may be coupled to a DC flushing port 3028. The DC flushing port 3028 may be in fluid communication with a lumen of the delivery catheter 3020 such that bodily fluid (e.g., blood) may be flushed from the lumen of the delivery catheter 3020 during implantation.

The distal end of the implant catheter 3030 may be coupled to the implant holder 3035 including a distal portion 3035a and a proximal portion 3035b joined by a connector, further described in FIGS. 40-42D. The implant holder 3035 may be configured to receive an implant between the distal portion 3035a and the proximal portion 3035b. The GC knob 3013 and the DC knob 3023 may be configured to rotate in a clockwise and counterclockwise, as described in further detail with respect to FIGS. 31A-31C and FIGS. 32A-32D, respectively. The handle assembly 3001 further includes the IC actuator 3032 including an atrial tether (AT) actuator 3042 and a ventricular tether (VT) actuator 3052, described further in FIGS. 37A-37B. The GC actuator 3012, the DC actuator 3022, and the IC actuator 3032 may be structurally and/or functionally similar to any of the GC actuator 112, 612, the DC actuator 122, 622, and the IC actuator 132, 632 previously described, and therefore certain details are not described in further detail herein.

FIGS. 30A-30B show the handle assembly 3001 including the GC actuator 3012 coupled to the guide catheter and a dilator 3015 configured to be disposed through a lumen of the guide catheter 3010, according to embodiments. As shown, the dilator 3015 is configured to extend through the GC actuator 3012 such that a proximal portion of the dilator 3015 extends beyond a proximal end of the GC actuator 3012. In some embodiments, the dilator may define a lumen configured to receive a guidewire therethrough. A distal portion of the dilator 3015 may include a tapered end, and a proximal portion of the dilator 3015 may have a diameter substantially equal to (or slightly smaller than) a diameter of the guide catheter 3010 and a distal end of the dilator 3015 may taper to a point such that the dilator 3015 creates an opening in the tissue corresponding to the diameter of the guide catheter 3010. The dilator 3015 may aid in insertion of the guide catheter 3010 during the procedure.

FIG. 31A shows a close-up view of the GC actuator 3012 including the GC knob 3013, the GC indicator 3014, and the GC flushing port 3018. Rotating the GC knob 3013 clockwise may bend a distal portion 3011 of the guide catheter 3010 (e.g., in the bending portion) in a first direction (e.g., toward the flushing port 3018) along a first plane, and rotating the GC knob 3013 counterclockwise may bend the distal portion of the guide catheter 3010 in a second direction (e.g., away from the flushing port 3018) opposite the first direction along the first plane. The clockwise and counterclockwise direction may correspond to when a user is positioned at a proximal end of the implant delivery system and facing toward the distal end of the implant delivery system. In some embodiments, a degree of rotation of the GC knob 3013 may bend the distal portion of the guide catheter 3010 to an angle α from the longitudinal axis. In some embodiments, a ratio of the degree of rotation of the GC knob 3013 to the angle α of the distal portion of the guide catheter 3010 from the longitudinal axis (e.g., the bending angle) may be referred to as a first ratio. In some embodiments, the first ratio is about 10:1, inclusive of all ranges and subranges therebetween. In some embodiments, the GC knob 3013 may turn a screw that in turn causes the distal end of the guide catheter 3010 to bend. In some embodiments, the first ratio of the rotation between the GC

61

knob **3013** and the bending angle of the guide catheter **3010** is defined by a pitch of the screw that is turned by the GC knob **3013**.

In some embodiments, the GC indicator **3014** may indicate the magnitude of bending at the distal portion **3011** of the guide catheter **3010**. The GC indicator **3014** may include a plurality of indicia **3014a** (e.g., numbers, symbols, colors, diagrams, etc.) and a movable indicator **3014b**. For example, the GC indicator **3014** may include numerical indicia between -5 to $+5$ (e.g., -5 , -4 , -3 , -2 , -1 , 0 , $+1$, $+2$, $+3$, $+4$, $+5$) and the movable indicator **3014b** may be configured to indicate to a user one of the indicia between -5 to $+5$ as the user rotates the GC knob **3013**. In some embodiments, each of the indicia may correspond to 10 degrees of rotation of the GC knob **3013**, and therefore, an angle α of 1 degree of bending of the distal portion **3011** of the guide catheter **3010**. While the indicia are shown between -5 to $+5$, it should be appreciated that any numerical range may be used to indicate the magnitude. In some embodiments, the GC scale **3014** may indicate the magnitude of bending in higher or lower resolution. In some embodiments, an entirety of the GC actuator **3012** may be configured to rotate clockwise and/or counterclockwise about the catheter system. Rotating the entire GC actuator **3012** clockwise may move the distal portion of the guide catheter **3010** in a third direction along a second plane substantially orthogonal to the first plane. Rotating the entirety of the GC actuator **3012** counterclockwise may move the distal portion of the guide catheter **3010** in a fourth direction opposite the third direction along the second plane.

FIG. 32A shows a first actuation of the DC actuator **3022**, according to embodiments. In some embodiments, the first actuation of the DC actuator **3022** may include rotation of the DC knob **3023**. In some embodiments, rotating the DC knob **3023** clockwise may bend a distal portion of the delivery catheter **3020** in a fifth direction along a third plane, as shown in FIGS. 32B-32D. In some embodiments, rotating the DC knob **3023** counterclockwise may bend the distal portion of the delivery catheter **3020** in the sixth direction opposite the fifth direction along the third plane. In some embodiments, the first plane and the third plane may be substantially equal. In some embodiments, the fifth direction may be substantially equal to the first direction, and the sixth direction may be substantially equal to the second direction. In some embodiments, a degree of rotation of the DC knob **3023** may correspond to an angle R the delivery catheter **3020** bends from the longitudinal axis of the delivery catheter **3020**. In some embodiments, a ratio of the degree of rotation of the DC knob **3023** and the angle R may be referred to as a second ratio. In some embodiments, the second ratio may be about 5:1, inclusive of all ranges and subranges therebetween. In some embodiments, the first ratio may be greater than the second ratio meaning that the precision for controlling the guide catheter **3010** may be coarser than a precision for controlling the delivery catheter **3020**. This is because, in some implementations and use cases, control over the guide catheter **3010** does not need to be as precise as control over the delivery catheter **3020** to allow the user to steer distal end into the desired configuration. In some embodiments, the DC knob **3023** may turn a screw that in turn causes the distal end of the delivery catheter **3020** to bend. In some embodiments, the second ratio of the rotation between the DC knob **3023** and the bending angle of the delivery catheter **3020** is driven by a pitch of the screw that is turned by the DC knob **3023**. In some embodiments, a number of turns of the DC knob **3023** (or the GC knob **3013**) may depend on a tension on the atrial and/or ventricular

62

tethers and/or a pitch of the screw. In some embodiments, the DC actuator **3022** may include a DC scale. In some embodiments, the DC scale may be structurally and/or functionally similar to the GC scale **3014**, and therefore, details of the DC scale are not described herein.

In some embodiments, an entirety of the DC actuator **3022** may be configured to rotate clockwise and/or counterclockwise about the catheter system, as shown in FIG. 33A. Rotating the entire DC actuator **3022** clockwise may move the distal portion of the delivery catheter **3020** in a seventh direction along a fourth plane substantially orthogonal to the third plane. Rotating the entirety of the DC actuator **3022** counterclockwise may move the distal portion of the delivery catheter **3020** in an eighth direction opposite the seventh direction along the fourth plane, as shown in FIGS. 33B-33C. In some embodiments, the second plane and the fourth plane may be substantially equal. In some embodiments, the seventh direction may be substantially equal to the third direction, and the eighth direction may be substantially equal to the fourth direction.

In some embodiments, the IC actuator **3032** and the DC actuator **3022** may be configured to move relative to one another in a proximal/distal direction to transition a distal end of the implant holder **3035** between a retracted configuration and an extended configuration. In some embodiments, distal movement of the IC actuator **3032** may move the implant holder **3035** distally, proximal movement of the IC actuator **3032** may move the retract the implant holder **3035**. Alternatively or additionally, proximal movement of the DC actuator **3022** may extend the implant holder **3035**, as shown in FIGS. 34A-34B, and distal movement of the DC actuator **3022** may retract the implant holder **3035**, as shown in FIGS. 34C-34D. In some embodiments, the DC actuator **3022** may be moved linearly in a distal/proximal direction to extend and/or retract the delivery catheter **3020**.

FIG. 35A shows a distal end of the catheter system in an extended configuration in which the delivery catheter **3020** extends distally from the guide catheter **3010**, and the implant catheter **3030** extends distally from the delivery catheter **3020** to expose the implant holder **3035**. FIG. 35B shows the distal end of the catheter system in a retracted configuration in which the delivery catheter **3020** is retracted into the lumen of the guide catheter **3010**, and the implant catheter **3030** is retracted into a lumen of the delivery catheter **3020** such that the implant holder **3035** is at least partially covered by the guide catheter **3010**. In some embodiments, the handle assembly **3001** may be configured to move proximally and/or distally (e.g., by a user via a handle assembly holder). The entire catheter system may move proximally in response to the handle assembly **3001** moving proximally. The entire catheter system may move distally in response to the handle assembly **3001** moving distally.

FIGS. 36A-36C show steerability maps of distal end of the implant delivery system relative to a mitral heart valve in response to actuation of the guide catheter actuator, the delivery catheter actuator, and the entire handle assembly, respectively. As shown in FIG. 36A, GC actuator + indicates clockwise rotation of the GC knob (e.g., GC knob **3013**), GC actuator - indicates counterclockwise rotation of the GC knob, GC actuator CW indicates clockwise rotation of the entire GC actuator (e.g., GC actuator **3012**) about the longitudinal axis of the device, and GC actuator CCW indicates counterclockwise rotation of the entire GC actuator about the longitudinal axis of the device. As described in FIGS. 31A-31B, the clockwise and counterclockwise direction correspond to when a user is positioned at a proximal

63

end of the implant delivery system and facing toward the distal end of the implant delivery system. As shown, rotation of the GC knob causes the distal end of the guide catheter to move along a first axis and rotation of the entirety of the GC actuator causes the distal end of the guide catheter to move along a second axis substantially orthogonal to the first axis. As shown, clockwise rotation of the GC knob causes the distal end of the implant delivery system to move posteriorly in the heart and counterclockwise rotation of the GC knob causes the distal end of the implant delivery system to move anteriorly in the heart. Clockwise rotation of the entire GC actuator causes the distal end of the implant delivery system to move laterally within the heart and counterclockwise rotation of the entire GC actuator causes the distal end to move medially within the heart.

As shown in FIG. 36B, DC actuator + indicates clockwise rotation of the DC knob (e.g., DC knob 3213), DC actuator – indicates counterclockwise rotation of the DC knob, DC actuator CW indicates clockwise rotation of an entirety of the DC actuator (e.g., DC actuator 3212) about the longitudinal axis of the device, and DC actuator CCW indicates counterclockwise rotation of the entire DC actuator about the longitudinal axis of the device. As shown, the DC actuator can control movement of the distal end of the DC catheter along three axes. In some embodiments, clockwise rotation of the DC knob may move the distal end of the implant delivery system posteriorly in the heart, counterclockwise rotation of the DC knob should move the distal end of the implant delivery system anteriorly in the heart. Clockwise rotation of the entire DC actuator causes the distal end of the implant delivery system to move laterally within the heart and counterclockwise rotation of the entire DC actuator causes the distal end to move medially within the heart. Additionally, the entire DC actuator may be moved proximally and distally (e.g., linearly). Moving the entire DC actuator distally results in the delivery catheter moving out of the guide catheter lumen. Moving the entire DC actuator proximally results in the distal end of the delivery catheter system into the lumen of the guide catheter.

In some embodiments, when the DC actuator knob is neutral (e.g., at “0”, or not rotated clockwise or counterclockwise), the distal end of the delivery catheter may be parallel to the mitral valve annulus. In some embodiments, when the DC actuator knob is rotated about 45 degrees clockwise, the distal end of the delivery catheter may extend toward a lateral commissure. In some embodiments, when the DC actuator knob is rotated about 90 degrees clockwise, the distal end of the delivery catheter may extend toward a medial commissure.

As shown in FIG. 36C, the implant delivery system can be moved proximally and/or distally. In some embodiments, moving the implant delivery system distally results in the distal end of the implant delivery system moving anterior-lateral. Moving the entire DC actuator proximally results in the implant delivery system moving posterior-medial.

FIGS. 37A-37B show actuation of an AT actuator 3042, according to embodiments. The AT actuator 3042 of the IC actuator 3030 may include a linear slider and/or a rotating actuator. In some embodiments, the AT actuator 3042 may include a locking mechanism configured to lock a linear position of the AT actuator 3042, thereby locking an amount of tension applied on the atrial tether and a position of the atrial arm. In some embodiments, rotating the AT actuator 3042 may lock and/or unlock the locking mechanism such that the AT actuator 3042 can be moved proximally and/or distally. For example, the AT actuator 3042 may be configured to rotate inward (e.g., counter-clockwise or towards a

64

centerline of the delivery system) and then move linearly in (e.g., about 5 mm to about 50 mm). In some embodiments, the AT actuator 3042 may be locked at any position by rotating the AT actuator 3042 outward (e.g., clockwise). In some embodiments, the AT actuator 3042 can be locked in the most proximal position.

In some embodiments, when the AT actuator 3042 is moved distally, tension is released on the atrial tether, and therefore the atrial arm of the implant. Releasing the tension on the atrial arm may allow the atrial arm to drop onto a portion of the leaflet that is disposed between the atrial arm and the ventricular arm (e.g., to clamp the leaflet). Further rotation inwards or distal movement of the AT actuator 3042 can introduce additional slack into the atrial tether, allowing the implant holder to be moved away from the implant when the implant is in its deployed state on the leaflet. When the AT actuator 3042 is rotated outwards (e.g., clockwise) and/or moved proximally the atrial tether, and therefore the atrial arm of the implant, may be tensioned such that the atrial arm of the implant is moved towards the implant holder (e.g., securing the atrial arm to an engagement surface of the implant holder). For example, when preparing open the ventricular arm, the AT actuator 3042 may be moved proximally to tension the atrial tether and secure the atrial arm to the implant holder. As another example, the AT actuator 3042 may be moved proximally to tension the atrial tether and secure the atrial arm to the implant holder before the ventricular tether is decoupled from the ventricular arm.

FIGS. 38A-38C show actuation of a ventricular tether (VT) actuator 3052 to open a ventricular arm of the implant away from an atrial arm of the implant, according to embodiments. The VT actuator 3052 may include a linear slider and/or a rotating actuator. The AT actuator 3042 and the VT actuator 3052 may allow the user to deliver the implant to the leaflet with precise control. For example, to deliver the leaflet, the following steps can be completed. In some embodiments, the VT actuator 3052 may include a rotating actuator configured to apply tension and/or reduce tension in the ventricular tether in response to rotation. For example, the VT actuator 3052 may be configured to apply tension to the ventricular tether in response to rotation of the VT actuator 3052 in a first direction (e.g., clockwise) to move the ventricular arm away from the atrial arm and open the implant. In some embodiments, the VT actuator 3052 may reduce tension to the ventricular tether in response to rotation of the VT actuator 3052 in a second direction (e.g., counterclockwise) to move the ventricular arm toward the atrial arm and close the implant. In some embodiments, a degree of rotation of the VT actuator 3052 may correspond to a distance the ventricular arm opens.

FIGS. 39A-39B show a handle assembly holder 3005 to help a user control the handle assembly 3001, according to embodiments. As shown, the handle assembly holder 3005 includes a base 3009a and a moveable portion 3009 coupled to a support axis 3008. The moveable portion 3009 may be configured to adjust an orientation of the support axis 3008 relative to a surface on which the base 3009 is disposed (e.g., a table). The support axis 3008 may be configured to hold the handle assembly 3001 of the implant delivery system such that the implant delivery system is disposed at the predetermined angle relative to the table. In some embodiments, the support axis 3008 may include a first control mechanism 3002a configured to hold and/or control a position of the GC actuator 3010, a second control mechanism 3003a configured hold and/or control a position of the DC actuator 3020, and a third control mechanism 3004a configured to hold and/or control a position of the IC actuator

3030. In some embodiments, the control mechanisms **3002a**, **3003a**, **3004a** may be configured to move along a longitudinal axis of the support axis **3008** (e.g., distally and/or proximally). In some embodiments, the control mechanism **3002a**, **3003a**, **3004a** may move along the support axis **3008** independently, such that the GC actuator **3010**, the DC actuator **3020**, and the IC actuator **3030** can be moved distally and/or proximally independent from one another. In some embodiments, the first control mechanism **3002a** may include a first actuator **3002b** (e.g., a knob, a slider), configured to receive an input from a user to move the first control mechanism **3002a**. Similarly, the second control mechanism **3003a** may include a second actuator **3003b** (e.g., a knob, a slider), and the third control mechanism **3004a** may include a third actuator **3004b** (e.g., a knob, a slider). In some embodiments, the first, second, and third control mechanisms **3002a**, **3003a**, **3004a**, may include a lock configured to lock a position of the control mechanisms **3002a**, **3003a**, **3004a**, and therefore the GC actuator **3010**, DC actuator **3020m** and IC actuator **3030** in place.

FIG. 40 is a schematic diagram of an implant holder **3135**, according to embodiments. In some embodiments, the implant holder **3135** may be structurally and/or functionally similar to the implant holder **335**, **835**, and therefore, certain details of the implant holder **3135** are not described again with respect to FIG. 40. The implant holder **3135** may include a proximal portion **3135b** and a distal portion **3135a** joined by a connector **3137** therebetween. The implant holder **3135** may be configured to receive an implant **3160** between the proximal portion **3135b** and the distal portion **3135a**. In some embodiments, a front side of the proximal portion **3135b** may include an engagement surface (e.g., an atrial plate stabilizing holder) configured to receive at least an atrial arm of the implant **3160**. In some embodiments, a first portion of the implant **3160** may be disposed on the engagement surface and a second portion of the implant (e.g., the leaflet enhancer) may be disposed between the proximal portion **3135b** and the distal portion **3135a**. In some embodiments, a back side of the implant holder **3135** opposite the front side may define an opening extending from the front side to the back side and disposed between the distal portion **3135a** and the proximal portion **3135b**. In some embodiments, a portion of the implant **3160** may be disposed in the opening such that a back side of the implant extends beyond (e.g., laterally beyond) the connector and/or the back side of the implant holder **3135**.

The proximal portion **3135b** of the implant holder **3135** may include one or more atrial channels **3136** through which one or more atrial tethers **3140** may extend. The atrial channel(s) **3136** may define an opening at or near the engagement surface of the proximal portion **3135b** such that the atrial tether(s) **3140** can be coupled to the atrial arm of the implant **3160**. The proximal portion **3135b** may further include a first ventricular channel or a first set of ventricular channels **3138b** through which one or more ventricular tethers **3150** may extend. The first set of ventricular channels **3138b** may define openings in the proximal portion **3135b** such that the ventricular tether(s) **3150** can extend distal to the proximal portion **3135b** and into a second ventricular channel or a second set of ventricular channels **3138a** in the distal portion **3135a**. The second set of ventricular channels **3138a** may receive the ventricular tether(s) **3150** through a first set of openings on the backside of the distal portion **3135a** (e.g., proximate to a central member of the implant **3160**) and a second set of openings on the front side of the distal portion **3135a** (e.g., near the arms of the implant **3160**) such that the ventricular tether(s) **3150** can extend proxi-

mally from the distal portion **3135a** and couple to a ventricular arm of the implant **3160**.

In contrast to some implementations of the implant holder (e.g., implant holder **835**), implant holder **3135** can have a separate proximal portion **3135b** and distal portion **3135a** (e.g., “the open back design”) such that the tethers are not constrained in the material of the implant holder **3135**. Therefore, the implant holder **3135** reduces friction on the atrial tether(s) **3140** and ventricular tether(s) **3150** and reduces difficulty in manipulating the tethers from the proximal end of the delivery system. A closed back design may require a sharper bend on the tethers, and therefore, may impart more strain on the tethers.

The open back design of implant holder **3135** may also improve ease of flushing out the atrial channel(s) **3136** and ventricular channel(s) **3138a**, **3138b** (e.g., due to the reduced length and reduced circuitry of the channel pathways) and reduces the likelihood of blood clots forming in the atrial and ventricular channels **3140**, **3150**, which can restrict the tethers from being pulled out from a proximal end of the delivery system to release the implant **3160**. In some embodiments, the open back design may enable easier flushing of fluid (e.g., saline) to remove bodily fluid (e.g., blood) and/or air from the atrial and ventricular channels **3140**, **3150**. In the body, if blood pressure is higher than system, blood will flow into the device through the atrial and ventricular channels **3140**, **3150**; therefore, the atrial and ventricular channels **3140**, **3150** should be flushed intermittently or continuously. In some embodiments, flushing can occur periodically with a predefined time interval. In some embodiments, flushing may be continuous. In some embodiments, flushing can occur in response to a measurement from a sensor (e.g., a measurement of blood pressure in the body, a measurement of the presence of blood in a lumen of the device, etc.). In some embodiments, flushing may occur in response to actuation by a user as needed.

The implant holder **3135** with the open back design may improve localization of the implant **3160** during implantation. For example, the implant holder **835** with the closed back design may appear as one structure during imaging (e.g., ultrasound), and therefore, distinguishing between the implant **3160** and the implant holder **835** may be difficult for a user. The implant holder **3135** with the open back design may appear as two separate or distinct portions with a gap or other visual demarcation therebetween when imaging such that the implant **3160** can be discernible from the implant holder **3135** during the procedure. In some embodiments, the proximal portion **3135b** and the distal portion **3135a** may be used as imaging landmarks (e.g., provide an axis) to help the user maneuver the distal end of the implant delivery system, and specifically the implant **3160**, in the desired orientation relative to the heart valve.

In some embodiments, the implant holder **3135** with the open back design may enable easier assessment of whether the implant **3160** is an appropriate size. For example, without the closed back, the implant holder **3135** fills less of the native gap during the implantation process and the implant **3160** can bias more toward the back of the implant holder **3135** (e.g., laterally in the body), whereas the implant holder **835** with the closed back would fill more of the native gap. In some embodiments, the implant **3160** may extend beyond the connector **3137** in a lateral direction (e.g., a direction perpendicular to the longitudinal axis of the distal end of the implant delivery system) such that a maximum width of the distal end of the implant delivery system is defined by the implant **3160**. In some embodiments, the implant holder **3135** may include a pair of connectors (e.g.,

67

hypotubes), and the implant **3160** may be configured to extend through a space between the connectors towards (or beyond) the back side of the implant **3160**. The backside **3161** of the implant **3160** may extend beyond the connector **3137**. In other words, the implant holder **3135** may have a first width **W1** (or cross-sectional area), and the implant **3160** may have a second width **W2** (or cross-sectional area) greater than the first width **W1** such that the maximum width of the distal end of the implant delivery system is defined by the second width. In some embodiments, the implant **3160** and the implant holder **3135** may be collectively configured to be delivered into the native valve gap (e.g., the orifice defined by the valve), and when the implant **3160** and implant holder **3135** are disposed in that gap, a maximum dimension (e.g., cross-sectional area, perimeter, or the like) in a direction spanning the gap between the leaflets is defined by the implant **3160**. In other words, the implant holder **3135** and implant **3160** collectively define a maximum dimension (e.g., width in the direction spanning the gap between the leaflet), and that maximum dimension is limited by and/or defined by the implant **3160**.

Because of the implant **3160** spanning wider (e.g., laterally) than the implant holder **3135**, the user may not need to (1) clamp the implant **3160** to the leaflet, (2) loosen (e.g., apply slack to) the atrial and/or ventricular tethers and/or (3) move the distal end of the implant delivery system away from the implant **3160** in order to assess the effectiveness of the implant **3160**. Therefore, the open back design may enable the user to assess whether the implant **3160** is addressing valve regurgitation without attaching the implant **3160** to the native leaflet. Instead, the distal end of the implant delivery system can remain in place while the user assesses whether the implant **3160** is treating the regurgitation. With the open back design, when the implant **3160** in the implant holder **3135** (e.g., with the tether(s) taut) is disposed in the opening of the heart valve, the implant **3160** in the implant holder **3135** may occupy a cross-sectional area that is equal to a cross-sectional area of the implant **3160** detached from the implant delivery system. Therefore, with the open back design, the user can more accurately assess whether the implant **3160** is the appropriate size.

In embodiments with the open back design, a method of implantation may include positioning the implant in an opening defined by the heart valve without capturing the free end of the native heart valve leaflet between the pair of arms of the implant; determining a level of valve regurgitation when the implant is positioned in the opening defined by the heart valve, after determining the level of valve regurgitation, transitioning the implant from its closed position to its open position; and transitioning the implant from the open position towards its closed position to clamp the free end of the native leaflet between the pair of arms.

In some embodiments, the distal portion **3135a** of the implant holder and the proximal portion of the implant holder **3135b** may be configured to move relative to one another. For example, the connector **3137** may be slidable, retractable, moveable, etc. In some embodiments, the connector **3137** may be one or more hypotubes with an adjustable length. In some embodiments, the distal portion **3135a** of the implant holder **3135** may be configured to move away from the proximal portion **3135b** to accommodate a larger implant, for example, and move toward the proximal portion **3135b** to accommodate a smaller implant. In some embodiments, the connector **3137** may extend from the implant holder **3135** proximally through the catheter system and to the proximal end of the delivery system (e.g., the handle

68

assembly). In some embodiments, the connector **3137** may be adjustable from the proximal end of the delivery system.

In some embodiments, a length of the connector **3137** may be adjustable intra-procedure. For example, the implant holder **3135** may be shortened during a turn toward the valve to reduce likelihood of the distal end of the delivery system colliding with any heart anatomy. In some embodiments, once the sharp turn into the heart is completed, the implant holder **3135** may be lengthened such that the implant **3160** is closer to the native valve leaflets. In some embodiments, the distal portion **3135a** of the implant holder **3135** may be extended distally to transition the implant **3160** to the open configuration rather than or in addition to pulling the ventricular tether(s) **3150**. In some embodiments, controlling a position of the distal portion **3135a** of the implant holder **3160** may improve controllability of opening the implant **3160**. For example, a distance of the distal portion **3135a** may be controlled incrementally with fine precision. In some embodiments, the ventricular tether(s) **3140** may be tensioned to transition the implant **3160** to the open configuration. In some embodiments, movement of the distal portion **3135a** may also allow the user to detangle the distal end of the delivery system from the chords below the leaflets.

In some embodiments, after the implant **3160** has been attached to the leaflet, the distal portion **3135a** and the proximal portion **3135b** may be moved away from one another to assess whether regurgitation has been improved in addition to or instead of applying slack to the atrial tether(s) **3140** and ventricular tether(s) **3150**. In some embodiments, the implant holder **3135** may include a material that improves visibility for imaging (e.g., a transparent or semi-transparent material). For example, the implant holder may include a polymer such as polyetherimide (e.g., ULTEM™). In some embodiments, the connector **3137** may be formed from or include any suitable biocompatible, fluorogenic and echogenic such as, for example, stainless steel, nitinol, etc. In some embodiments, the implant **3160** may be monolithically constructed, while in some embodiments, the implant **3160** may be formed of separate components and then coupled together. In some such embodiments, for example, the distal portion **3135a**, the proximal portion **3135b**, and the connector **3237** may all be formed separately and then coupled together via any suitable means (e.g., the proximal portion **3135b** being coupled to the distal portion **3135a** via the connector **3237**).

FIGS. 41A-41D shows an implant holder **3235** extending distal to a delivery catheter **3220**, the implant holder **3235** including a proximal portion **3235b** and a distal portion **3235a** adjoined by a connector **3237**, according to embodiments. In some embodiments, the implant holder **3235** may be structurally and/or functionally similar to the implant holder **335**, **835**, **3135**, and therefore, certain details of the implant holder **3135** are not described again with respect to FIGS. 41A-41D. As shown, the distal portion **3235a** and the proximal portion **3235b** define space therebetween configured to receive the implant **3260**. The proximal portion **3235a** defines one or more atrial channels **3236** through which one or more atrial tethers extend and a first or first set of ventricular channels **3238b** through which one or more ventricular tethers **3250** extend. In some embodiments, the proximal portion **3235a** includes an engagement portion **3239** configured to receive an atrial arm **3262** of the implant **3260**. The ventricular tether(s) **3150** may extend from the proximal portion **3235b**, outside of the implant holder **3235**, and into a second or second set of ventricular channels **3238a**. The ventricular tether(s) **3150** may extend from the distal portion **3235a** to a proximal end of the ventricular arm

3264, as shown. In some embodiments, the connector **3237** may include one or more hypotubes extending from the proximal portion **3135b** to the distal portion **3135a**. In some embodiments, the connector **3237** may include a pair of hypotubes.

FIGS. **42A-42F** and FIG. **44** graphical and flowchart representations, respectively, of a method of reducing heart valve regurgitation using an implant. The method can be applicable to any of the implant delivery systems and implants described herein. As shown, method **4600** may optionally include selecting an implant from a plurality of implants **4608**. For example, a plurality of implants corresponding to a variety of potential patients may be provided, and a physician may select an implant that may be appropriately sized for a patient's parameters (e.g., age, gender, type of heart valve, size of heart valve, severity of regurgitation, etc.). The implant may be releasably coupled to a delivery device (e.g., the implant holder) in the delivery configuration, as described previously. The method **4600** may further include advancing the implant in the delivery configuration to a heart valve using the implant delivery system at **4610**, as can be seen in FIG. **42A**. The heart valve may include the mitral valve, tricuspid valve, aortic valve, pulmonary valve, or portions thereof (e.g., leaflet, chordae tendineae, annulus). As shown, a distal end of the implant delivery system navigates from the atrium, through the mitral valve, and into the ventricle, as shown in FIG. **42B**.

Once positioned near the valve, the first and second arms of the implant may be separated at **4612**. The first and second arms may be separated using one or more elongate members that extend through the implant delivery system and a respective portion of the implant (e.g., one or more of the arms). For example, the physician may manipulate an actuator (e.g., press a button, pull a trigger) of the implant delivery system to apply a tensile force to the one or more elongate members (e.g., pulls a suture). The one or more elongate members may be coupled to the first and/or second arms such that the applied tensile force may overcome the biasing force of the central member that connects to each of the first and second arms. The tensile force may be of a magnitude sufficient to achieve a separation angle such that the implant may receive heart valve tissue between the first and second arms. The first arm may be positioned on a first side of the heart valve and the second arm may be positioned on a second side of the heart valve at **4620**, as shown in FIG. **42C-42D**. In some variations, the first segment or arm may be coupled to a leaflet surface on the first side, such as an atrial surface of a mitral valve leaflet. The second segment or arm may be coupled to a leaflet surface on the second side, such as a ventricular surface of the mitral valve leaflet. In further variations, the first and second segments or arms may each be coupled to sides of the tricuspid valve, aortic valve, and/or pulmonary valve.

In some embodiments, a position of the first arm and the second arm relative to one another may be determined using visualization markers on the first arm and the second arm. For example, at least one visualization marker (e.g., radiopaque marker) may be coupled to each of the first and second arms such that a physician may indirectly observe the configuration (e.g., open or closed) of the implant via a suitable imaging modality.

Upon a determination that a placement of the implant is acceptable, at least one of the first and second arms may be released to couple the implant to one or more of a native leaflet and chordae tendineae of the heart valve at **4630**. The implant may be coupled to the heart valve tissue via (i) the compressive force generated by the central member and

applied by the first and second arms and/or (ii) the friction force generated by the friction elements thereof. In some variations, the compressive and/or friction force may correspond to a pull-out force. The pull-out force may comprise the force (e.g., a tensile force acting along a longitudinal dimension of the implant) required to move the implant without first separating the first and second segments or arms. Unintended movement (e.g., movement without first separating the first and second segments or arms) of the implant while one or more of the compressive and friction forces may be applied by the implant to heart valve tissue may be referred to as dehiscence. In some variations, the compressive and friction forces may, alone or in combination, be multiples greater than the pull-out force, such as about 2 times greater, about 100 times greater, including about 5 times greater, about 10 times greater, about 20 times greater, about 30 times greater, about 40 times greater, or about 50 times greater.

The implant may optionally be repositioned relative to the heart valve at **4640**. Repositioning the implant may comprise separating the first and second arms, which may be achieved by applying a tensile force to the one or more elongate members routed through one or more of the first and second arms. Separating the arms may non-destructively (e.g., atraumatically) decouple the arms, including the friction elements, from the heart valve tissue such that the implant may move without scraping or otherwise damaging the heart valve tissue. The implant may then be moved to a second position, which may be observed and/or confirmed via the visualization markers previously described. The second position may be at a different location of the same heart valve to which the implant was previously coupled or, in some variations, may be a location on a different heart valve. With the arms still separated, the implant may receive heart valve tissue associated with the second position. The arms may then be released to recouple the implant to the heart valve. Recoupling the implant may comprise applying the same or similar friction and compressive forces as were applied during the first coupling. The implant may be repeatedly repositioned until the implant is appropriately positioned to effectively treat a cardiac condition, such as heart valve regurgitation. The elongate members routed through the implant may be cut or otherwise decoupled from the implant such that the elongate members and/or delivery device may be retracted from the patient, as can be seen in FIGS. **42E**. Optionally, in some variations, a valve replacement procedure (e.g., transcatheter mitral valve replacement) may be performed on the same heart valve coupled by the implant. For example, the valve replacement procedure may be performed to the same heart valve coupled by the implant without removing the coupled implant before, during, or after the procedure.

In further variations, more than one implant may be coupled to the heart valve, as illustrated in FIG. **42F**. For example, a first implant may be coupled to A1, a second implant may be coupled to P2, and a third implant may be coupled to A3. More than one treatment site may be treated to maximize the efficacy of the implants described herein. That is, a plurality of implants may be required to effectively reduce heart valve regurgitation. The plurality of implants may be implanted during the same surgical procedure, such that the same delivery device may be advanced to insert an implant, retracted, and subsequently advanced to insert an additional implant, or may be implanted during different surgical procedures. In some variations, the implant may optionally be removed from the heart valve at **4650**. Similar to the steps described previously, the first and second arms

71

may be separated via a tensile force applied to one or more elongate members routed through one or more of the first and second arms. The implant may then be recoupled to the delivery device such that the delivery device and implant may be removed from the patient's body

FIGS. 43A-43D illustrate a method for delivering an implant to a heart valve using an implant delivery system to treat heart valve regurgitation, according to embodiments. The method can be applicable to any of the implant delivery systems and implants described herein. A distal end of the guide catheter is advanced through the inferior vena cava into the right atrium and across the atrial septum into the left atrium, as shown in FIG. 43A. The delivery catheter and the implant catheter including the implant coupled thereto can be advanced through the guide catheter and advanced further into the left atrium (e.g., approximately 2 cm) into the left atrium, as shown in FIG. 43B. The delivery catheter and the implant catheter are steered toward the mitral valve, as shown in FIG. 43C. As shown in FIG. 43D, the implant catheter can be further advanced distally to position the implant relative to a leaflet of the mitral valve. The implant can then be disposed around a portion of the leaflet of the mitral valve, and the effectiveness of the implant for treating heart valve regurgitation can be tested before fully deploying the implant and removing the catheters from the heart.

FIG. 45 is a flow chart diagram showing an example method for delivering an implant to reduce heart valve regurgitation, according to embodiments. The method can be applicable to any of the implant delivery systems and implants described herein. The method includes inserting a steerable guide catheter having an implant catheter slidably and rotatably disposed therein through a vasculature of a patient such that a distal end of the guide catheter is disposed in an atrium of the heart, the implant being in its closed position and removably coupled to a distal end portion of the implant catheter during the inserting, at 4710. At 4720, with the guide catheter disposed within the atrium, the implant catheter may be advanced relative to the guide catheter to expose the implant from the guide catheter within the atrium. The implant can then be transitioned from a closed position to an open position, at 4730. At 4740, the method 4700 may optionally include rotating and/or steering (or deflecting) the implant catheter relative to the guide catheter to align the implant with a native valve leaflet. At 4750, the implant can be aligned in its open position to dispose the free end of the native leaflet within a space defined by the implant. For example, the implant can be opened such that a portion of heart tissue is positioned between the pair of arms of the implant. With the free end of the native leaflet disposed within the space defined by the implant, the implant can be transitioned from the open position to the closed position to clamp the free end of the native leaflet between the pair of arms, at 4760. With the implant clamped to the native leaflet, the implant catheter and guide catheter can be withdrawn from the patient at 4770. In some embodiments, withdrawing the guide catheter from the patient includes decoupling the elongate members (e.g., the atrial tether(s) and ventricular tether(s)) from the implant by pulling the atrial tethers and ventricular tethers from a proximal end of the implant delivery system. For example, a portion of each of the atrial tether and ventricular tether near a distal end of the implant delivery system may be severed or unlooped and each of the atrial tethers and ventricular tethers may be pulled proximally from the proximal end of the implant delivery system. In some embodiments, the implant catheter can be withdrawn proximally

72

into a lumen of the guide catheter, and the guide catheter with the implant catheter disposed therein can be withdrawn from the heart.

FIG. 46 is a flow chart for an example method of manufacture of an implant, according to embodiments. As shown, the implant, including the implant body and plates described herein, may be cut from a substantially flat sheet of material and subsequently folded, rolled, and/or bent into a 3-dimensional shape 4810. The initially flat configuration of the implant may facilitate faster and more affordable manufacturing. For example, the substantially flat sheet may be cut via laser cutting, waterjet cutting, plasma cutting, and/or handheld tin snips according to a pre-determined design of the implant. The pre-determined design of the implant may correspond to parameters associated with potential patients. Accordingly, a plurality of designs may be generated that correspond to a variety of patients. The cut-out implant body, which may still be flat, may then be further processed to eliminate any sharp edges via, for example, sanding, brushing, and/or deburring 4820. Eliminating sharp edges may prevent damaging tissue when advancing to or otherwise placing the implant at a treatment location and/or coupling the implant to heart valve tissue. The still-flat cut-out may then be set into a pre-set multi-dimensional configuration (e.g., the first configuration) described above 4830. In some embodiments, the implant body may be heat treated in order to be shape set. The flat sheet may comprise a thickness that may facilitate the manufacturing steps described (e.g., cutting, bending), such as about 50 microns to about 500 microns (including all values and sub-ranges therein). In some variations, the flat sheet may have a thickness of about 250 microns. Furthermore, the material used to form the implant body may be a pliable material to facilitate performing the manufacturing steps described. For example, the material may be a metal (e.g., nitinol, titanium, aluminum, gold, silver, alloys thereof) or a plastic (e.g., polypropylene, polyvinyl chloride, polyethylene, polyurethane). The material may further be determined by the shape-setting, shape-memory and/or super-elastic properties thereof. For example, the implant described herein may be capable of returning to the pre-set configuration after elastic deformation. In this way, the implant may advantageously return to the desired shape after being released from the delivery device or, for example, after impingement from a native leaflet. In some embodiments, the first plate and the second plate may be set into a predetermined configuration at 4840. In some variations, the first plate may be attached to the first arm of the implant body 4850. Similarly, the second plate may be attached to the second arm of the implant body 4860. In yet further variations, at least one visualization marker may be attached to the first arm (e.g., via welding) 4870. Similarly, at least one visualization marker may be attached to the second arm (e.g., via welding) 4880. Then, at least one cover may be attached to the implant body 4890. In some variations, a first cover may be coupled to the first arm, second arm, and/or central member, and a second cover may be coupled to the support members extending from the first arm.

EXAMPLES

The following examples are offered to illustrate, but not to limit, the claimed invention.

Example 1: Pull-Out Force

A device as described herein was coupled to an anterior leaflet or a posterior leaflet in 5 pig mitral valves to evaluate

the force required for dehiscence and/or pull-out. Dehiscence was defined as motion of the implant by at least 2 mm from its implanted position. Pull-out was defined as complete removal of the implant from the leaflet. Graded weights were suspended from the implant with a non-extensible suture until dehiscence and/or pull-out occurred. The average weight at which the implant moved from its implanted location by 2 mm was 500 g, which corresponds to 4.9N force. None of the tested configurations led to pull out of the implant. Accordingly, the force required to remove the implant was significantly higher than real physiological conditions the implant may experience.

Example 2: Compressive Force

A device as described herein was tested to measure the compressive force imposed on mitral valve tissue by the two segments or arms. A custom made load cell setup was designed to measure the compressive force that the implant imposes on native mitral valve leaflets of different thicknesses. The implant imposed a minimum of 200 mN compressive force, which increased linearly as leaflet thickness increased. Upon performing theoretical calculations of the hemodynamic forces imposed by blood flow and transmitral pressure, the measured force was 40× greater than the calculated force associated with dehiscence. Accordingly, the compressive force imposed by the implant on the native mitral valve was sufficient for firm attachment without any dehiscence.

Example 3: Patient Populations

Three clinically relevant valve pathologies were studied—atrial FMR (aFMR), where the primary contributor to loss of valve coaptation is severe annular dilatation (seen in patients with chronic atrial fibrillation); ischemic FMR (iFMR), where the contributor to loss of valve coaptation is annular dilatation combined with focal displacement of the postero-medial papillary muscle due to an underlying postero-lateral or inferior myocardial infarction; dilated cardiomyopathy FMR (dFMR), where the contributor to loss of valve coaptation is annular dilatation combined with displacement of both papillary muscles due to concentric dilatation of the left ventricle from various etiologies. A left heart simulator was used. Pig mitral valves were mounted into this simulator, with the annulus mounted into a system that enables dilatation, and the papillary muscles mounted onto a system that allows for spatial displacement/relocation. The simulator was then used to induce aFMR, iFMR or dFMR. Mitral valve hemodynamics were measured with calibrated pressure transducers, and flow was measured with calibrated electromagnetic flow probes. At baseline, which is the valve in its healthy/non-diseased configuration, regurgitant fraction was 0%. With the implant deployed onto the healthy/non diseased valve, the regurgitant fraction was $1.03 \pm 0.86\%$, which is negligible. In the atrial FMR condition (aFMR), significant rise in regurgitation fraction was observed with annular dilatation, at $14.78 \pm 3.10\%$. With the implant deployed, aFMR was significantly reduced to $6.79 \pm 1.89\%$ ($p=0.001$). In the ischemic FMR condition (iFMR), the regurgitant fraction before the repair was $20.52 \pm 2\%$ and was reduced with use of the implant to $6.63 \pm 2.15\%$ ($p<0.0001$). In the dilated cardiomyopathy FMR condition (dFMR), the regurgitant fraction before the repair was $21.93 \pm 7.2\%$, which was significantly reduced by use of the implant to $7.12 \pm 2.8\%$. The results are figuratively depicted in FIG. 16. Accordingly, Quantitative measurement

of mitral regurgitation in different clinically relevant anatomical states that cause FMR demonstrate that the implant can very effectively treat aFMR, iFMR, and dFMR.

While various inventive embodiments have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the inventive embodiments described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications and anatomy (e.g., intracranial and extracranial vascular structure) for which the inventive teachings is/are used. Those skilled in the art will recognize or be able to ascertain using no more than routine experimentation, many equivalents to the specific inventive embodiments described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto; inventive embodiments may be practiced otherwise than as specifically described and claimed. Inventive embodiments of the present disclosure are directed to each individual feature, system, article, material, kit, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the inventive scope of the present disclosure.

Also, various inventive concepts may be embodied as one or more methods, of which an example has been provided. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments.

As used herein, the terms “about” and/or “approximately” when used in conjunction with numerical values and/or ranges generally refer to those numerical values and/or ranges near to a recited numerical value and/or range. In some instances, the terms “about” and “approximately” may mean within 10% of the recited value. For example, in some instances, “about 100 [units]” may mean within 10% of 100 (e.g., from 90 to 110). The terms “about” and “approximately” may be used interchangeably.

Any and all references to publications or other documents, including but not limited to, patents, patent applications, articles, webpages, books, etc., presented anywhere in the present application, are herein incorporated by reference in their entirety. Moreover, all definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

The specific examples and descriptions herein are exemplary in nature and embodiments may be developed by those skilled in the art based on the material taught herein without departing from the scope of the present invention.

The invention claimed is:

1. A system, comprising:

an implant configured to be coupled to a native heart valve leaflet of a heart of a patient to reduce valve regurgitation, the implant including (1) an atrial arm and a

75

ventricular arm collectively movable between a closed position and an open position and defining a space sufficient to capture a free end of the native leaflet between the atrial arm and the ventricular arm, and (2) a leaflet enhancer;

an implant catheter including an implant holder at a distal end thereof, the implant holder including a distal portion and a proximal portion collectively defining an opening therebetween the proximal portion and the distal portion connected by a connector extending across a portion of the opening, the front side of the implant holder including an engagement surface, a first portion of the implant configured to be disposed in the opening between the distal and the proximal portion, a second portion of the implant configured to abut the engagement surface,

the proximal portion of the implant holder defining a pair of atrial channels configured to slidably receive an atrial tether, the atrial tether configured to be removably coupled to the atrial arm of the implant,

the proximal portion of the implant holder defining a first pair of ventricular channels configured to slidably receive a first portion of a ventricular tether, the distal portion of the implant holder defining a second pair of ventricular channels separate from the first pair of ventricular channels and configured to slidably receive a second portion of the ventricular tether such that the ventricular tether extends through the first pair of ventricular channels, outside of the implant holder, and through the second pair of ventricular channels, the ventricular tether configured to be removably coupled to the ventricular arm of the implant.

2. The system of claim 1, wherein a portion of the leaflet enhancer is configured to extend laterally beyond the connector and a perimeter defined by the implant holder.

3. The system of claim 2, wherein the leaflet enhancer is deformable such that the portion of the leaflet enhancer can extend beyond the connector.

4. The system of claim 1, wherein tension applied to a proximal portion of the atrial tether moves the atrial arm of the implant toward the engagement surface.

5. The system of claim 4, wherein tension is released from the proximal portion of the ventricular tether to move the ventricular arm of the implant closer to the atrial arm of the implant, thereby transitioning the implant from the open position towards the closed position.

6. The system of claim 1, wherein tension applied to a proximal portion of the ventricular tether moves the ventricular arm of the implant away from the atrial arm of the implant, thereby transitioning the implant towards the open position.

7. The system of claim 1, wherein the distal portion of the implant holder includes an atraumatic distal end.

8. The system of claim 1, wherein the pair of atrial channels include a pair of atrial openings in the proximal portion of the implant holder,

the first pair of ventricular channels include a first pair of ventricular openings in the proximal portion of the implant holder,

the second pair of ventricular channels include a second pair of ventricular openings on a back side of the distal portion of the implant holder and a third pair of ventricular openings on a front side of the distal portion of the implant holder.

9. The system of claim 1, wherein the pair of atrial channels and the first pair of ventricular channels are con-

76

figured to receive a flushing fluid from a fluid source to prevent blood clots from forming therein.

10. The system of claim 1, wherein the connector includes one or more hypotubes.

11. The system of claim 1, wherein the atrial arm of the implant includes an atrial suture loop disposed thereon and the ventricular arm of the implant includes a ventricular suture loop disposed thereon, the atrial tether configured to be routed through the atrial suture loop to operably couple to the atrial tether to the atrial arm and the ventricular tether configured to be routed through the ventricular suture loop to operably couple the ventricular tether to the ventricular arm.

12. The system of claim 1, wherein the implant holder includes a transparent material.

13. The system of claim 1, wherein a distance between the proximal end and the distal end of the implant holder is adjustable.

14. A system, comprising:
an implant configured to be delivered to a native heart valve leaflet of a heart of a patient to reduce valve regurgitation through a gap within the native heart valve, the implant including (1) an atrial arm and a ventricular arm collectively movable between a closed position and an open position and defining a space sufficient to capture a free end of the native leaflet B between the atrial arm and the ventricular arm, the ventricular arm being configured to move laterally relative to the atrial arm to define the space; and (2) a leaflet enhancer

an implant catheter including an implant holder at a distal end thereof, the implant holder configured to hold the implant to form a distal assembly, the implant holder defining a pair of atrial channels configured to slidably receive an atrial tether, the atrial tether configured to be removably coupled to the atrial arm of the implant, the implant holder defining a first pair of ventricular channels and a second pair of ventricular channels separate from the first pair of ventricular channels, the first pair of ventricular channels and the second pair of ventricular channels configured to slidably receive a ventricular tether, the ventricular tether configured to be removably coupled to the ventricular arm of the implant,

a maximum lateral dimension of the distal assembly being defined by the implant when the implant is in the closed position.

15. The system of claim 14, wherein tension applied to a proximal portion of the ventricular tether moves the ventricular arm of the implant away from the atrial arm of the implant, thereby transitioning the implant towards the open position,

tension released from the proximal portion of the ventricular tether moves the ventricular arm of the implant closer to the atrial arm of the implant, thereby transitioning the implant from the open position towards the closed position.

16. The system of claim 14, wherein the implant catheter is configured to be manipulated to position the implant, when the implant is coupled to the implant catheter in the closed position, near the gap within the native heart valve leaflet such that a level of regurgitation can be determined by performing at least one of fluoroscopy imaging or echocardiography.

17. The system of claim 14, wherein a level of valve regurgitation can be determined when the implant is positioned near the gap within the native heart valve leaflet

77

without capturing the free end of the native leaflet between the atrial arm and the ventricular arm.

18. The system of claim 14, wherein the implant holder includes a distal portion and proximal portion separate from the distal portion, the distal portion and the proximal portion defining an opening therebetween and connected by a connector extending across the opening, a portion of the implant configured to be disposed in the opening.

19. The system of claim 18, wherein the portion of the implant configured to be disposed in the opening is a first portion, a front side of the proximal portion of the implant holder includes an engagement surface, a second portion of the implant configured to abut the engagement surface.

20. The system of claim 18, wherein a portion of the leaflet enhancer is configured to extend beyond the connector and a perimeter defined by the implant holder so that the maximum lateral dimension of the leaflet enhancer is greater than a maximum lateral dimension of the implant holder.

21. The system of claim 20, wherein the leaflet enhancer is deformable such that the portion of the leaflet enhancer can extend beyond the connector.

22. The system of claim 18, wherein the connector includes one or more hypotubes.

23. The system of claim 14, wherein the implant holder includes a transparent material.

24. A method of coupling an implant to a native heart valve leaflet of a heart valve of a patient to reduce valve regurgitation, the implant including (1) a pair of arms movable between an at-rest closed position and a biased open position, the pair of arms in the biased open position defining a space therebetween sufficient to capture a free end of the native leaflet between the pair of arms, and (2) a leaflet enhancer, the method comprising:

positioning the implant in an opening defined by the heart valve without capturing the free end of the native heart valve leaflet between the pair of arms;

determining a level of valve regurgitation when the implant is positioned in the opening defined by the heart valve;

after determining the level of valve regurgitation, transitioning the implant from its closed position to the biased open position; and

transitioning the implant from the biased open position towards its closed position to clamp the free end of the native leaflet between the pair of arms.

78

25. The method of claim 24, wherein the method further comprises:

inserting a steerable guide catheter having an implant catheter slidably and rotatably disposed therein through a vasculature of the patient such that a distal end of the guide catheter is disposed in an atrium of the heart, the implant being in its closed position and removably coupled to a distal end portion of the implant catheter during the inserting; and

with the guide catheter disposed within the atrium, advancing the implant catheter relative to the guide catheter to expose the implant within the atrium.

26. The method of claim 25, wherein a cross-sectional area of the leaflet enhancer is greater than or equal to a distal end portion of the implant catheter such that the level of valve regurgitation can be determined without capturing the free end of the native leaflet between the pair of arms.

27. The method of claim 25, wherein the implant catheter includes an implant holder disposed on a distal end thereof and configured to hold the implant, the implant holder including:

a distal portion and a proximal portion collectively defining an opening therebetween extending from a front side of the implant holder to a back side of the implant holder, the proximal portion and the distal portion connected by a connector extending across the opening, a portion of the implant configured to be disposed in the opening between the distal portion and the proximal portion.

28. The method of claim 27, wherein determining the level of valve regurgitation includes performing fluoroscopy imaging, the implant being discernable from the implant holder when the fluoroscopy imaging is performed.

29. The method of claim 24, wherein the determining the level of valve regurgitation includes performing at least one of fluoroscopy imaging of the heart or echocardiography while the implant is positioned in the opening defined by the heart valve.

30. The method of claim 24, wherein the pair of arms includes a first arm and a second arm and the implant catheter includes a first tether coupled to the first arm and a second tether coupled to the second arm.

31. The method of claim 30, wherein tension is not removed from the first tether and the second tether during the determining the level of valve regurgitation.

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